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CHAPTER

6

CONIC SECTION

Conic Sections or simply conics are the curves obtained by cutting a right circular cone by a plane.

If the cone is cut by a plane perpendicular to the axis of the cone, then the section is a circle.

Definition of a Circle

(Lahore Board 2009, 2011)

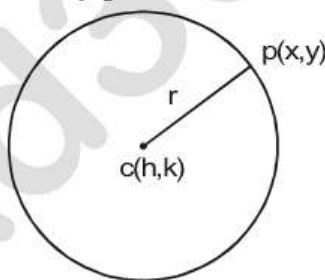
The set of all the points in the plane that are equally distant from a fixed point is called a circle.

The fix point is called center of circle. The distance from a center to any point of circle is called radius of circle.

Equation of Circle

If $c(h, k)$ be center of a circle and $p(x, y)$ be any point on the circle.

' r ' is distance from center to any point of circle i.e. r is radius of circle.



By distance formula

$$|cp| = \sqrt{(x-h)^2 + (y-k)^2}$$

$$r = \sqrt{(x-h)^2 + (y-k)^2}$$

$$(\because |cp| = r)$$

$$r^2 = (x-h)^2 + (y-k)^2 \quad \dots\dots\dots (I)$$

is an equation of circle of standard form.

If center of circle is origin i.e; $c(0, 0)$ then equation of circle becomes

$$r^2 = (x-0)^2 + (y-0)^2$$

$$r^2 = x^2 + y^2$$

Point Circle:

If $r = 0$ then circle is called point circle.


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Parametric Equations of Circle:

$$x = r \cos \theta, \quad y = r \sin \theta$$

General Form of an equation of Circle

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle g, f, c being constants.

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$x^2 + 2gx + g^2 + y^2 + 2fy + f^2 = g^2 + f^2 - c \quad (\because \text{Adding } g^2 \text{ and } f^2 \text{ on both sides})$$

$$(x^2 + 2gx + g^2) + (y^2 + 2fy + f^2) = g^2 + f^2 - c$$

$$(x + g)^2 + (y + f)^2 = g^2 + f^2 - c$$

$$(x - (-g))^2 + (y - (-f))^2 = g^2 + f^2 - c$$

Which is standard form of an equation of circle, where center is $(-g, -f)$ and radius is $\sqrt{g^2 + f^2 - c}$.



EXERCISE 6.1

Q.1: In each of following find an equation of circle with

- (a) Center at (5, -2) and radius 4.
(Lahore Board 2009)

Solution:

The equation of circle by standard form is

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(x - 5)^2 + (y - (-2))^2 = (4)^2$$

$$(x - 5)^2 + (y + 2)^2 = 16$$

$$x^2 + 25 - 10x + y^2 + 4 + 4y - 16 = 0$$

$$x^2 + y^2 - 10x + 4y + 13 = 0$$

- (b) Center at $(\sqrt{2}, -3\sqrt{3})$ radius $2\sqrt{2}$

Solution:

The equation of circle by standard form is

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(x - \sqrt{2})^2 + (y - (-3\sqrt{3}))^2 = (2\sqrt{2})^2$$

$$x^2 + 2 - 2\sqrt{2}x + y^2 + 27 + 6\sqrt{3}y = 8$$

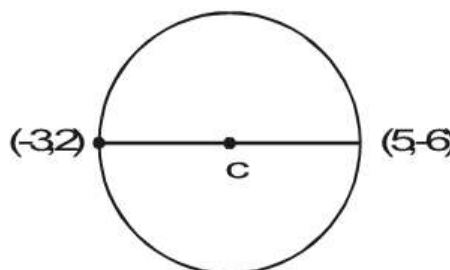
$$x^2 + y^2 - 2\sqrt{2}x + 6\sqrt{3}y + 29 - 8 = 0$$

$$x^2 + y^2 - 2\sqrt{2}x + 6\sqrt{3}y + 21 = 0$$



(c) ends of a diameter at $(-3, 2)$ & $(5, -6)$

Solution:



Center = Mid point of ends of diameter

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Center} = \left(\frac{-3 + 5}{2}, \frac{2 - 6}{2} \right) = \left(\frac{2}{2}, \frac{-4}{2} \right) = (1, -2)$$

Radius = Distance from center to any one point of circle

$$= \sqrt{(1 + 3)^2 + (-2 - 2)^2}$$

$$= \sqrt{16 + 16}$$

$$= \sqrt{32}$$

The equation of circle by standard form

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(x - 1)^2 + (y - (-2))^2 = (\sqrt{32})^2$$

$$(x - 1)^2 + (y + 2)^2 = 32$$

$$x^2 + 1 - 2x + y^2 + 4 + 4y - 32 = 0$$

$$x^2 + y^2 - 2x + 4y - 27 = 0$$

Q.2: Find center & radius of the circle with the given equation

(a) $x^2 + y^2 + 12x - 10y = 0$

(Lahore Board 2010)

Solution:

General form of an equation of circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Compare it with $x^2 + y^2 + 12x - 10y = 0$

$$2g = 12, \quad 2f = -10, \quad c = 0$$

$$g = 6, \quad f = -5, \quad c = 0$$

We know that

Center of general form is $(-g, -f) = (-6, 5)$

$$\text{Radius of circle is} = \sqrt{g^2 + f^2 - c}$$

$$= \sqrt{(6)^2 + (-5)^2 - 0}$$



$$= \sqrt{36 + 25}$$

$$= \sqrt{61}$$

(b) $5x^2 + 5y^2 + 14x + 12y - 10 = 0$

Solution:

$$5x^2 + 5y^2 + 14x + 12y - 10 = 0$$

Dividing throughout by 5

$$\frac{5x^2}{5} + \frac{5y^2}{5} + \frac{14x}{5} + \frac{12y}{5} - \frac{10}{5} = 0$$

$$x^2 + y^2 + \frac{14x}{5} + \frac{12y}{5} - 2 = 0$$

Comparing with general form of an equation of circle

$$2g = \frac{14}{5}, \quad 2f = \frac{12}{5}, \quad c = -2$$

$$g = \frac{7}{5}, \quad f = \frac{6}{5}, \quad c = -2$$

$$\text{Center of circle} = (-g, -f) = \left(-\frac{7}{5}, -\frac{6}{5}\right)$$

$$\begin{aligned} \text{Radius of circle} &= \sqrt{(g^2 + f^2 - c)} = \sqrt{\left(\frac{7}{5}\right)^2 + \left(\frac{6}{5}\right)^2 - (-2)} \\ &= \sqrt{\frac{49}{25} + \frac{36}{25} + 2} \\ &= \sqrt{\frac{49 + 36 + 50}{25}} = \sqrt{\frac{135}{25}} \\ &= \frac{\sqrt{135}}{5} \text{ Ans.} \end{aligned}$$

(c) $x^2 + y^2 - 6x + 4y + 13 = 0$ (Lahore Board 2009)

Solution:

$$x^2 + y^2 - 6x + 4y + 13 = 0$$

Comparing it with general form of an equation of circle

$$2g = -6, \quad 2f = 4, \quad c = 13$$

$$g = -3, \quad f = 2, \quad c = 13$$

$$\text{Center of circle} = (-g, -f) = (3, -2)$$

$$\begin{aligned} \text{Radius of circle} &= \sqrt{g^2 + f^2 - c} \\ &= \sqrt{(-3)^2 + (2)^2 - 13} \\ &= \sqrt{9 + 4 - 13} \end{aligned}$$



$$= \sqrt{13 - 13}$$

$$= 0$$

(d) $4x^2 + 4y^2 - 8x + 12y - 25 = 0$

Solution:

$$4x^2 + 4y^2 - 8x + 12y - 25 = 0$$

Dividing throughout by 4

$$x^2 + y^2 - 2x + 3y - \frac{25}{4} = 0$$

Comparing it with general form of an equation of circle

$$2g = -2 \quad 2f = 3 \quad c = -\frac{25}{4}$$

$$g = -1, \quad f = \frac{3}{2}$$

$$\text{Center of Circle} = (-g, -f) = (1, -\frac{3}{2})$$

$$\begin{aligned} \text{Radius of Circle} &= \sqrt{g^2 + f^2 - c} \\ &= \sqrt{(-1)^2 + \left(\frac{3}{2}\right)^2 - \left(-\frac{25}{4}\right)} \\ &= \sqrt{1 + \frac{9}{4} + \frac{25}{4}} = \sqrt{\frac{4 + 9 + 25}{4}} \\ &= \sqrt{\frac{38}{4}} = \frac{\sqrt{38}}{2} \end{aligned}$$

Q.3: Write an equation of the circle that passes through the given points.

(a) A(4, 5), B(-4, -3), C(8, -3)

(Lahore Board 2009, 2011)

Solution:

When the circle passes through different points we use general equation of circle.

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (I)$$

For A (4, 5) we have

$$(4)^2 + (5)^2 + 2g(4) + 2f(5) + c = 0$$

$$16 + 25 + 8g + 10f + c = 0$$

$$41 + 8g + 10f + c = 0 \quad (i)$$

For B (-4, -3) we have

$$(-4)^2 + (-3)^2 + 2g(-4) + 2f(-3) + c = 0$$

$$16 + 9 - 8g - 6f + c = 0$$

$$25 - 8g - 6f + c = 0 \quad (ii)$$



For C (8, -3) we have

$$(8)^2 + (-3)^2 + 2g(8) + 2f(-3) + c = 0$$

$$64 + 9 + 16g - 6f + c = 0$$

$$73 + 16g - 6f + c = 0 \quad \text{(iii)}$$

Subtracting (ii) from (i)

$$41 + 8g + 10f + c = 0$$

$$-25 - 8g - 6f + c = 0$$

$$16 + 16g + 16f = 0$$

$$1 + g + f = 0 \quad \text{(iv)}$$

$$1 - 2 + f = 0$$

$$1 - 2 + f = 0$$

$$-1 + f = 0 \Rightarrow \boxed{f = 1}$$

Subtracting (iii) from (ii)

$$25 - 8g - 6f + c = 0$$

$$-73 + 16g - 6f + c = 0$$

$$-48 - 24g = 0$$

$$2 + g = 0 \quad \boxed{g = -2}$$

Put in (iv)

Now putting values in (i)

$$41 + 8(-2) + 10(1) + c = 0$$

$$41 - 16 + 10 + c = 0$$

$$35 + c = 0$$

$$\boxed{c = -35} \text{ Substitute all value in (I)}$$

$$x^2 + y^2 + 2(-2)x + 2(1)y - 35 = 0$$

$$x^2 + y^2 - 4x + 2y - 35 = 0$$

(b) A(-7, 7) B(5, -1) C(10, 0)

Solution:

General equation of circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \text{(I)}$$

For A(-7, 7) we have $(-7)^2 + (7)^2 + 2g(-7) + 2f(7) + c = 0$

$$49 + 49 - 14g + 14f + c = 0$$

$$98 - 14g + 14f + c = 0 \quad \text{..... (i)}$$

For B(5, -1) $(5)^2 + (-1)^2 + 2g(5) + 2f(-1) + c = 0$

$$25 + 1 + 10g - 2f + c = 0$$

$$26 + 10g - 2f + c = 0 \quad \text{..... (ii)}$$

For C(10, 0) $(10)^2 + (0)^2 + 2g(10) + 2f(0) + c = 0$

$$100 + 20g + c = 0 \quad \text{..... (iii)}$$



Subtracting (ii) from (i)

$$\begin{array}{r}
 98 - 14g + 14f + c = 0 \\
 - 26 \mp 10g \mp 2f \pm c = 0 \\
 \hline
 72 - 24g + 16f = 0 \quad (\text{iv}) \\
 8(9 - 3g + 2f) = 0 \\
 9 - 3g + 2f = 0 \quad (\text{vi})
 \end{array}$$

$$\boxed{g = -5} \quad \text{Put in (vi)}$$

Put in (iii)

$$\begin{array}{r}
 100 + 20(-5) + c = 0 \\
 100 - 100 + c = 0 \\
 \boxed{c = 0}
 \end{array}$$

Substitute all values in (I)

$$\begin{array}{l}
 x^2 + y^2 + 2(-5)x + 2(-12)y + 0 = 0 \\
 x^2 + y^2 - 10x - 24y = 0 \quad \text{Ans.}
 \end{array}$$

(c) $A(a, 0)$, $B(0, b)$, $C(0, 0)$ **Solution:**

General equation of circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (\text{I})$$

For $A(a, 0)$ we have

$$\begin{array}{r}
 a^2 + (0)^2 + 2g(a) + 2f(0) + c = 0 \\
 a^2 + 2ag + c = 0 \quad (\text{i})
 \end{array}$$

For $B(0, b)$

$$b^2 + 2fb + c = 0 \quad (\text{ii})$$

For $C(0, 0)$

$$(0)^2 + (0)^2 + 2g(0) + 2f(0) + c = 0$$

Subtracting (iii) from (ii)

$$\begin{array}{r}
 26 + 10g - 2f + c = 0 \\
 - 100 \pm 20g \pm c = 0 \\
 \hline
 - 74 - 10g - 2f = 0 \quad (\text{v}) \\
 \text{Adding (v) \& (vi)}
 \end{array}$$

$$\begin{array}{r}
 - 74 - 10g - 2f = 0 \\
 9 - 3g + 2f = 0 \\
 \hline
 - 65 - 13g = 0 \\
 - 65 = 13g
 \end{array}$$

$$9 - 3(-5) + 2f = 0$$

$$9 + 15 + 2f = 0$$

$$2f = -24 \quad \boxed{f = -12}$$



$$\Rightarrow \boxed{c = 0}$$

Putting value of c in (i) and (ii) we have

$$\begin{array}{l|l} a^2 + 2ag = 0 & b^2 + 2fb = 0 \\ a(a + 2g) = 0 & b(b + 2f) = 0 \\ a + 2g = 0 & b + 2f = 0 \\ 2g = -a & \end{array}$$

$$\Rightarrow \boxed{g = \frac{-a}{2}}$$

$$\boxed{f = \frac{-b}{2}}$$

Substituting all values in (I) we have

$$x^2 + y^2 + 2\left(\frac{-a}{2}\right)x + 2\left(\frac{-b}{2}\right)y + c = 0$$

$$x^2 + y^2 - ax - by + c = 0 \quad \text{Ans}$$

(d) **A(5, 6) , B(-3, 2) , C(3, -4)**
(Gujranwala Board 2006)

Solution:

General equation of circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \text{(I)}$$

For A (5, 6) we have

$$\begin{aligned} (5)^2 + (6)^2 + 2g(5) + 2f(6) + c &= 0 \\ 25 + 36 + 10g + 12f + c &= 0 \\ 61 + 10g + 12f + c &= 0 \end{aligned} \quad \text{(i)}$$

For B(-3, 2)

$$\begin{aligned} (-3)^2 + (2)^2 + 2g(-3) + 2f(2) + c &= 0 \\ 9 + 4 - 6g + 4f + c &= 0 \\ 13 - 6g + 4f + c &= 0 \end{aligned} \quad \text{(ii)}$$

For C(3, -4)

$$\begin{aligned} (3)^2 + (-4)^2 + 2g(3) + 2f(-4) + c &= 0 \\ 9 + 16 + 6g - 8f + c &= 0 \\ 25 + 6g - 8f + c &= 0 \end{aligned} \quad \text{(iii)}$$

Subtracting (ii) from (i) & (iii) from (ii)

$$\begin{array}{rcl} 61 + 10g + 12f + c & = & 0 \\ - 13 - 6g + 4f + c & = & 0 \\ \hline 48 + 16g + 8f & = & 0 \end{array} \quad \begin{array}{rcl} 13 - 6g + 4f + c & = & 0 \\ - 25 + 6g - 8f + c & = & 0 \\ \hline - 12 - 12g + 12f & = & 0 \end{array}$$

$$\begin{array}{rcl} 48 + 16g + 8f & = & 0 \\ 6 + 2g + f & = & 0 \dots \text{(iv)} \quad (\text{Dividing by 8}) \end{array} \quad \begin{array}{rcl} - 12 - 12g + 12f & = & 0 \\ 1 + g - f & = & 0 \dots \text{(v)} \end{array}$$

Adding (iv) & (v)

$$\begin{array}{rcl} 6 + 2g + f & = & 0 \\ 1 + g - f & = & 0 \\ \hline \end{array}$$



$$7 + 3g = 0$$

$$\boxed{g = -\frac{7}{3}} \text{ Put in (v)} \quad 1 - \frac{7}{3} - f = 0$$

$$\frac{3-7}{3} = f$$

$$\boxed{\frac{-4}{3} = f}$$

Put in (ii)

$$13 - 6\left(\frac{-7}{3}\right) + 4\left(\frac{-4}{3}\right) + c = 0$$

$$13 + 14 - \frac{16}{3} + c = 0$$

$$65 + 3c = 0 \quad \boxed{c = -\frac{65}{3}} \text{ substitute all values in (I)}$$

$$x^2 + y^2 + 2\left(\frac{-7}{3}\right)x + 2\left(\frac{-4}{3}\right)y - \frac{65}{3} = 0$$

$$x^2 + y^2 - \frac{14}{3}x - \frac{8}{3}y - \frac{65}{3} = 0 \quad \text{Ans}$$

Q.4: In each of the following, find an equation of the circle passing through

(a) A(3, -1) B(0, 1) and having center at $4x - 3y - 3 = 0$

Solution:

We know that General equation of circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \text{(I)}$$

$$\text{For A(3, -1)} \quad (3)^2 + (-1)^2 + 2g(3) + 2f(-1) + c = 0$$

$$9 + 1 + 6g - 2f + c = 0$$

$$10 + 6g - 2f + c = 0 \quad \text{(i)}$$

$$\text{For B(0, 1)} \quad (0)^2 + (1)^2 + 2g(0) + 2f(1) + c = 0$$

$$1 + 2f + c = 0 \quad \text{(ii)}$$

Subtracting (ii) from (i)

$$10 + 6g - 2f + c = 0$$

$$- 1 \pm 0g \pm 2f \pm c = 0$$

$$9 + 6g - 4f = 0 \quad \text{(iii)}$$

Since center $(-g, -f)$ lies at $4x - 3y - 3 = 0$

we have $4(-g) - 3(-f) - 3 = 0$



$$-4g + 3f - 3 = 0 \quad \dots\dots (iv)$$

Multiplying equation (iii) by 3 and (iv) by 4 and adding

$$\begin{array}{rcl} 27 + 18g - 12f & = & 0 \\ -12 - 16g + 12f & = & 0 \\ \hline 15 + 2g & = & 0 \\ 2g & = & -15 \end{array}$$

$$g = \frac{-15}{2} \text{ Put in (iii)}$$

$$\begin{aligned} 9 + 6\left(\frac{-15}{2}\right) - 4f &= 0 \\ 9 - 45 - 4f &= 0 \\ -36 &= 4f \end{aligned}$$

$$f = -9 \text{ Put in (ii)}$$

$$1 + 2(-9) + c = 0$$

$$1 - 18 + c = 0 \quad \boxed{c = 17}$$

Substitute values in (I)

$$\begin{aligned} x^2 + y^2 + 2\left(\frac{-15}{2}\right)x + 2(-9)y + 17 &= 0 \\ x^2 + y^2 - 15x - 18y + 17 &= 0 \quad \text{Ans.} \end{aligned}$$

(b) A (-3, 1) with radius 2 and center at $2x - 3y + 3 = 0$

Solution:

With center (h, k) and radius r, we know equation of standard form of circle is

$$(x - h)^2 + (y - k)^2 = r^2$$

For A(-3, 1) $(-3 - h)^2 + (1 - k)^2 = r^2$

$$9 + h^2 + 6h + 1 + k^2 - 2k - 4 = 0$$

$$h^2 + k^2 + 6h - 2k + 6 = 0 \quad (i)$$

Since center (h, k) lies at $2x - 3y + 3 = 0$

$$2h - 3k + 3 = 0$$

$$\Rightarrow h = \frac{3k - 3}{2} \quad (ii)$$

Put in (i)

$$\left(\frac{3k - 3}{2}\right)^2 + k^2 + 6\left(\frac{3k - 3}{2}\right) - 2k + 6 = 0$$



$$\frac{9k^2 + 9 - 18k}{4} + k^2 + 9k - 9 - 2k + 6 = 0$$

$$9k^2 + 9 - 18k + 4k^2 + 36k - 36 - 8k + 24 = 0$$

$$13k^2 + 10k - 3 = 0$$

$$13k^2 + 13k - 3k - 3 = 0$$

$$(13k - 3)(k + 1) = 0$$

$$k = -1$$

$$k = \frac{3}{13}$$

Putting $k = -1$ in (ii)

$$h = \frac{3(-1) - 3}{2}$$

$$= \frac{-3 - 3}{2}$$

$$= \frac{-6}{2}$$

$$= -3$$

Required equation of circle

$$(x + 3)^2 + (y + 1)^2 = 4$$

$$\left(x + \frac{15}{13}\right)^2 + \left(y - \frac{3}{13}\right)^2 = 4 \text{ Ans.}$$

Putting $k = \frac{3}{13}$ in (ii)

$$h = \frac{3\left(\frac{3}{13}\right) - 3}{2}$$

$$= \frac{\frac{9}{13} - 3}{2}$$

$$= \frac{9 - 39}{26}$$

$$= \frac{-30}{26}$$

$$= \frac{-15}{13}$$

(c) A(5, 1) and tangent to the line $2x - y - 10 = 0$ at B(3, -4)

Solution:

We know that General Equation of circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ (I)

$$\text{For A (5, 1)} \quad (5)^2 + (1)^2 + 2g(5) + 2f(1) + c = 0$$

$$25 + 1 + 10g + 2f + c = 0$$

$$26 + 10g + 2f + c = 0 \quad (i)$$

$$\text{For B (3, -4)} \quad (3)^2 + (-4)^2 + 2g(3) + 2f(-4) + c = 0$$

$$9 + 16 + 6g - 8f + c = 0$$

$$25 + 6g - 8f + c = 0 \quad (ii)$$

Subtracting (ii) from (i)

$$26 + 10g + 2f + c = 0$$

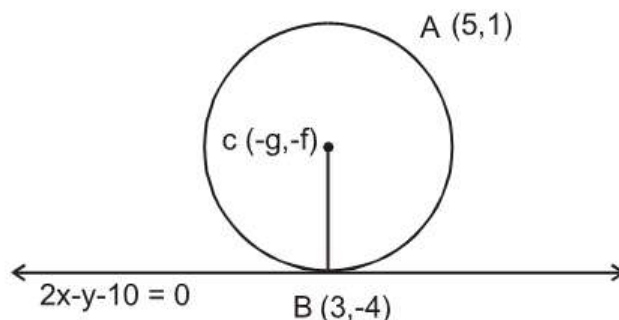
$$- 25 + 6g - 8f + c = 0$$

$$1 + 4g + 10f = 0 \quad (iii)$$



$$m_1 = \text{Slope of BC} = \frac{-f+4}{-g-3} = \frac{+(f-4)}{+(g+3)} = \frac{f-4}{g+3}$$

$$m_2 = \text{Slope of line } 2x - y - 10 = 0 \text{ is } \frac{-\text{coefficient of } x}{\text{coefficient of } y} \\ = -\frac{2}{-1} = 2$$



Since lines are perpendicular

So $m_1 m_2 = -1$

$$\frac{f-4}{g+3} \times 2 = -1$$

$$2f - 8 = -g - 3$$

$$-5 + g + 2f = 0 \quad \text{(iv)}$$

Now solving (iii) and (iv)

Multiply equation (iv) by 4 and subtracting from (iii)

$$1 + 4g + 10f = 0$$

$$\mp 20 \pm 4g \pm 8f = 0$$

$$21 + 2f = 0$$

$$\boxed{f = -\frac{21}{2}} \text{ Put in (iv)}$$

$$-5 + g + 2\left(-\frac{21}{2}\right) = 0$$

$$-26 + g = 0$$

$$\boxed{g = 26}$$

Putting values of g & f in (i)

$$26 + 10(26) + 2\left(-\frac{21}{2}\right) + c = 0$$

$$26 + 260 - 21 + c = 0$$

$$\boxed{c = -265}$$



Substitute all values in (I)

$$x^2 + y^2 + 2(26)x + 2\left(\frac{-21}{2}\right)y - 265 = 0$$

$$x^2 + y^2 + 52x - 21y - 265 = 0 \quad \text{Ans}$$

(d) A (1, 4), B (-1, 8) and tangent to the line $x + 3y - 3 = 0$

Solution:

We know that standard form of an equation of circle is $(x - h)^2 + (y - k)^2 = r^2$ (I)

For A (1, 4) $(1 - h)^2 + (4 - k)^2 = r^2$ (i)

For B (-1, 8) $(-1 - h)^2 + (8 - k)^2 = r^2$ (ii)

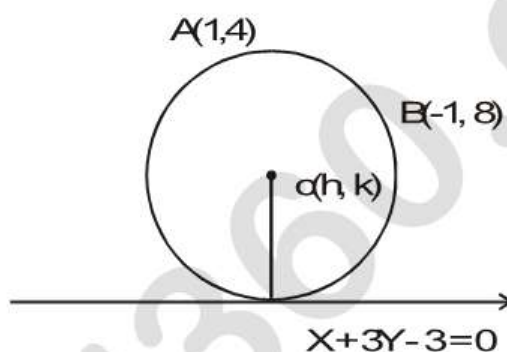
Subtracting $(1 - h)^2 - (-1 - h)^2 + (4 - k)^2 - (8 - k)^2 = 0$

$$1 + h^2 - 2h - 1 - h^2 - 2h + 16 + k^2 - 8k - 64 - k^2 + 16k = 0$$

$$-4h + 8k - 48 = 0 \Rightarrow -4(h - 2k + 12) = 0$$

$$\Rightarrow h - 2k + 12 = 0$$

$$h = 2k - 12 \quad \text{(iii)}$$



Since Circle (I) is tangent to the line $x + 3y - 3 = 0$

By perpendicular distance formula

$$r = \frac{|1(h) + 3k - 3|}{\sqrt{(1)^2 + (3)^2}}$$

$$r = \frac{h + 3k - 3}{\sqrt{10}} \quad \text{squaring} \quad r^2 = \frac{(h + 3k - 3)^2}{10} \quad \text{(A)}$$

$$10r^2 = h^2 + 9k^2 + 9 + 6hk - 18k - 6h$$

Using (ii)

$$10[1 + h^2 - 2h + 16 + k^2 - 8k] = h^2 + 9k^2 + 9 + 6hk - 18k - 6h$$

$$10 + 10h^2 - 20h + 160 + 10k^2 - 80k - h^2 - 9k^2 - 9 - 6hk + 18k + 6h = 0$$

$$9h^2 + k^2 - 14h - 62k - 64k + 161 = 0 \quad \text{(iv)}$$

Using (iii) in (iv) we have

$$9(2k - 12)^2 + k^2 - 14(2k - 12) - 62k - 6k(2k - 12) + 161 = 0$$



$$36k^2 + 1296 - 432k + k^2 - 28k + 168 - 62k - 12k^2 + 72k + 161 = 0$$

$$25k^2 - 450k + 1625 = 0$$

$$25(k^2 - 18k + 65) = 0$$

$$k^2 - 18k + 65 = 0$$

$$(k - 5)(k - 13) = 0$$

$$k = 5$$

$$k = 13$$

If $k = 5$

$$h = 2(5) - 12$$

$$= 10 - 12$$

$$= -2$$

If $k = 13$

$$h = 2(13) - 12$$

$$= 26 - 12$$

$$= 14$$

Putting values of h and k in "A"

$$r^2 = \left(\frac{-2 + 15 - 3}{10} \right)^2$$

$$r^2 = \frac{(10)^2}{10} = 10$$

$$r^2 = \left(\frac{14 + 39 - 3}{10} \right)^2$$

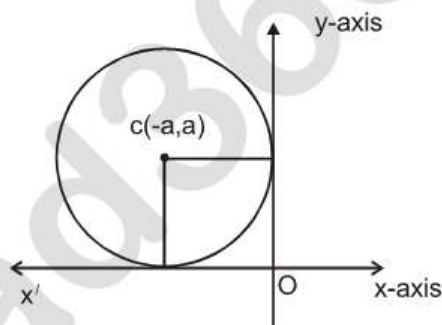
$$r^2 = \frac{(50)^2}{10} = \frac{2500}{10}$$

Required equations of circle are

$$(x + 2)^2 + (y - 5)^2 = 10 \quad \text{and} \quad (x - 14)^2 + (y - 13)^2 = 250 \quad \text{Ans}$$

Q.5: Find an equation of a circle of radius a and lying in the second quadrant such that it is tangent to both the axes.

Solution:



As the circle is in 2nd quadrant and it is tangent to both the axes.

Therefore its center is $(-a, a)$ and radius is a . Equation of circle by standard form is

$$(x - h)^2 + (y - k)^2 = r^2$$

i.e; $(x + a)^2 + (y - a)^2 = a^2$

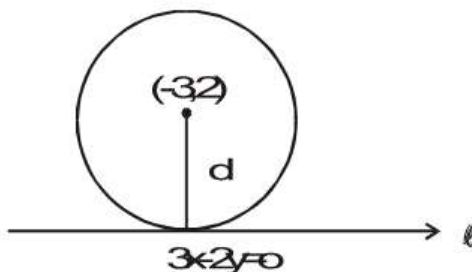
$$x^2 + a^2 + 2ax + y^2 + a^2 - 2ay - a^2 = 0$$

$$x^2 + y^2 + 2ax - 2ay + a^2 = 0 \quad \text{Ans}$$



Q.6: Show that the lines $3x - 2y = 0$ and $2x + 3y - 13 = 0$ are tangents to the circle $x^2 + y^2 + 6x - 4y = 0$.

Solution:



Given circle is $x^2 + y^2 + 6x - 4y = 0$

Center is $(-g, -f) = (-3, 2)$ ($\because 2g = 6, 2f = -4$)

Radius $= \sqrt{g^2 + f^2 - c} = \sqrt{9 + 4} = \sqrt{13}$

The given two lines will be tangent to the given circle, if we find distance from the center of the circle to the given lines and it will equal to the radius of circle.

(i) Now by perpendicular distance formula for line $3x - 2y = 0$

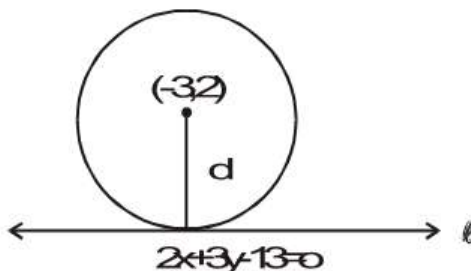
$$\begin{aligned} d &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\ &= \frac{|3(-3) - 2(2)|}{\sqrt{(3)^2 + (-2)^2}} \\ &= \frac{|-9 - 4|}{\sqrt{9 + 4}} \\ &= \frac{|-13|}{\sqrt{13}} \\ &= \frac{13}{\sqrt{13}} = \sqrt{13} \end{aligned}$$

Which show that $d = \text{radius}$

Hence $3x - 2y = 0$ is tangent to circle.

(ii) Now we will take the second line

$$2x + 3y - 13 = 0$$



$$\begin{aligned}
 d &= \frac{|2(-3) + 3(2) - 13|}{\sqrt{(2)^2 + (3)^2}} \\
 &= \frac{|-6 + 6 - 13|}{\sqrt{13}} \\
 &= \frac{|-13|}{\sqrt{13}} = \frac{13}{\sqrt{13}} = \sqrt{13}
 \end{aligned}$$

Which is equal to radius $2x + 3y - 13 = 0$ is also tangent to given circle.

Q.7: Show that the circles $x^2 + y^2 + 2x - 2y - 7 = 0$ and $x^2 + y^2 - 6x + 4y + 9 = 0$ touch externally.

Solution:

Given circles are

$$x^2 + y^2 + 2x - 2y - 7 = 0 \quad \text{and} \quad x^2 + y^2 - 6x + 4y + 9 = 0$$

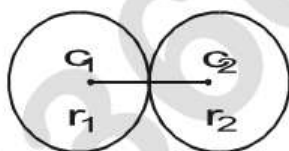
$$\text{Center} = C_1 = (-1, 1) \quad \text{center} = C_2 = (3, -2)$$

$$\text{Radius} = r_1 = \sqrt{1 + 1 + 7} = \sqrt{9} = 3 \quad \text{Radius} = r_2 = \sqrt{9 + 4 + 9} = \sqrt{4} = 2$$

The two circles will touch each other externally if $r_1 + r_2 = |C_1 C_2|$

Note:

The two circle touch externally if their centers distance is equal to sum of their radius.



$$3 + 2 = \sqrt{(3 + 1)^2 + (-2 - 1)^2}$$

$$5 = \sqrt{16 + 9} \Rightarrow 5 = 5 \quad \text{Hence proved.}$$

Q.8: Show that the circles $x^2 + y^2 + 2x - 8 = 0$ and $x^2 + y^2 - 6x + 6y - 46 = 0$ touch internally.

Solution:

Given, circles are

$$x^2 + y^2 + 2x + 0y - 8 = 0 \quad \text{and} \quad x^2 + y^2 - 6x + 6y - 46 = 0$$

$$\text{Center } C_1 = (-1, 0)$$

$$\text{Center} = C_2 = (3, -3)$$

$$\text{Radius} = r_1 = \sqrt{1 + 0 - (-8)}$$

$$\text{Radius} = r_2 = \sqrt{9 + 9 - (-46)}$$

$$= \sqrt{1 + 8}$$

$$= \sqrt{9 + 9 + 46}$$

$$= \sqrt{9}$$

$$= \sqrt{64}$$

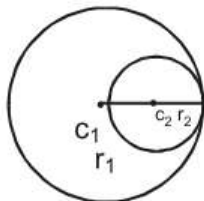
$$r_1 = 3$$

$$r_2 = 8$$



Recall

Two circles touch internally if their centers distance is equal to difference of their radii.



The two circles will touch internally

$$\text{if } r_2 - r_1 = |C_1 C_2|$$

$$8 - 3 = \sqrt{(3 + 1)^2 + (-3 - 0)^2}$$

$$5 = \sqrt{16 + 9}$$

$$5 = \sqrt{25}$$

$$5 = 5$$

Hence proved.

Q.9: Find equation of the circle of radius 2 and tangent to the line $x - y - 4 = 0$ at $A(1, -3)$.

Solution: Let (h, k) be center of circle with radius 2. Equation of circle by standard form is $(x - h)^2 + (y - k)^2 = r^2$ (I)

For $A(1, -3)$ and $r = 2$ (I) becomes

$$(1 - h)^2 + (-3 - k)^2 = 4$$

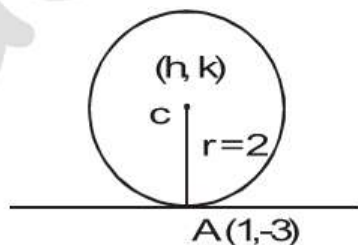
$$1 + h^2 - 2h + 9 + k^2 + 6k - 4 = 0$$

$$h^2 + k^2 - 2h + 6k + 6 = 0 \quad (i)$$

Slope of line $x - y - 4 = 0$ is

$$m_1 = \frac{-\text{coefficient of } x}{\text{coefficient of } y} = -\frac{1}{-1}$$

$$m_1 = 1$$



$$\text{Slope of AC} = m_2 = \frac{k + 3}{h - 1} \quad \left(\because \frac{y_2 - y_1}{x_2 - x_1} \right)$$

Since lines are perpendicular so

$$m_1 m_2 = -1$$



$$1 \times \frac{k+3}{h-1} = -1$$

$$k+3 = -h+1$$

$$k = -h-2 \quad \text{Put in (i)}$$

$$h^2 + (-h-2)^2 - 2h + 6(-h-2) + 6 = 0$$

$$h^2 + h^2 + 4 + 4h - 2h - 6h - 12 + 6 = 0$$

$$2h^2 - 4h - 2 = 0$$

$$h^2 - 2h - 1 = 0$$

$$h = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2(1)}$$

$$= \frac{2 \pm \sqrt{4+4}}{2}$$

$$= \frac{2 \pm \sqrt{8}}{2}$$

$$= \frac{2 \pm 2\sqrt{2}}{2}$$

$$= \frac{2[1 \pm \sqrt{2}]}{2}$$

$$h = 1 \pm \sqrt{2}$$

$$\text{At } h = 1 + \sqrt{2}$$

$$k = -1 - \sqrt{2} - 2$$

$$k = -3 - \sqrt{2}$$

$$\text{At } h = 1 - \sqrt{2}$$

$$\text{At } k = -1 + \sqrt{2} - \sqrt{2}$$

$$k = -3 + \sqrt{2}$$

Equations of circles are

$$(x - 1 - \sqrt{2})^2 + (y + 3 - \sqrt{2})^2 = 4 \quad \& \quad (x - 1 + \sqrt{2})^2 + (y + 3 - \sqrt{2})^2 = 4$$

Tangents

A tangent to a curve is a line that touches the curve without cutting through it.

Let $y = f(x)$ or $f(x, y) = 0$

$\frac{dy}{dx}$ is slope of tangent at any point P (x, y) to the curve.

Normal

The perpendicular to the tangent line is called normal to the curve.

Note

Slope of normal is negative reciprocal of slope of tangent.



EXERCISE 6.2

Q.1: Write down equations of the tangent and normal to the circle.

- (i) $x^2 + y^2 = 25$ at (4, 3)
(Lahore Board 2011)

Solution:

$$\begin{aligned} x^2 + y^2 &= 25 \\ x^2 + y^2 - 25 &= 0 \end{aligned}$$

Compare it with

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\Rightarrow g = 0, f = 0, c = -25$$

Equation of tangent line is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$4x + 3y + 0 + 0 - 25 = 0$$

$$\boxed{4x + 3y - 25 = 0}$$

Equation of normal line is

$$(y - y_1)(x_1 + g) = (x - x_1)(y_1 + f)$$

$$(y - 3)(4 + 0) = (x - 4)(3 + 0)$$

$$4y - 12 = 3x - 12$$

$$3x - 4y - 12 + 12 = 0$$

$$\boxed{3x - 4y = 0}$$

- (b) $x^2 + y^2 = 25$ at $(5 \cos \theta, 5 \sin \theta)$

Solution:

Equation of tangent line is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$5 \cos \theta x + 5 \sin \theta y + 0 + 0 - 25 = 0$$

$$5 \cos \theta x + 5 \sin \theta y - 25 = 0$$

$$5 (\cos \theta x + \sin \theta y - 5) = 0$$

$$\boxed{x \cos \theta + y \sin \theta - 5 = 0}$$

Equation of normal line is

$$(y - y_1)(x_1 + g) = (x - x_1)(y_1 + f)$$

$$(y - 5 \sin \theta)(5 \cos \theta + 0) = (x - 5 \cos \theta)(5 \sin \theta + 0)$$



$$5 \cos \theta y - 25 \sin \theta \cos \theta = 5 \sin \theta x - 25 \sin \theta \cos \theta$$

$$5 \sin \theta x - 5 \cos \theta y = 0$$

$$\boxed{x \sin \theta - y \cos \theta = 0}$$

(ii) $3x^2 + 3y^2 + 5x - 13y + 2 = 0$ at $(1, \frac{10}{3})$

Solution:

$$3x^2 + 3y^2 + 5x - 13y + 2 = 0$$

$$3 \left(x^2 + y^2 + \frac{5}{3}x - \frac{13}{3}y + \frac{2}{3} \right) = 0$$

$$x^2 + y^2 + \frac{5}{3}x - \frac{13}{3}y + \frac{2}{3} = 0$$

Compare it with

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\Rightarrow 2g = \frac{5}{3}, \quad 2f = \frac{-13}{3}, \quad C = \frac{2}{3}$$

$$g = \frac{5}{6}, \quad f = \frac{-13}{6}$$

Equation of tangent at $(1, \frac{10}{3})$

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$x + \frac{10}{3}y + \frac{5}{6}(x + 1) - \frac{13}{6}(y + \frac{10}{3}) + \frac{2}{3} = 0$$

$$x + \frac{10}{3}y + \frac{5}{6}x + \frac{5}{6} - \frac{13}{6}y - \frac{65}{9} + \frac{2}{3} = 0$$

$$\frac{18x + 60y + 15x + 15 - 39y - 130 + 12}{18} = 0$$

$$\boxed{33x + 21y - 103 = 0}$$

Equation of normal

$$(y - y_1)(x_1 + g) = (x - x_1)(y_1 + f)$$

$$\left(y - \frac{10}{3}\right)\left(1 + \frac{5}{6}\right) = (x - 1)\left(\frac{10}{3} + \frac{13}{6}\right)$$

$$\left(y - \frac{10}{3}\right)\left(\frac{11}{6}\right) = (x - 1)\left(\frac{20 - 13}{6}\right)$$

$$11y - \frac{11}{3} = 7x - 7$$

$$\frac{33y - 110}{3} = 7x - 7$$

$$33y - 110 = 21x - 21$$

$$21x - 33y - 21 + 110 = 0$$

$$\boxed{21x - 33y + 89 = 0}$$



Q.2: Write down equations of the tangent and normal to the circle $4x^2 + 4y^2 - 16x + 24y - 117 = 0$ at the points on circle whose abscissa is -4 .

Solution:

Given

$$4x^2 + 4y^2 - 16x + 24y - 117 = 0 \quad \text{at } x = -4$$

To find "y" put $x = -4$ in (I)

$$4(-4)^2 + 4y^2 - 16(-4) + 24y - 117 = 0$$

$$64 + 4y^2 + 64 + 24y - 117 = 0$$

$$4y^2 + 24y + 11 = 0$$

$$4y^2 + 22y + 2y + 11 = 0$$

$$2y(2y + 11) + 1(2y + 11) = 0$$

$$(2y + 11)(2y + 1) = 0$$

$$\Rightarrow 2y + 11 = 0 \quad 2y + 1 = 0$$

$$y = \frac{-11}{2} \quad y = \frac{-1}{2}$$

Thus the points on the circle are $(-4, \frac{-11}{2})$ & $(-4, \frac{-1}{2})$

$$4\left(x^2 + y^2 - 4x + 6y - \frac{117}{4}\right) = 0$$

$$x^2 + y^2 - 4x + 6y - \frac{117}{4} = 0$$

Compare it with

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\Rightarrow 2g = -4, \quad 2f = 6, \quad c = \frac{-117}{4}$$

$$g = -2, \quad f = 3$$

Equation of tangent at $(-4, \frac{-1}{2})$

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$-4x - \frac{1}{2}y - 2(x - 4) + 3(y - \frac{1}{2}) - \frac{117}{4} = 0$$

$$-4x - \frac{y}{2} - 2x + 8 + 3y - \frac{3}{2} - \frac{117}{4} = 0$$

$$\frac{-16x - 2y - 8x + 32 + 12y - 6 - 117}{4} = 0$$

$$-24x + 10y - 91 = 0$$

$$-(24x - 10y + 91) = 0$$

$$\boxed{24x - 10y + 91 = 0}$$

Equation of normal at $(-4, \frac{-1}{2})$

$$(y - y_1)(x_1 + g) = (x - x_1)(y_1 + f)$$

$$\left(y + \frac{1}{2}\right)(-4 - 2) = (x + 4)\left(\frac{-1}{2} + 3\right)$$

$$\left(y + \frac{1}{2}\right)(-6) = (x + 4)\left(\frac{5}{2}\right)$$

$$-6y - 3 = \frac{5x + 20}{2}$$

$$-12y - 6 = 5x + 20$$

$$5x + 12y + 20 + 6 = 0$$

$$\boxed{5x + 12y + 26 = 0}$$



Equation of tangent at $(-4, \frac{-11}{2})$

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$-4x - \frac{11}{2}y - 2(x - 4) + 3(y - \frac{11}{2}) - \frac{117}{4} = 0$$

$$-4x - \frac{11}{2}y - 2x + 8 + 3y - \frac{33}{2} - \frac{117}{4} = 0$$

$$\frac{-16x - 22y - 8x + 32 + 12y - 66 - 117}{4} = 0$$

$$-24x - 10y - 151 = 0$$

$$-(24x + 10y + 151) = 0$$

$$\boxed{24x + 10y + 151 = 0}$$

Equation of normal at $(-4, \frac{-11}{2})$

$$(y - y_1)(x_1 + g) = (x - x_1)(y_1 + f)$$

$$\left(y + \frac{11}{2}\right)(-4 - 2) = (x + 4)\left(\frac{-11}{2} + 3\right)$$

$$\left(y + \frac{11}{2}\right)(-6) = (x + 4)\left(\frac{-5}{2}\right)$$

$$-6y - 33 = \frac{-5x - 20}{2}$$

$$-12y - 66 = -5x - 20$$

$$5x - 12y - 66 + 20 = 0$$

$$\boxed{5x - 12y - 46 = 0}$$

Q.3: Check the position of the point (5, 6) with respect to the circle.

(i) $x^2 + y^2 = 81$

(Lahore Board 2009, 2010)

Solution:

Given $x^2 + y^2 - 81 = 0$ (I)

Put (5, 6) in L.H.S of (I)

$$= (5)^2 + (6)^2 - 81$$

$$= 25 + 36 - 81$$

$$= -20 < 0 \quad (-ve)$$

then (5, 6) lies inside the circle.

(ii) $2x^2 + 2y^2 + 12x - 8y + 1 = 0$

(Lahore Board 2011)

Solution:

Given $2x^2 + 2y^2 + 12x - 8y + 1 = 0$ (I)

Put (5, 6) in L.H.S of (i)

$$= 2(5)^2 + 2(6)^2 + 12(5) - 8(6) + 1$$

$$= 50 + 72 + 60 - 48 + 1$$

$$= 135 > 0 \quad (+ve)$$

Then (5, 6) lies outside the circle.



Q.4: Find length of the tangent drawn from the point $(-5, 4)$ to the circle $5x^2 + 5y^2 - 10x + 15y - 131 = 0$
(Lahore Board 2009)

Solution:

$$\text{Given } 5x^2 + 5y^2 - 10x + 15y - 131 = 0$$

Dividing throughout by 5

$$x^2 + y^2 - 2x + 3y - \frac{131}{5} = 0$$

$$\begin{aligned} \text{Length of Tangent} &= \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + C} \\ &= \sqrt{(-5)^2 + (4)^2 - 2(-5) + 3(4) - \frac{131}{5}} \\ &= \sqrt{25 + 16 + 10 + 12 - \frac{131}{5}} \\ &= \sqrt{63 - \frac{131}{5}} \\ &= \sqrt{\frac{315 - 131}{5}} \\ &= \sqrt{\frac{184}{5}} \quad \text{Ans.} \end{aligned}$$

Q.5: Find the length of the chord cut off from the line $2x + 3y = 13$ by the circle $x^2 + y^2 = 26$.

Solution:

$$\text{Given line } 2x + 3y = 13 \quad \dots\dots (i)$$

$$\text{Circle is } x^2 + y^2 = 26 \quad \dots\dots (ii)$$

$$\text{From (i) } y = \frac{13 - 2x}{3} \quad \dots\dots (iii)$$

Put in (ii)

$$x^2 + \left(\frac{13 - 2x}{3}\right)^2 = 26$$

$$x^2 + \frac{169 + 4x^2 - 52x}{9} = 26$$

$$9x^2 + 169 + 4x^2 - 52x = 236$$

$$13x^2 - 52x - 65 = 0$$

$$x^2 - 4x - 5 = 0 \quad (\text{Dividing throughout by 5})$$

$$x^2 - 5x + x - 5 = 0$$

$$x(x - 5) + 1(x - 5) = 0$$



$$(x - 5)(x + 1) = 0$$

$$x = 5 \quad x = -1$$

$$\text{If } x = 5 = y = \frac{13 - 2(5)}{3}$$

$$y = \frac{13 - 10}{3}$$

$$y = \frac{3}{3} = 1$$

$$\text{and if } x = -1 \quad y = \frac{13 - 2(-1)}{3}$$

$$y = \frac{13 + 2}{3} = \frac{15}{3}$$

$$y = 5$$

Hence points of intersection are A (5, 1) & B (-1, 5)

$$\begin{aligned} \text{Required Length of chord} &= |AB| = \sqrt{(-1 - 5)^2 + (5 - 1)^2} \\ &= \sqrt{36 + 16} \\ &= \sqrt{52} \\ &= 2\sqrt{13} \quad \text{Ans} \end{aligned}$$

Q.6: Find the coordinates of the points of intersection of line $x + 2y = 6$ with the circle $x^2 + y^2 - 2x - 2y - 39 = 0$

Solution:

$$\text{Line is } x + 2y = 6 \quad \dots\dots (i)$$

$$\text{Circle is } x^2 + y^2 - 2x - 2y - 39 = 0 \quad \dots\dots (ii)$$

$$\text{From (i) } x = 6 - 2y \quad \dots\dots (iii) \quad \text{Put in (ii)}$$

$$(6 - 2y)^2 + y^2 - 2(6 - 2y) - 2y - 39 = 0$$

$$36 + 4y^2 - 24y + y^2 - 12 + 4y - 2y - 39 = 0$$

$$5y^2 - 22y - 15 = 0$$

$$5y^2 - 25y + 3y - 15 = 0$$

$$5y(y - 5) + 3(y - 5) = 0$$

$$\Rightarrow y - 5 = 0 \quad 5y + 3 = 0$$

$$y = 5 \quad \& \quad y = \frac{-3}{5}$$

$$\text{if } y = 5 \quad x = 6 - 2(5) \quad (\text{By iii})$$

$$x = 6 - 10$$

$$x = -4$$

$$\text{if } y = \frac{-3}{5} \quad x = 6 - 2\left(\frac{-3}{5}\right) \quad (\text{By iii})$$

$$x = 6 + \frac{6}{5} \Rightarrow x = \frac{36}{5}$$

Hence points of intersection are



$$(-4, 5) \quad \& \quad \left(\frac{36}{5}, \frac{-3}{5}\right) \quad \text{Ans.}$$

Q.7 Find equations of the tangents to the circle $x^2 + y^2 = 2$

(i) Parallel the $x - 2y + 1 = 0$

Solution:

Let required tangent $y = mx + c \rightarrow$ (i)

Given circle is $x^2 + y^2 = 2 \Rightarrow r^2 = 2$

Given line is $x - 2y + 1 = 0$

Slope of line $= m = -\frac{\text{coefficient of } x}{\text{coefficient of } y} = -\frac{1}{-2} = \frac{1}{2}$

Since the tangent line is parallel to this line so $m = \frac{1}{2}$

We know that condition of tangency for circle is

$$c^2 = r^2 (1 + m^2)$$

$$c^2 = 2 \left(1 + \frac{1}{4}\right)$$

$$= 2 \left(\frac{5}{4}\right) = \frac{10}{4}$$

$$\Rightarrow c = \pm \frac{\sqrt{10}}{2} \quad \text{Substitute values in (i)}$$

$$y = \frac{1}{2}x \pm \frac{\sqrt{10}}{2}$$

$$y = \frac{x \pm \sqrt{10}}{2}$$

$$2y = x \pm \sqrt{10}$$

$$x - 2y \pm \sqrt{10} = 0 \quad \text{Required equations of tangent.}$$

(ii) Perpendicular to the line $3x + 2y = 6$

Solution:

Given circle $x^2 + y^2 = 2 \Rightarrow r^2 = 2$

Given line $3x + 2y = 6$

$$\therefore \text{Slope of line} = \frac{-\text{coeff of } x}{+\text{coeff of } y} = -\frac{3}{2}$$

But since tangent line is perpendicular to this line so its slope will be $= \frac{-1}{m} = \frac{2}{3} = m$

We know that condition of tangency of circle is



$$c^2 = r^2 (1 + m^2)$$

$$c^2 = 2 \left(1 + \frac{4}{9}\right)$$

$$c^2 = 2 \left(\frac{13}{9}\right) = \frac{26}{9}$$

$$c = \pm \frac{\sqrt{26}}{3}$$

$$y = \frac{2x \pm \sqrt{26}}{3} \Rightarrow 2x - 3y \pm \sqrt{26} = 0 \quad \text{Ans.}$$

Required equations of tangents are

$$y = mx + c$$

$$y = \frac{2}{3}x \pm \frac{\sqrt{26}}{3}$$

$$y = \frac{2x \pm \sqrt{26}}{3}$$

$$2x - 3y \pm \sqrt{26} = 0 \quad \text{Ans}$$

Q.8: Find equations of tangent drawn from

(i) $(0, 5)$ to $x^2 + y^2 = 16$

Solution:

Given circle $x^2 + y^2 = 16$

$\Rightarrow r^2 = 16 \quad r = 4$ & Center $(0, 0)$

Let the tangent drawn from the $P(0, 5)$ point $(0, 5)$ to the circle touch circle at point (x_1, y_1)

$\therefore$ Given circle becomes

$$x_1^2 + y_1^2 = 16 \quad (1)$$

$$\text{Now } m_1 = \text{Slope of PA} = \frac{y_1 - 5}{x_1 - 0} = \frac{y_1 - 5}{x_1}$$

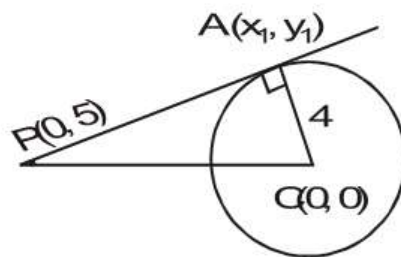
$$m_2 = \text{Slope of CA} = \frac{y_1 - 0}{x_1 - 0} = \frac{y_1}{x_1}$$

Since two lines are perpendicular $\therefore m_1 \times m_2 = -1$

$$\frac{y_1 - 5}{x_1} \times \frac{y_1}{x_1} = -1$$

$$y_1^2 - 5y_1 = -x_1^2$$

$$x_1^2 + y_1^2 = 5y_1 \quad (2)$$



$$16 = 5y_1 \quad (\text{using 1})$$

$$\Rightarrow y_1 = \frac{16}{5} \quad \text{Put in (2)}$$

$$x_1^2 + \frac{256}{25} = 5 \left(\frac{16}{5} \right)$$

$$x_1^2 = 16 - \frac{256}{25}$$

$$x_1^2 = \frac{400 - 256}{25}$$

$$= \frac{144}{25} \Rightarrow x_1 = \pm \frac{12}{5}$$

We have two points $\left(\frac{12}{5}, \frac{16}{5}\right)$ & $\left(-\frac{12}{5}, \frac{16}{5}\right)$

Now $m_1 = \text{slope of line PA} = \frac{y_1 - 5}{x_1}$ at $\left(\frac{12}{5}, \frac{16}{5}\right)$

$$\text{Now } m_1 = \frac{\frac{16}{5} - 5}{\frac{12}{5}}$$

$$= \frac{\frac{16 - 25}{5}}{\frac{12}{5}} = \frac{-9}{12} = \frac{-3}{4}$$

Equation of tangent at point $\left(\frac{12}{5}, \frac{16}{5}\right)$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{16}{5} = \frac{-3}{4} \left(x - \frac{12}{5} \right)$$

$$\frac{5y - 16}{5} = \frac{-3}{4} \left(\frac{5x - 12}{5} \right)$$

$$20y - 64 = -15x + 36$$

$$15x + 20y = 100 \quad \text{Ans.}$$

Next, $m_1 = \text{Slope of line PA at point } \left(-\frac{12}{5}, \frac{16}{5}\right)$



$$m_1 = \frac{\frac{16}{5} - 5}{\frac{-12}{5}} = \frac{16 - 25}{-12} = \frac{-9}{-12} = \frac{3}{4}$$

Equation of tangent at point $\left(\frac{-12}{5}, \frac{16}{5}\right)$ is given by

$$y - \frac{16}{5} = \frac{3}{4} \left(x + \frac{12}{5}\right)$$

$$\frac{5y - 16}{5} = \frac{3}{4} \left(\frac{5x + 12}{5}\right)$$

$$20y - 64 = 15x + 36$$

$$15x - 20y + 100 = 0 \quad \text{Ans}$$

(ii) Find equation of tangents drawn from $(-1, 2)$ to the circle $x^2 + y^2 + 4x + 2y = 0$.

Solution:

$$\text{Given } x^2 + y^2 + 4x + 2y = 0 \quad \dots\dots\dots (1)$$

$$\text{Center} = \left(-\frac{4}{2}, -\frac{2}{2}\right) = (-2, -1)$$

Let tangent drawn from $(-1, 2)$ to the circle touch the circle at the point (x_1, y_1) then becomes

(I)

$$x_1^2 + y_1^2 + 4x_1 + 2y_1 = 0 \quad \dots\dots\dots (2)$$

$$m_1 = \text{Slope of PA} = \frac{y_1 - 2}{x_1 + 1}$$

$$m_2 = \text{Slope of CA} = \frac{y_1 + 1}{x_1 + 2}$$

Since two lines are perpendicular so

$$m_1 m_2 = -1$$

$$\frac{y_1 - 2}{x_1 + 1} \times \frac{y_1 + 1}{x_1 + 2} = -1$$

$$y_1^2 + y_1 - 2y_1 - 2 = -(x_1^2 + 3x_1 + 2)$$

$$y_1^2 - y_1 - 2 + x_1^2 + 3x_1 + 2 = 0$$

$$x_1^2 + y_1^2 - 3x_1 - y_1 = 0 \quad \dots\dots\dots (3)$$

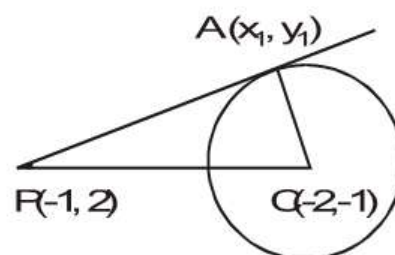
$$-x_1^2 \pm y_1^2 \pm 4x_1 \pm 2y_1 = 0 \quad \text{(By using - 2)}$$

$$-x_1 - 3y_1 = 0$$

$$\Rightarrow x_1 = -3y_1 \quad (4) \quad \text{Put in (3)}$$

$$9y_1^2 + y_1^2 - 9y_1 - y_1 = 0$$

$$10y_1^2 - 10y_1 = 0$$



$$10y_1(y_1 - 1) = 0$$

$$\Rightarrow y_1 = 0, \quad y_1 = 1$$

$$\text{If } y_1 = 0 \quad x_1 = 0 \quad (\text{Using -4}) \text{ if } y_1 = 1, x_1 = -3$$

Required points of tangency are (0, 0) & (-3, 1)

At Point (0, 0)

$$m_1 = \text{Slope of (PA)} = \frac{-2}{1} = -2$$

Equation of tangent at point (0, 0)

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -2(x - 0)$$

$$y = -2x$$

$$2x + y = 0 \quad \text{Ans}$$

At Point (-3, 1)

$$m_1 = \text{Slope of (PA)} = \frac{1-2}{-3+1} = \frac{-1}{-2}$$

Equation of tangent at point (-3, 1)

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{2}(x + 3)$$

$$2y - 2 = x + 3$$

$$x - 2y + 5 = 0 \quad \text{Ans}$$

Q.8 (iii): $(-7, -2)$ to $(x+1)^2 + (y-2)^2 = 26$

Solution:

$$\text{Given circle } (x+1)^2 + (y-2)^2 = 26$$

$$\text{Center} = (-1, 2)$$

Let tangent drawn from point $(-7, -2)$ to the circle touch it at point (x_1, y_1) .

Given circle becomes

$$(x_1 + 1)^2 + (y_1 - 2)^2 = 26$$

$$x_1^2 + 1 + 2x_1 + y_1^2 + 4 - 4y_1 - 26 = 0$$

$$x_1^2 + y_1^2 + 2x_1 - 4y_1 - 21 = 0 \quad \dots\dots\dots (i)$$

$$\text{Now } m_1 = \text{Slope of PA} = \frac{y_1 + 2}{x_1 + 7}$$

$$m_2 = \text{Slope of CA} = \frac{y_1 - 2}{x_1 + 1}$$

Since lines are perpendicular

$$\text{So } m_1 m_2 = -1$$

$$\frac{y_1 + 2}{x_1 + 7} \times \frac{y_1 - 2}{x_1 + 1} = -1$$

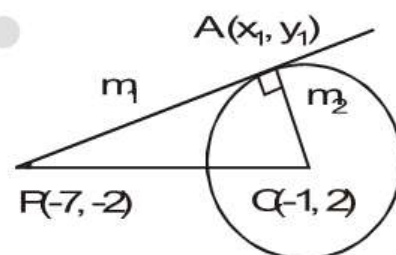
$$y_1^2 - 4 = -x_1^2 - 8x_1 - 7$$

$$x_1^2 + y_1^2 + 8x_1 + 3 = 0 \quad \dots\dots\dots (ii)$$

$$\text{Subtracting} \quad -x_1^2 \pm y_1^2 \pm 2x_1 \mp 21 \mp 4y_1 = 0$$

$$6x_1 + 4y_1 + 24 = 0$$

$$4y_1 = -24 - 6x_1$$



$$y_1 = \frac{-2(12 + 3x_1)}{4}$$

$$= \frac{-(12 + 3x_1)}{2}$$

Put in (ii)

$$x_1^2 + \frac{(12 + 3x_1)^2}{4} + 8x_1 + 3 = 0$$

$$x_1^2 + \frac{144 + 9x_1^2 + 72x_1}{4} + 8x_1 + 3 = 0$$

$$4x_1^2 + 144 + 9x_1^2 + 72x_1 + 32x_1 + 12 = 0$$

$$13x_1^2 + 104x_1 + 156 = 0$$

$$13(x_1^2 + 8x_1 + 12) = 0$$

$$x_1^2 + 8x_1 + 12 = 0$$

$$x_1^2 + 6x_1 + 2x_1 + 12 = 0$$

$$(x_1 + 2)(x_1 + 6) = 0 \Rightarrow x_1 = -2 \quad \& \quad x_1 = -6$$

$$\text{if } x_1 = -2 \quad ; \quad y_1 = -\left(\frac{3(-2) + 12}{2}\right) = -\left(\frac{-6 + 12}{2}\right) = -3$$

$$\text{if } x_1 = -6 \quad ; \quad y_1 = -\left(\frac{3(-6) + 12}{2}\right) = -\left(\frac{-18 + 12}{2}\right) = 3$$

Then points of tangency are $(-2, -3)$ & $(-6, 3)$

At Point $(-2, -3)$

$$m_1 = \text{Slope of PA} = \frac{-3 + 2}{-2 + 7} = \frac{-1}{5}$$

Equation of tangent at point $(-2, -3)$ is

$$y - y_1 = m(x - x_1)$$

$$y + 3 = \frac{-1}{5}(x + 2)$$

$$5(y + 3) = -(x + 2)$$

$$5y + 15 + x + 2 = 0$$

$$x + 5y + 17 = 0$$

At Point $(-6, 3)$

$$m_1 = \text{Slope of (PA)} = \frac{3 + 2}{-6 + 7} = \frac{5}{1} = 5$$

Equation of tangent at point $(-6, 3)$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = 5(x + 6)$$

$$y - 3 = 5x + 30$$

$$5x - y + 33 = 0 \quad \text{Ans}$$

Q.9: Find an equation of the chord of contact of the tangents drawn from $(4, 5)$ to the circle $2x^2 + 2y^2 - 8x + 12y + 21 = 0$

Solution:

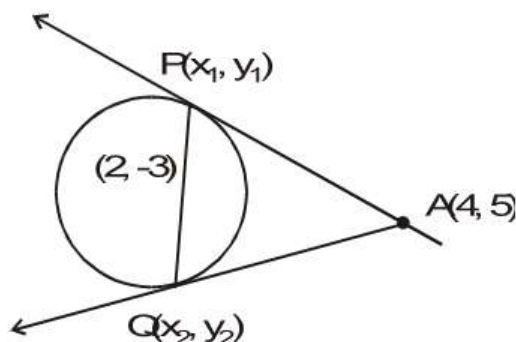
$$\text{Given } 2x^2 + 2y^2 - 8x + 12y + 21 = 0$$

Dividing throughout by 2



$$x^2 + y^2 - 4x + 6y + \frac{21}{2} = 0$$

Now Let points of contact of the two tangents be $P(x_1, y_1)$ & $Q(x_2, y_2)$ An equation of the tangent at $P(x_1, y_1)$ is



$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0 \quad \dots\dots\dots (1)$$

$$\text{Since } (-g, -f) = (2, -3)$$

$$g = -2 \quad f = 3 \quad \text{Put in I}$$

$$xx_1 + yy_1 - 2(x + x_1) + 3(y + y_1) + \frac{21}{2} = 0 \quad \dots\dots\dots (2)$$

Since it passes through (4, 5)

$$4x_1 + 5y_1 - 2(4 + x_1) + 3(5 + y_1) + \frac{21}{2} = 0$$

$$4x_1 + 5y_1 - 8 - 2x_1 + 15 + 3y_1 + \frac{21}{2} = 0$$

$$2x_1 + 8y_1 + 7 + \frac{21}{2} = 0$$

$$4x_1 + 16y_1 + 14 + 21 = 0$$

$$4x_1 + 16y_1 + 35 = 0 \quad \dots\dots\dots (i)$$

Similarly for point $Q(x_2, y_2)$, we have

$$4x_2 + 16y_2 + 35 = 0 \quad \dots\dots\dots (ii)$$

(i) & (ii) Show that both the points $P(x_1, y_1)$ & $Q(x_2, y_2)$ lie on $4x + 16y + 35 = 0$ and so it is the required equation of the chord of contact.



EXERCISE 6.3

Q.1: Prove that normal lines of a circle pass through the center of the circle.
(Lahore Board 2009)

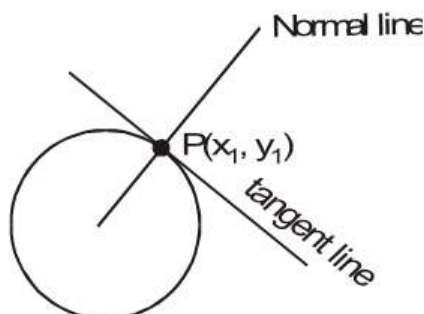
Solution:

Let us consider a circle with center $(0, 0)$ and radius r .
Therefore equation of circle is



$$x^2 + y^2 = r^2$$

Diff. w.r.t. 'x'



$$2x + 2y \frac{dy}{dx} = 0$$

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = \frac{-2x}{2y} = \frac{-x}{y}$$

$$m = \left. \frac{dy}{dx} \right|_{(x_1, y_1)} = \text{Slope of tangent} = -\frac{x_1}{y_1}$$

$$m_1 = \text{Slope of normal} = \frac{1}{-\frac{x_1}{y_1}} = \frac{y_1}{x_1} \quad (-\text{ve reciprocal})$$

Thus equation of the normal line passing through $P(x_1, y_1)$ is given by

$$y - y_1 = m_1 (x - x_1)$$

$$y - y_1 = \frac{y_1}{x_1} (x - x_1) \quad \dots\dots\dots (2)$$

$$x_1 y - x_1 y_1 = x y_1 - x_1 y_1$$

$$x_1 y = y_1 x$$

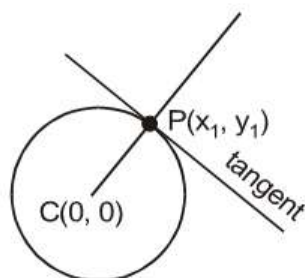
Clearly center of circle $(0, 0)$ satisfy the above equation. Hence normal lines of circles passing through the center of the circle.

Q.2: Prove that the straight line drawn from the center of a circle perpendicular to a tangent passes through the point of tangency.

Solution:

Equation of circle with center $(0, 0)$ & radius r is given by

$$x^2 + y^2 = r^2$$



Diff. w.r.t. 'x'

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$m = \text{Slope of tangent} = \frac{dy}{dx} = -\frac{x_1}{y_1}$$

$$\text{Since the required line is perpendicular therefore } m_1 = \frac{1}{-\frac{x_1}{y_1}} = \frac{y_1}{x_1}$$

The equation of the straight line perpendicular to the tangent through (0, 0).

$$y - y_1 = m' (x - x_1)$$

$$y - 0 = \frac{y_1}{x_1} (x - 0)$$

$$x_1 y = y_1 x$$

Thus the straight line drawn from the center and perpendicular to the tangent passes through the point of tangency.

Q.3: Prove that the mid point of the hypogenous of a right-angled triangle is the circum center of the triangle.

Solution:

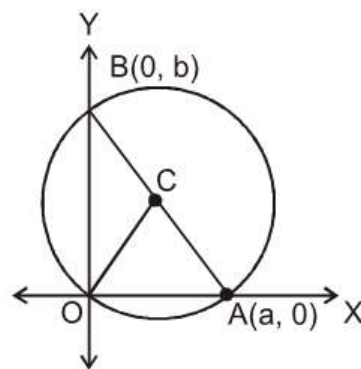
Let OAB be a right angle triangle with

$|OA| = a$ and $|OB| = b$.

Since 'c' be the mid point of $\overline{AB}$.

$\therefore$ By ratio formula coordinates of C are

$$\left(\frac{a}{2}, \frac{b}{2}\right)$$



$$|CA| = \sqrt{\left(\frac{a}{2} - a\right)^2 + \left(\frac{b}{2} - 0\right)^2} = \sqrt{\left(-\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} \dots\dots\dots (1)$$

$$|CB| = \sqrt{\left(\frac{a}{2} - 0\right)^2 + \left(\frac{b}{2} - b\right)^2} = \sqrt{\left(\frac{a}{2}\right)^2 + \left(-\frac{b}{2}\right)^2} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} \dots\dots\dots (2)$$

$$|CO| = \sqrt{\left(\frac{a}{2} - 0\right)^2 + \left(\frac{b}{2}\right)^2} = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} \dots\dots\dots (3)$$

From equations (1), (2) and (3)



$$|CA| = |CB| = |CO|$$

Shows the mid point of hypotenuse of a right triangle is the circum center of the triangle.

Q.4: Prove that the perpendicular dropped from a point of circle on a diameter is a mean proportional between the segments into which it divides the diameter.

Solution:

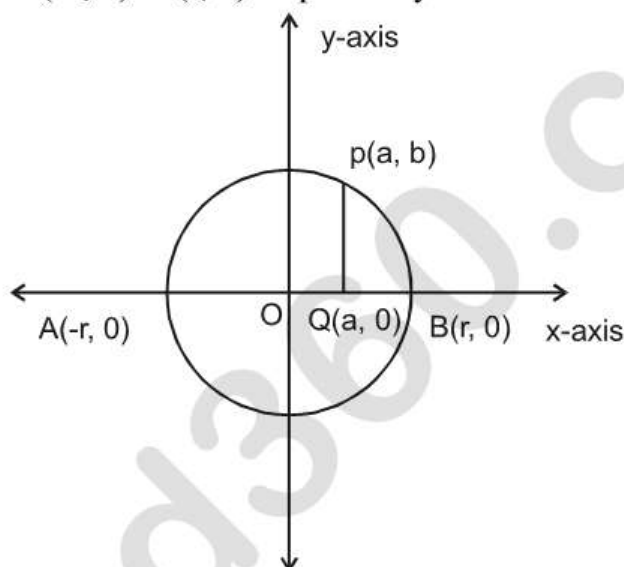
Let us consider a circle with center (0, 0) and radius r.

$$\therefore \text{Equation of circle is } x^2 + y^2 = r^2 \dots\dots\dots (1)$$

Let p (a, b) be any point on the circle then (1) becomes

$$a^2 + b^2 = r^2 \dots\dots\dots (2)$$

From point P drop a perpendicular on the diameter AB at point Q. Therefore coordinates of A & B are (-r, 0) & (r, 0) respectively.



we known that definition of mean proportional.

$$x : M :: M : y$$

$$xy = M^2 \rightarrow (3)$$

$$|PQ| = \sqrt{(a-a)^2 + (0-b)^2} = \sqrt{0+b^2} = \sqrt{b^2} = b$$

$$|AQ| = \sqrt{[a-(-r)]^2 + (0-0)^2} = \sqrt{(a+r)^2} = \sqrt{a+r} = r+a$$

$$|QB| = \sqrt{(r-a)^2 + (0-0)^2} = \sqrt{r-a^2} = r-a$$

(3) becomes

$$|AQ| |QB| = |PQ|^2$$

$$(r+a)(r-a) = b^2$$

$$r^2 - a^2 = b^2$$



$$b^2 = b^2$$

Hence proved.

Parabola: (Lahore Board 2009)

Let L be a fixed line in a plane and F be a fixed point not on the line L. Suppose |PM| denotes the distance of a point P(x, y) from the line L. The set of all points P in the plane such that

$$\frac{|PF|}{|PM|} = 1 \quad \text{is called a parabola}$$

Where fixed line L is directrix.

fixed point F is called Focus

STANDARD FORMS OF PARABOLA

- (i) $y^2 = 4ax$
- (ii) $y^2 = -4ax$
- (iii) $x^2 = 4ay$
- (iv) $x^2 = -4ay$

Axis of Parabola

The line through the focus and perpendicular to the directrix is called axis of Parabola.

Vertex

The point where the axis meets the parabola is called vertex.

Chord

A line joining two distinct points on a parabola is called chord of the parabola.

Focal Chord

A chord, which passes through focus is called focal chord.

Latusrectum

The focal chord perpendicular to the axis of the parabola is called latusrectum of the parabola.

Note:

In standard form vertex is at origin (0, 0).

If vertex is not at origin then equations of parabola become

- (i) $(y - k)^2 = 4a(x - h)$
- (ii) $(y - k)^2 = -4a(x - h)$
- (iii) $(x - h)^2 = 4a(y - k)$
- (iv) $(x - h)^2 = -4a(y - k)$



EXERCISE 6.4

Q.1: Find the focus, vertex and directrix of the parabola. Sketch its graph.

(i) $y^2 = 8x$ (Gujranwala Board 2007)

Solution:

$$y^2 = 8x$$

As standard form is

$$y^2 = 4ax$$

$$4a = 8 \Rightarrow \boxed{a = 2}$$

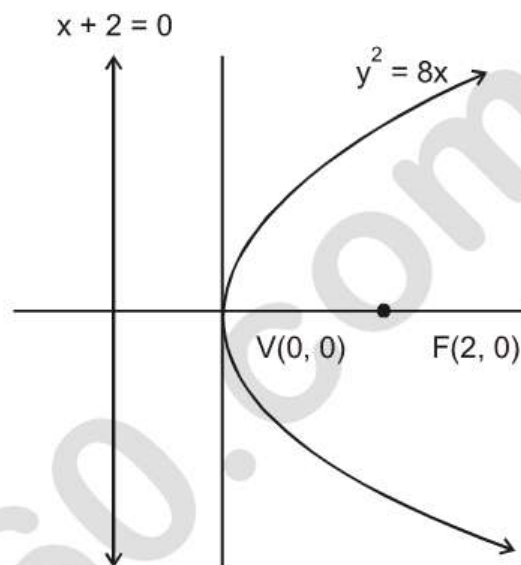
$$\text{Focus} = (a, 0) = (2, 0)$$

$$\text{Vertex} = (0, 0) = (0, 0)$$

$$\text{Directrix } x = -a$$

$$x = -2$$

$$\Rightarrow x + 2 = 0$$



(ii) $x^2 = -16y$

Solution:

$$x^2 = -16y$$

As standard form is

$$x^2 = -4ay$$

$$\Rightarrow -4a = -16 \Rightarrow$$

$$\boxed{a = +4}$$

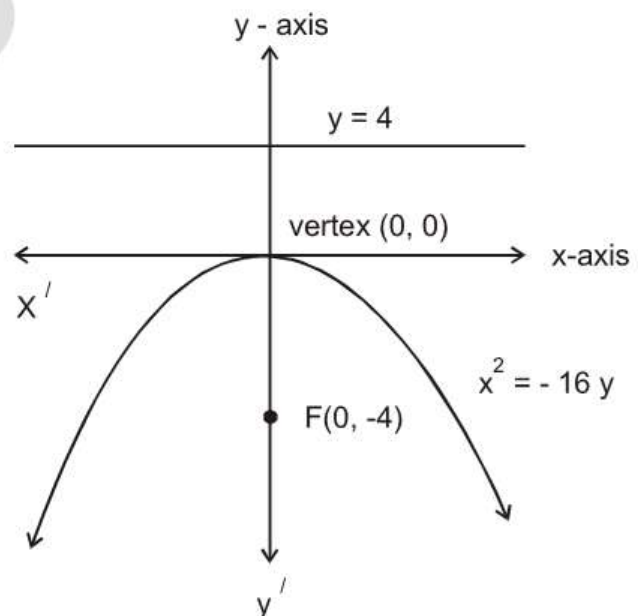
$$\text{Focus} = (0, -a)$$

$$\text{Focus} = (0, -4)$$

$$\text{Vertex} = (0, 0)$$

$$\text{Directrix } y = a$$

$$y = 4$$



(iii) $x^2 = 5y$

Solution:

$$x^2 = 5y$$

As standard form is

$$x^2 = 4ay$$

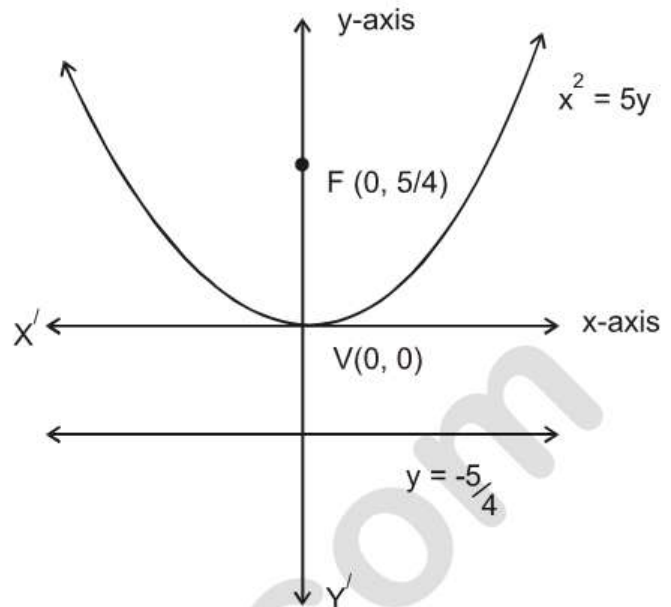
$$4a = 5 \Rightarrow \boxed{a = \frac{5}{4}}$$

$$\text{Focus} = (0, a) = (0, \frac{5}{4})$$

$$\text{Vertex} = (0, 0)$$

$$\text{Directrix } y = -a$$

$$y = -\frac{5}{4}$$



(iv) $y^2 = -12x$

Solution:

$$y^2 = -12x$$

As standard form is

$$y^2 = -4ax$$

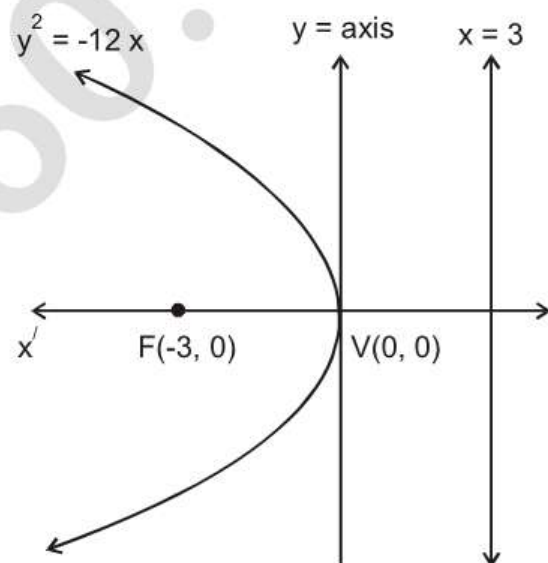
$$4a = 12 \Rightarrow \boxed{a = 3}$$

$$\text{Focus} = (-a, 0) = (-3, 0)$$

$$\text{Vertex} = (0, 0)$$

$$\text{Directrix } x = a$$

$$x = 3$$



(v) $x^2 = 4(y - 1)$

Solution:

$$x^2 = 4(y - 1)$$

$$(x - 0)^2 = 4(y - 1) \dots\dots\dots (1)$$

$$\text{Let } x - 0 = X, \quad y - 1 = Y$$

(1) Becomes

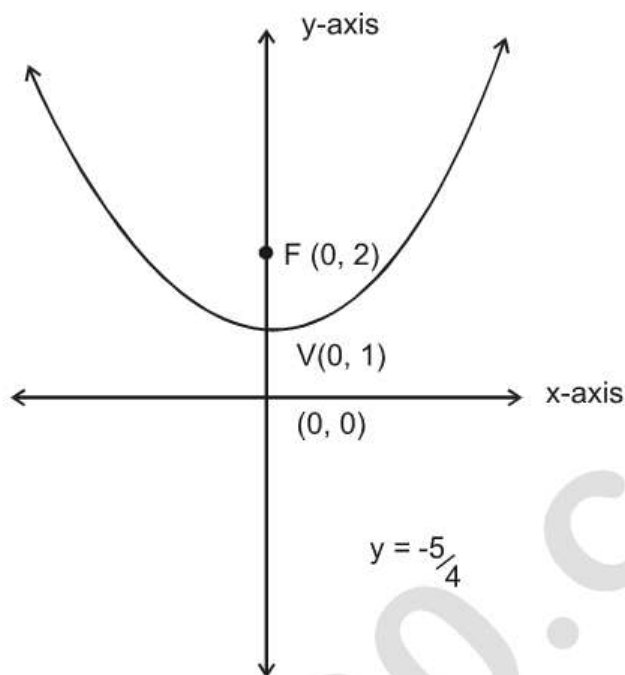
$$x^2 = 4Y$$



As standard form is

$$x^2 = 4ay$$

$$4a = 4 \quad \boxed{a = 1}$$



$$\text{Focus} = (0, a)$$

$$(X, Y) = (0, 1)$$

$$(x, y - 1) = (0, 1)$$

$$x = 0 \quad y - 1 = 1$$

$$x = 0 \quad y = 2$$

$$\text{Focus} = (0, 2)$$

For the vertex put $X = 0, Y = 0$

$$x - 0 = 0, \quad y - 1 = 0$$

$$x = 0, \quad y = 1$$

$$\text{Vertex} = (0, 1)$$

Directrix $Y = -a$

$$y - 1 = -1$$

$$y = -1 + 1$$

$$\boxed{y = 0}$$

$$\text{(vi)} \quad y^2 = -8(x - 3)$$

Solution:

$$y^2 = -8(x - 3)$$



$$(y - 0)^2 = -8(x - 2) \quad \dots\dots\dots (1)$$

Let $y - 0 = Y$, $x - 3 = X$

(1) Becomes

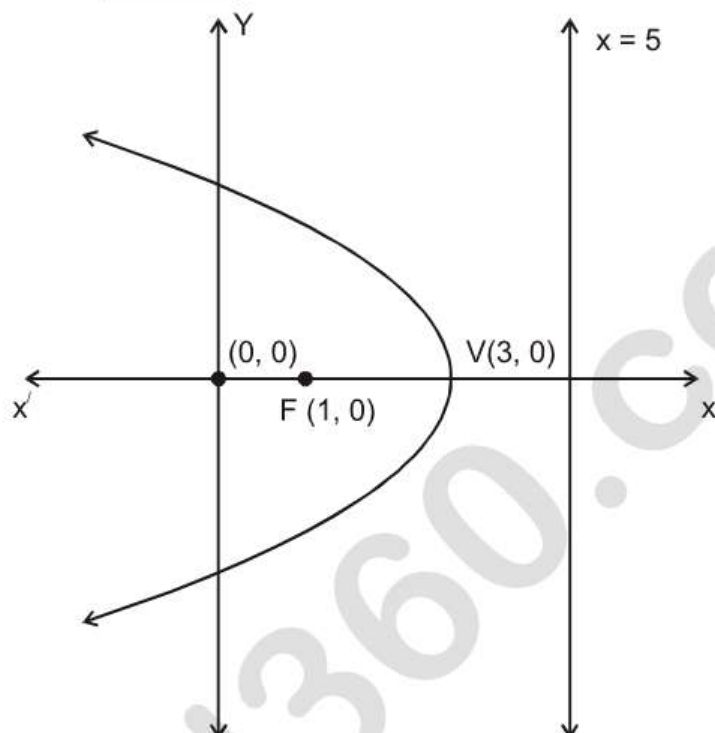
$$Y^2 = -8X$$

As standard form is

$$y^2 = -4ax$$

$$4a = 8$$

$$a = 2$$



$$\text{Focus} = (-a, 0)$$

$$(X, Y) = (-2, 0)$$

$$(x - 3, y - 0) = (-2, 0)$$

$$x - 3 = -2, \quad y - 0 = 0$$

$$x = -2 + 3, \quad y = 0$$

$$x = 1, \quad y = 0$$

$$\text{Focus} = (1, 0)$$

For the vertex

$$\text{Put } X = 0, Y = 0$$

$$x - 3 = 0, \quad y - 0 = 0$$

$$x = 3, \quad y = 0$$

$$\text{Vertex} = (3, 0)$$

$$\text{Directrix } X = a$$



$$x - 3 = 2$$

$$x = 5$$

(vii) $(x - 1)^2 = 8(y + 2)$ (Lahore Board 2009)

Solution:

$$(x - 1)^2 = 8(y + 2) \quad (i)$$

$$\text{Let } x - 1 = X, \quad y + 2 = Y$$

(i) Becomes $X^2 = 8Y$

As standard form is

$$x^2 = 4ay$$

$$4a = 8$$

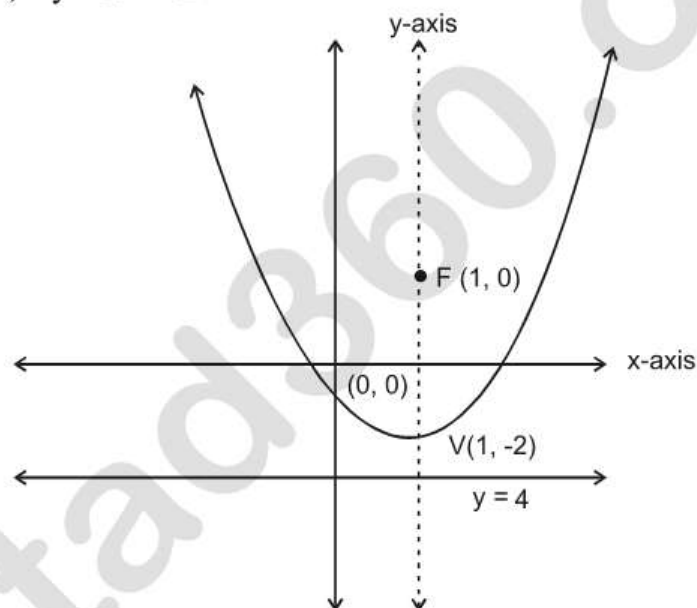
$$a = 2$$

$$\text{Focus} = (0, a)$$

$$(X, Y) = (0, 2)$$

$$(x - 1, y + 2) = (0, 2)$$

$$x - 1 = 0, \quad y + 2 = 2$$



$$x = 1, \quad y = 2 - 2$$

$$x = 1, \quad y = 0$$

$$\text{Focus} = (1, 0)$$

For the vertex put $X = 0, \quad Y = 0$

$$x - 1 = 0, \quad y + 2 = 0$$

$$x = 1, \quad y = -2$$

$$\text{Vertex} = (1, -2)$$



directrix $Y = a$

$$y + 2 = -2$$

$$y = -2 - 2$$

$$\boxed{y = -4}$$

(viii) $y = 6x^2 - 1$

Solution:

$$y = 6x^2 - 1$$

$$6x^2 = y + 1$$

$$x^2 = \frac{1}{6}(y + 1)$$

$$\Rightarrow (x - 0)^2 = \frac{1}{6}(y + 1) \quad \dots\dots\dots (i)$$

Let $x - 0 = X$ $y + 1 = Y$

(i) Becomes $X^2 = \frac{1}{6}Y$

$$x^2 = 4ay$$

As standard form is

$$4a = \frac{1}{6} \Rightarrow \boxed{a = \frac{1}{24}}$$

Focus = $(0, a)$

$(X, Y) = (0, \frac{1}{24})$

$(x - 0, y + 1) = (0, \frac{1}{24})$

$x - 0 = 0, y + 1 = \frac{1}{24}$

$x = 0, y = \frac{1}{24} - 1$

$y = \frac{-23}{24}$

$F = (0, \frac{-23}{24})$

For the vertex

Put $X = 0, Y = 0$

$x - 0 = 0, y + 1 = 0$

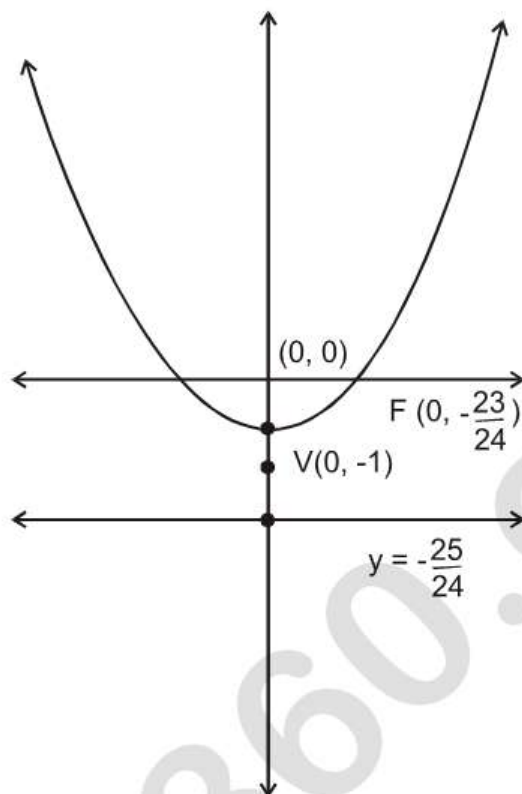
$x = 0, y = -1$



$$\boxed{\text{Vertex} = (0, -1)}$$

$$\text{Directrix } Y = -a$$

$$y + 1 = -\frac{1}{24} \quad y = -\frac{1}{24} - 1 = \boxed{-\frac{25}{24} = y}$$



(ix) $x + 8 - y^2 + 2y = 0$ (Lahore Board 2011)

Solution:

$$x + 8 - y^2 + 2y = 0$$

$$y^2 - 2y = x + 8$$

$$y^2 - 2y + 1 = x + 8 + 1$$

$$(y - 1)^2 = x + 9 \quad (i)$$

$$\text{Let } y - 1 = Y, \quad x + 9 = X$$

(i) becomes

$$Y^2 = X$$

As standard form is

$$y^2 = 4ax$$

$$4a = 1$$

$$\boxed{a = \frac{1}{4}}$$



$$\text{Focus} = (a, 0)$$

$$(X, Y) = \left(\frac{1}{4}, 0\right)$$

$$(x+9, y-1) = \left(\frac{1}{4}, 0\right)$$

$$x+9 = \frac{1}{4}, \quad y-1 = 0$$

$$x = \frac{1}{4} - 9 \quad y = 1$$

$$x = \frac{-35}{4}$$

$$\boxed{\text{Focus} \left(\frac{-35}{4}, 1\right)}$$

For the vertex put $X = 0, \quad Y = 0$

$$x+9 = 0, \quad y-1 = 0$$

$$x = -9, \quad y = 1$$

Required $\boxed{\text{Vertex} = (-9, 1)}$

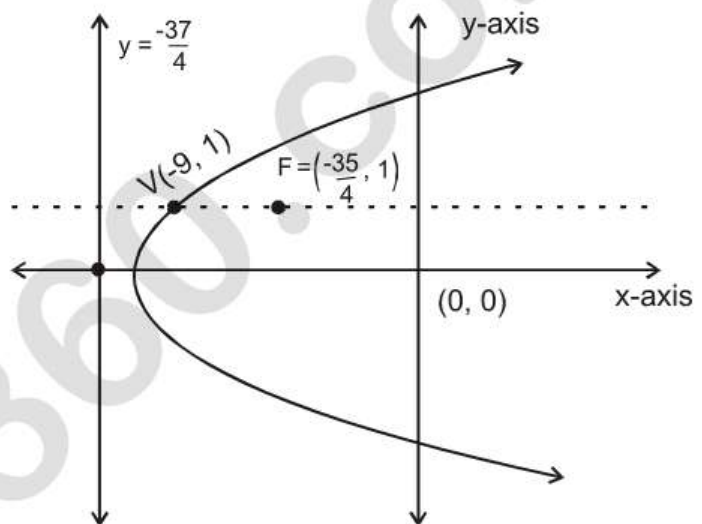
Directrix is

$$X = -a$$

$$x+9 = -\frac{1}{4}$$

$$x = -9 - \frac{1}{4}$$

$$\boxed{x = \frac{-37}{4}}$$



(x) $x^2 - 4x - 8y + 4 = 0$ (Lahore Board 2011)

Solution:

$$x^2 - 4x = 8y - 4 = 0$$

$$x^2 - 4x + 4 = 8y - 4 + 4$$

$$(x-2)^2 = 8y \quad \dots (i)$$

Let $x-2 = X \quad y-0 = Y$

(i) becomes

$$X^2 = 8Y$$



As standard form is

$$x^2 = 4ay$$

$$4a = 8$$

$$a = 2$$

$$\text{Focus} = (0, a)$$

$$(X, Y) = (0, 2)$$

$$(x - 2, y - 0) = (0, 2)$$

$$x - 2 = 0, \quad y - 0 = 2$$

$$x = 2, \quad y = 2$$

$$\text{Focus} = (2, 2)$$

For the vertex

$$\text{Put } X = 0, \quad Y = 0$$

$$(X, Y) = (0, 0)$$

$$(x - 2, y - 0) = (0, 0)$$

$$x - 2 = 0, \quad y - 0 = 0$$

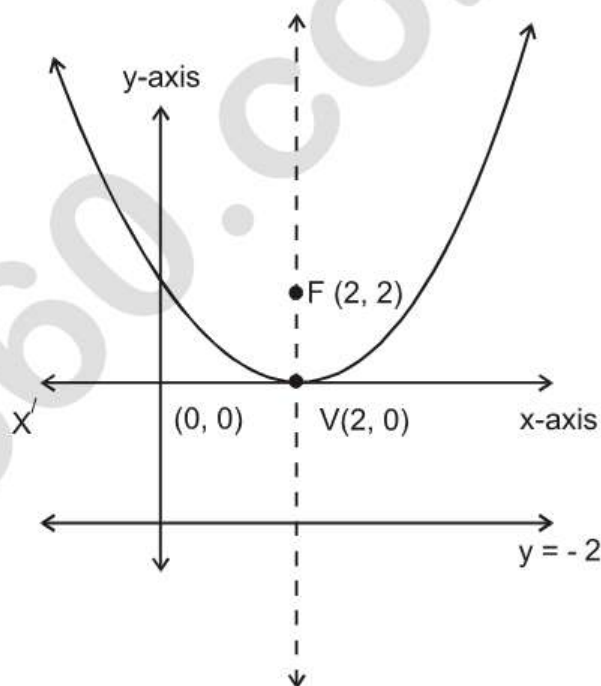
$$x = 2, \quad y = 0$$

$$\text{Vertex} = (2, 0)$$

$$\text{Directrix } Y = -a$$

$$y - 0 = -2$$

$$y = -2$$



Q.2: Write an equation of the Parabola with given elements.

(i) **Focus** $(-3, 1)$; **Directrix** $x = 3$

Solution:

$$\text{Given } F = (-3, 1)$$

$$\& \text{ directrix } x + 0y - 3 = 0$$

Let $P(x, y)$ be any point on the Parabola. Then $|PM|$ = Length of Perpendicular from $P(x, y)$ to the directrix L .



$$|PM| = \frac{|x + 0y - 3|}{\sqrt{(1)^2 + (0)^2}}$$

By definition of Parabola

$$|PF| = |PM|$$

or $|PF|^2 = |PM|^2$

$$\begin{aligned} \Rightarrow (x+3)^2 + (y-1)^2 &= (x+0y-3)^2 \\ x^2 + 9 + 6x + y^2 + 1 - 2y &= x^2 + 9 - 6x \\ y^2 - 2y + 1 &= -12x \\ (y-1)^2 &= -12x \quad \text{Ans.} \end{aligned}$$

(ii) Focus (2, 5) ; directrix $y = 1$

Solution:

Given $F = (2, 5)$

directrix $0x + y - 1 = 0$

Let $P(x, y)$ be any point on the Parabola.

Then $|PM|$ = Length of Perpendicular from $P(x, y)$ to directrix

$$|PM| = \frac{|0x + y - 1|}{\sqrt{(0)^2 + (1)^2}} = y - 1$$

Now, by definition of Parabola

$$|PF| = |PM|$$

$\Rightarrow |PF|^2 = |PM|^2$

$$\begin{aligned} (x-2)^2 + (y-5)^2 &= (y-1)^2 \\ x^2 + 4 - 4x + y^2 + 25 - 10y &= y^2 + 1 - 2y \\ x^2 - 4x - 8y + 28 &= 0 \quad \text{Ans} \end{aligned}$$

(iii) Focus (-3, 1) ; directrix $x - 2y - 3 = 0$

Solution:

Given Focus $(-3, 1)$

directrix $x - 2y - 3 = 0$

Let $P(x, y)$ be any point on the Parabola

Then $|PM|$ = distance or length of Perpendicular from $P(x, y)$ to the directrix.

$$|PM| = \frac{|x - 2y - 3|}{\sqrt{(1)^2 + (-2)^2}} = \frac{(x - 2y - 3)}{\sqrt{5}}$$

By definition of Parabola

$$|PF| = |PM| \Rightarrow |PF|^2 = |PM|^2$$

$$\Rightarrow (x+3)^2 + (y-1)^2 = \frac{(x-2y-3)^2}{5}$$



$$\begin{aligned}
 5[x^2 + 9 + 6x + y^2 + 1 - 2y] &= x^2 + 4y^2 + 9 - 4xy + 12y - 6x \\
 5x^2 + 45 + 30x + 5y^2 - 10y - x^2 - 4y^2 - 9 + 4xy - 12y + 6x &= 0 \\
 4x^2 + y^2 + 4xy + 36x - 22y + 36 &= 0 \quad \text{Ans.}
 \end{aligned}$$

(iv) Focus (1, 2); Vertex (3, 2)

Solution:

Given Focus = (1, 2) , Vertex = (3, 2)

We know that a = distance between focus & vertex

$$a = \sqrt{(3-1)^2 + (2-2)^2} = \sqrt{4+0} = 2$$

Required equation of Parabola

$$(y-k)^2 = -4a(x-h)$$

$$(y-2)^2 = -4(2)(x-3)$$

$$y^2 + 4 - 4y = -8x + 24$$

$$y^2 - 4y + 8x - 20 = 0 \quad \text{Ans.}$$

(v) Focus (-1, 0) ; Vertex (-1, 2)

Solution:

F = (-1, 0) , V = (-1, 2)

a = distance between focus to vertex

$$= \sqrt{(-1+1)^2 + (2-0)^2} = 2$$

Required equation of Parabola is

$$(x-h)^2 = -4a(y-k)$$

$$(x+1)^2 = -4(2)(y-2)$$

$$x^2 + 1 + 2x = -8y + 16$$

$$x^2 + 2x + 8y - 15 = 0 \quad \text{Ans.}$$

(vi) Directrix $x = -2$; Focus (2, 2)

Solution:

Given F = (2, 2) directrix $x + 0y + 2 = 0$

$|PM|$ = distance or length of perpendicular from p (x, y) to the directrix.

$$|PM| = \frac{|x + 0y + 2|}{\sqrt{1^2 + 0^2}} = x + 2$$

By definition of Parabola

$$|PF| = |PM|$$

$$\Rightarrow |PF|^2 = |PM|^2$$

$$\Rightarrow (x-2)^2 + (y-2)^2 = (x+2)^2$$

$$x^2 + 4 - 4x + y^2 + 4 - 4y = x^2 + 4 + 4x$$

$$y^2 - 4y - 8x + 4 = 0 \quad \text{Ans}$$



(vii) Directrix $y = 3$; Vertex $(2, 2)$

Solution:

$$\text{Directrix } 0x + y - 3 = 0 \quad V = (2, 2)$$

We know that a = distance between directrix and vertex

$$= a = \frac{|0(2) + 1(2) - 3|}{\sqrt{0^2 + 1^2}} = \frac{|2 - 3|}{\sqrt{1}} = |-1| = 1$$

Since the directrix is above the vertex,

Therefore equation of Parabola is $(x - h)^2 = -4a(y - k)$

$$(x - 2)^2 = -4(1)(y - 2)$$

$$x^2 + 4 - 4x = -4y + 8$$

$$x^2 + 4 - 4x + 4y - 8 = 0$$

$$x^2 - 4x + 4y - 4 = 0$$



(viii) **Directrix $y = 1$, Length of latusrectum is 8.0 and opens downward.**

Solution:

Given $4a = 8$ $a = 2$

As Parabola opens downward, so its equation is of the form

$$(x - h)^2 = -4a(y - k) \dots\dots\dots (1)$$

We know that vertex is below the directrix $y = 1$

So y - coordinate of the vertex is $y + a$

$$1 = y + 2 \Rightarrow y = -1 \quad \text{i.e.;} \quad \boxed{k = -1}$$

with $a = 2$ & $k = -1$ equation (1) becomes

$$(x - h)^2 = -4(2)(y + 1)$$

$$x^2 + h^2 - 2hx = -8y - 8$$

$$x^2 + h^2 - 2hx + 8y + 8 = 0 \quad \text{Ans.}$$

(ix) **Axis $y = 0$, through $(2, 1)$ & $(11, -2)$**

Solution:

As axis $y = 0$, so required equation of the parabola is $(y - k)^2 = 4a(x - h)$ (1)
because of the axis of Parabola is x -axis & $y = 0$ so $k = 0$

$\therefore$ with $k = 0$ equation (1) becomes

$$y^2 = 4a(x - h) \dots\dots\dots (2)$$

Since the para-bola passes through the points $(2, 1)$ & $(11, -2)$ equation (2) becomes

For $(2, 1)$

$$1 = 4a(2 - h)$$

$$1 = 8a - 4ah$$

For $(11, -2)$

$$4 = 4a(11 - h)$$

$$4 = 44a - 4ah \quad (4)$$

Subtracting (3) from (4) we have

$$4 = 44a - 4ah$$

$$-1 = -8a + 4ah$$

$$3 = 36a$$

$$\boxed{a = \frac{1}{12}}$$

Put in (3)

$$1 = 8\left(\frac{1}{12}\right) - 4\left(\frac{1}{12}\right)h$$

$$1 = \frac{8}{12} - \frac{4}{12}h$$



$$\begin{aligned}
 1 &= \frac{8-4h}{12} \Rightarrow 12 = 8-4h \\
 &\Rightarrow 4 = -4h \\
 &\Rightarrow \boxed{h = -1}
 \end{aligned}$$

Equation (2) becomes

$$y^2 = 4 \times \frac{1}{12} (x+1)$$

$$\boxed{3y^2 = x+1} \quad \text{Ans.}$$

(x) Axis parallel to y-axis. The points (0, 3) (3, 4) & (4, 11) lie on the graph.

Solution:

As axis of parabola parallel to y – axis, so its equation will be

$$(x-h)^2 = 4a(y-k) \quad \dots\dots\dots (1)$$

As points (0, 3), (3, 4) and (4, 11) lies on the parabola (1) so $(0-h)^2 = 4a(3-k)$

$$h^2 = 12a - 4ak \quad \dots\dots\dots (2)$$

$$\begin{aligned}
 \text{For (3, 4)} \quad (3-h)^2 &= 4a(4-k) \\
 9 + h^2 - 6h &= 16a - 4ak \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{For (4, 11)} \quad (4-h)^2 &= 4a(11-k) \\
 16 + h^2 - 8h &= 44a - 4ak \quad (4)
 \end{aligned}$$

Subtracting (2) from (3)

$$\begin{array}{rcl}
 9 + h^2 - 6h & = & 16a - 4ak \\
 - h^2 & = & -12a + 4ak \\
 \hline
 9 - 6h & = & 4a \quad (5)
 \end{array}$$

Subtracting (2) from (4)

$$\begin{array}{rcl}
 16 + h^2 - 8h & = & 44a - 4ak \\
 - h^2 & = & -12a + 4ak \\
 \hline
 16 - 8h & = & 32a \\
 8(2-h) & = & 32a \\
 2-h & = & 4a \quad (6)
 \end{array}$$

Subtracting (6) from (5)

$$\begin{array}{rcl}
 9 - 6h & = & 4a \\
 - 2 + h & = & -4a \\
 \hline
 7 - 5h & = & 0 \\
 \Rightarrow 7 & = & 5h \\
 \boxed{\frac{7}{5} = h} & & \text{Put in 5}
 \end{array}$$

$$9 - 6h = 4a$$



$$9 - 6\left(\frac{7}{5}\right) = 4a$$

$$9 - \frac{42}{5} = 4a$$

$$\frac{45 - 42}{5} = 4a \quad \Rightarrow \quad \frac{3}{5} = 4a$$

$$\boxed{a = \frac{3}{20}}$$

Put in (2)

$$\left(\frac{7}{5}\right)^2 = 12\left(\frac{3}{20}\right) - 4\left(\frac{3}{20}\right)k$$

$$\frac{49}{25} = \frac{36}{20} - \frac{12}{20}k$$

$$\frac{49}{25} = \frac{36 - 12k}{20}$$

$$196 = 180 - 60k$$

$$60k = -16 \quad \Rightarrow \quad k = \frac{-16}{60} = \boxed{\frac{-4}{5} = k}$$

Substituting all values in (1)

$$\left(x - \frac{7}{5}\right)^2 = 4\left(\frac{3}{20}\right)\left(y + \frac{4}{5}\right)$$

$$\left(x - \frac{7}{5}\right)^2 = \frac{3}{5}\left(y + \frac{4}{5}\right) \quad \text{Ans.}$$

Q.3: Find an equation of the Parabola having its focus at the origin and directrix Parallel to

(i) the x-axis

Solution:

Given $F = (0, 0)$

Directrix Parallel to x-axis

$$0x + y - h = 0$$

Let $P(x, y)$ be any point on the Parabola such that

$$|PF| = |PM|$$

$$\Rightarrow |PF|^2 = |PM|^2$$

$$(x - 0)^2 + (y - 0)^2 = \left(\frac{|0x + y - h|}{\sqrt{1^2 + 0^2}}\right)^2$$

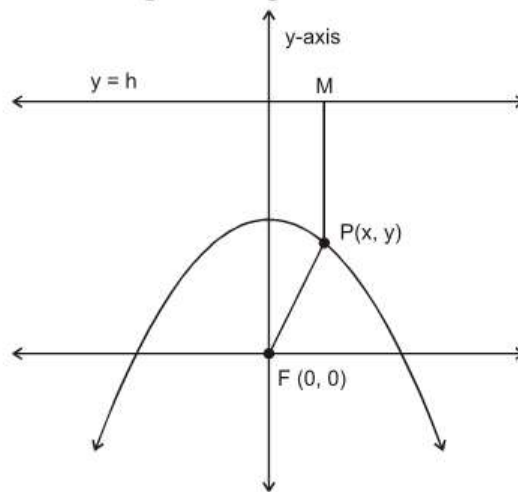
$$x^2 + y^2 = (y - h)^2$$

$$x^2 + y^2 = y^2 + h^2 - 2yh$$



$$x^2 + 2hy - h^2 = 0$$

Required Equation



(ii) The y-axis.

Solution:

Given $F = (0, 0)$

Directrix Parall to y – axis

$$x + 0y - h = 0$$

Let $P(x, y)$ be any point on the parabola such that

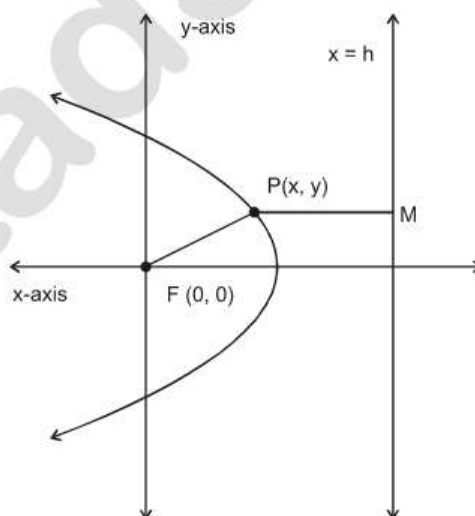
$$\Rightarrow |PF| = |PM|$$

$$\Rightarrow |PF|^2 = |PM|^2$$

$$(x - 0)^2 + (y - 0)^2 = (x + 0y - h)^2$$

$$x^2 + y^2 = x^2 + h^2 - 2xh$$

$$y^2 + 2xh - h^2 = 0 \quad \text{required equation.}$$



Q.4: Show that the parabola $(x \sin \alpha - y \cos \alpha)^2 = 4a(x \cos \alpha + y \sin \alpha)$ has focus at $(a \cos \alpha, a \sin \alpha)$ and its directrix is $x \cos \alpha + y \sin \alpha + a = 0$.

Solution:

Here Focus = $(a \cos \alpha, a \sin \alpha)$

directrix $M = x \cos \alpha + y \sin \alpha + a = 0$

Let $P(x, y)$ be any point on the Parabola, such that

$$|PF| = |PM|$$

$$\Rightarrow |PF|^2 = |PM|^2$$

$$(x - a \cos \alpha)^2 + (y - a \sin \alpha)^2 = \frac{(x \cos \alpha + y \sin \alpha + a)^2}{\sin^2 \alpha + \cos^2 \alpha}$$

$$x^2 + a^2 \cos^2 \alpha - 2ax \cos \alpha + y^2 + a^2 \sin^2 \alpha - 2ay \sin \alpha = x^2 \cos^2 \alpha + y^2 \sin^2 \alpha + a^2 + 2xy \sin \alpha \cos \alpha + 2ay \sin \alpha + 2ax \cos \alpha$$

$$x^2 - x^2 \cos^2 \alpha + y^2 - y^2 \sin^2 \alpha + a^2 (\cos^2 \alpha + \sin^2 \alpha) = a^2 + 2xy \sin \alpha \cos \alpha + 2ay \sin \alpha + 2ax \cos \alpha$$

$$x^2 (1 - \cos^2 \alpha) + y^2 (1 - \sin^2 \alpha) + a^2 - 2xy \sin \alpha \cos \alpha = a^2 + 4ay \sin \alpha + 4ax \cos \alpha$$

$$x^2 \sin^2 \alpha + y^2 \cos^2 \alpha - 2xy \sin \alpha \cos \alpha = 4ay \sin \alpha + 4ax \cos \alpha$$

$$(x \sin \alpha - y \cos \alpha)^2 = 4a(x \cos \alpha + y \sin \alpha) \quad \text{Hence proved.}$$

Q.5: Show that the ordinate at any point P of the Parabola is mean Proportional between the length of the Latusrectum and the abscissa of P.

Solution:

Let $P(x, y)$ be any point on the Parabola then the equation of Parabola is

$$y^2 = 4ax$$

$$y \cdot y = 4ax$$

$$\frac{y}{4a} = \frac{x}{y}$$

$$\Rightarrow \frac{x}{y} = \frac{y}{4a}$$

$$\Rightarrow x : y :: y : 4a$$

$$\Rightarrow \text{abscissa} : \text{ordinate} :: \text{ordinate} : \text{Length of Latusrectum}$$

Hence proved.

Q.6: A comet has a Parabolic orbit with the earth at the focus. When the comet is 150,000 km from the earth, the line joining the comet and the earth makes an angle of 30° with the axis of Parabola. How close will the comet come to the earth?

Solution:

Let focus be taken at origin, then $F = (0, 0)$



and directrix $x + 0y + 2a = 0$

Let $P(x, y)$ be any point on the Parabola such that

$$|PF| = |PM|$$

$$\Rightarrow |PF|^2 = |PM|^2$$

$$(x - 0)^2 + (y - 0)^2 = (x + 0y + 2a)^2$$

$$x^2 + y^2 = (x + 2a)^2 \quad \dots\dots (1)$$

Now, from right angled triangle EQP

By Pythagoras theorem

$$(150000)^2 = x^2 + y^2$$

$$\text{Putting in (1)} \quad (150000)^2 = (x + 2a)^2$$

$$x + 2a = 150000 \quad \dots\dots (2)$$

& we know that $\cos \alpha = \frac{\text{base}}{\text{hypotenous}}$

$$\cos 30^\circ = \frac{x}{150000} \Rightarrow x = \frac{\sqrt{3} \times 150000}{2} = \sqrt{3} \times 75000$$

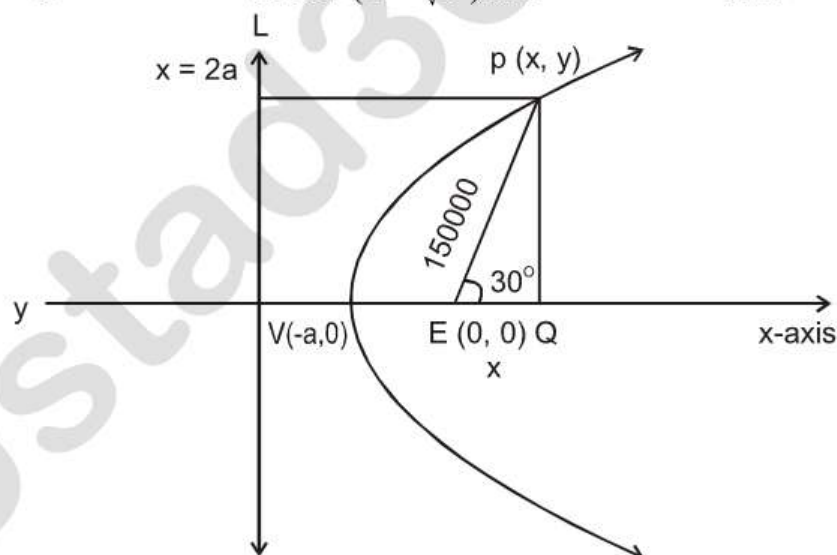
Put in (2)

$$75000 \sqrt{3} + 2a = 150000$$

$$2a = 150000 - 75000 \sqrt{3}$$

$$a = \frac{75000}{2} (2 - \sqrt{3})$$

$$a = 37500 (2 - \sqrt{3}) \text{ km} \quad \text{Ans}$$



Q.7: Find an equation of the Parabola formed by the cables of a suspension bridge whose span is a m and the verticle height of the supporting towers is bm .

Solution:

We know that an equation of Parabolla is

$$x^2 = 4ay \quad (1)$$

Since the point $P \left(\frac{a}{2}, b \right)$ lies on the parabola (1)

$\therefore$ (1) becomes

$$\left(\frac{a}{2} \right)^2 = 4a'b$$

$$\frac{a^2}{4} = 4a'b$$

$$\frac{a^2}{4} = a'(4b)$$

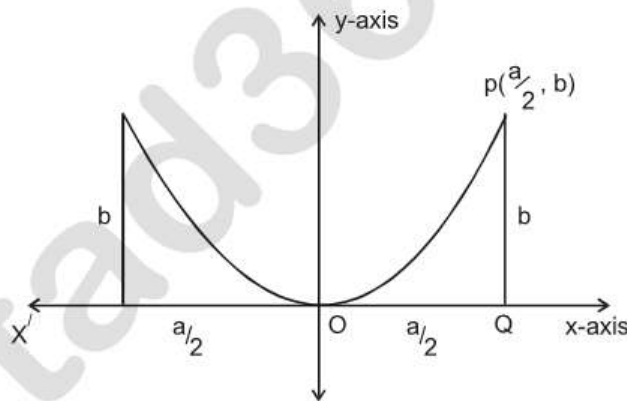
$$a' = \frac{a^2}{4} \times \frac{1}{4b} = \frac{a^2}{16b}$$

Now putting value of a' in (1)

$$x^2 = 4 \left(\frac{a^2}{16b} \right) y = \frac{a^2}{4b} y$$

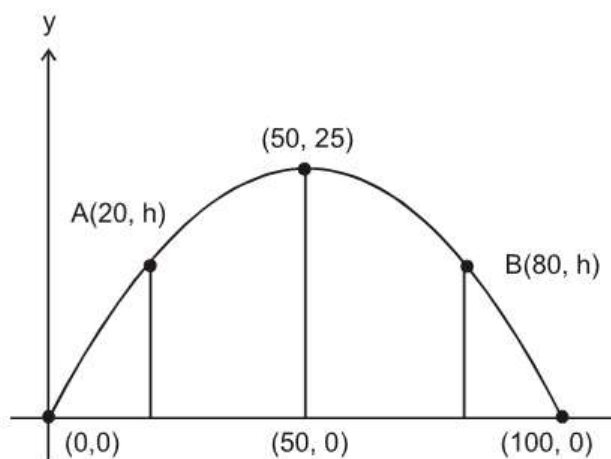
$$x^2 = \frac{a^2 y}{4b}$$

Ans



Q.8: A Parabolic arch has a 100 m base and height 25m. Find the height of the arch at the point 30 m from the center of the base.

Solution:



From the equation of Parabola is $(x - h)^2 = 4a (y - k)$ (1)

With vertex $V = (50, 25)$ (1) becomes

$$(x - 50)^2 = 4a(y - 25) \quad \text{..... (2)}$$

Since origin $0(0, 0)$ lies on parabola

(2) becomes

$$(0 - 50)^2 = 4a(0 - 25)$$

$$2500 = -100a$$

$$\boxed{a = -25}$$

Putting value of a in (2) we have

$$(x - 50)^2 = 4(-25)(y - 25) \quad \text{..... (3)}$$

Since point $A(20, h)$ also lies on parabola

(3) becomes

$$(20 - 50)^2 = -100(h - 25)$$

$$900 = -100h + 2500$$

$$100h = 2500 - 900$$

$$100h = 1600$$

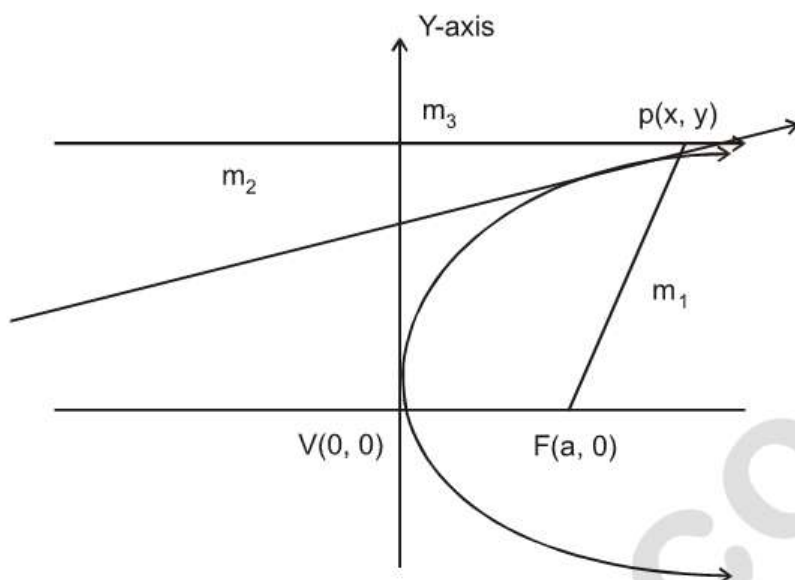
$$h = 16 \text{ m}$$

Ans



Q.9: Show that the tangent at any point P of a parabola makes equal angles with the line PF and the line through P Parallel to the axis of the Parabola, F being focus.

Solution:



We know that equation of parabola is

$$y^2 = 4ax \quad (1)$$

Diff. w.r.t 'x'

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{4a}{2y}$$

$$\frac{dy}{dx} = \frac{2a}{y}$$

$$\therefore m_2 = \text{Slope of tangent}$$

$$= \frac{dy}{dx} = \frac{2a}{y}$$

$$m_1 = \text{Slope of line between F and P}$$

$$m_1 = \frac{y-0}{x-a} = \frac{y}{x-a}$$

we know that slope of x – axis = 0

$$\therefore m_3 = \text{Slope of line parallel to x – axis is also}$$

$$\text{Now } \tan \theta_1 = \frac{m_2 - m_3}{1 + m_2 m_3}$$



$$\begin{aligned}\tan \theta_1 &= \frac{\frac{2a}{y} - a}{1 + \left(\frac{2a}{y}\right)(0)} \\ &= \frac{\frac{2a}{y}}{1} = \frac{2a}{y} \\ \theta_1 &= \tan^{-1} \left(\frac{2a}{y} \right) \quad \dots (2)\end{aligned}$$

$$\begin{aligned}\text{Next } \tan \theta_2 &= \frac{m_1 - m_2}{1 + m_1 m_2} \\ &= \frac{\frac{y}{x-a} - \frac{2a}{y}}{1 + \frac{y}{x-a} \times \frac{2a}{y}} \\ &= \frac{\frac{y^2 - 2ax + 2a^2}{xy - ay + 2ay}}{\frac{y^2 - 2ax + 2a^2}{xy + ay}} \quad \dots (3)\end{aligned}$$

Using (1)

$$\begin{aligned}\tan \theta &= \frac{4ax - 2ax + 2a^2}{xy + ay} = \frac{2ax + 2a^2}{xy + ay} = \frac{2a(x+a)}{y(x+a)} \\ \tan \theta_2 &= \frac{2a}{y} \\ \theta_2 &= \tan^{-1} \frac{2a}{y} \quad \dots (4)\end{aligned}$$

From (2) & (4)

$$\theta_1 = \theta_2 \quad \text{Hence Proved.}$$

Ellipse

Let $0 < e < 1$ and F be a fixed point and L be a fixed line not containing F. Let P(x, y) be a point in the plane and |PM| be the perpendicular distance of P from L. The set of all the points P such that

$$\frac{|PF|}{|PM|} = e \quad \text{is called an Ellipse.}$$

e is eccentricity, F is focus, L is directrix.

Standard Form of or Ellipse (Lahore Board 2010)



$$(i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (ii) \quad \frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

Special case of an Ellipse

Circle is a special case of an Ellipse. In circle 'e' = 0

Parametric Equations of an Ellipse

$x = a \cos \theta$, $y = b \sin \theta$ are Parametric Equations of Ellipse.

Important points about an Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(1) Eccentricity

$$e^2 = \frac{a^2 - b^2}{a^2}$$

$$(ae)^2 = a^2 - b^2$$

$$c^2 = a^2 - b^2$$

(2) Foci $(\pm ae, 0)$ or $(\pm c, 0)$

(3) Length of major axis = $2a$

(4) Length of minor axis = $2b$

(5) Equations of directrix x : $x = \pm \frac{a}{e}$

(6) Length of latus rectum = $\frac{2b^2}{a}$

(7) Center $(0, 0)$

(8) Vertices $(\pm a, 0)$

(9) Covertices $(0, \pm b)$

Note:

If center is other than $(0, 0)$ then equations of ellipse becomes

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

(1) Eccentricity

$$e^2 = \frac{a^2 - b^2}{a^2}$$

$$(ae)^2 = a^2 - b^2$$

$$c^2 = a^2 - b^2 \text{ where } c = ae$$

(2) Foci $(0, \pm ae)$ or $(0, \pm c)$

(3) Length of major axis = $2a$

(4) Length of minor axis = $2b$

(5) Equations of directrix y : $y = \pm \frac{a}{e}$

(6) Length of latus rectum = $\frac{2b^2}{a}$

(7) Center $(0, 0)$

(8) Vertices $(0, \pm a)$

(9) Covertices $(\pm b, 0)$

$$\& \quad \frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$



EXERCISE 6.5

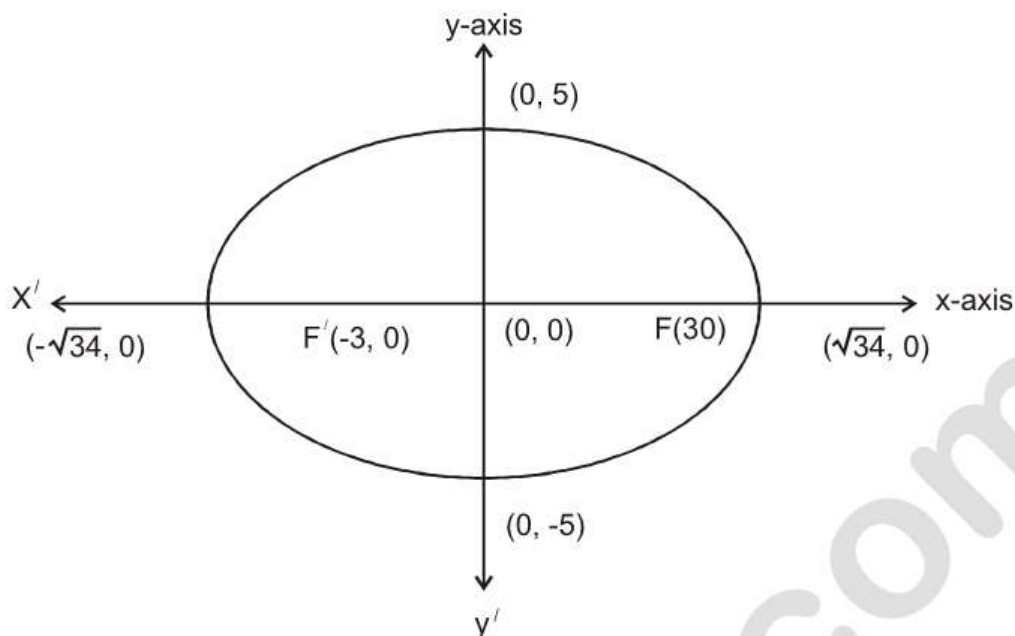
Q.1: Find an equation of the Ellipse with given data and sketch its graph.

- (i) Foci $(\pm 3, 0)$ and minor axis of length 10.
(Lahore Board 2009)

Solution:



Given $(\pm ae, 0) = (\pm 3, 0)$



$$ae = 3$$

$$\Rightarrow c = 3$$

$$\text{Also } 2b = 10 \Rightarrow \boxed{b = 5}$$

We know that

$$c^2 = a^2 - b^2$$

$$(3)^2 = a^2 - (5)^2$$

$$9 = a^2 - 25$$

$$9 + 25 = a^2$$

$$a^2 = 34 \Rightarrow \boxed{a = \pm \sqrt{34}}$$

Required equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{x^2}{34} + \frac{y^2}{25} = 1$$

Here vertices are

$$= (\pm a, 0)$$

$$= (\pm \sqrt{34}, 0)$$

Covertices are $(0, \pm b)$

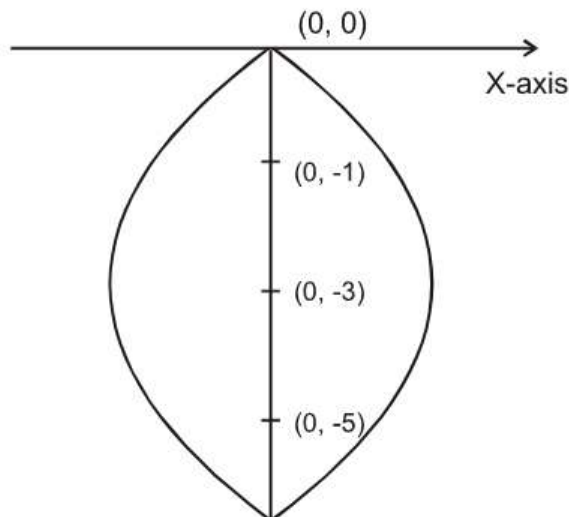
$$= (0, \pm 5)$$



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(ii) Foci $(0, -1)$ & $(0, -5)$ and major axis of length 6.

Solution:



$$\begin{aligned}\text{We know that center} &= \text{mid point of foci} \\ &= \left(\frac{0+0}{2}, \frac{-1-5}{2} \right) \\ &= (0, -3)\end{aligned}$$

Also we know that

C = distance between centre and focus

$$C = \sqrt{(0-0)^2 + (-3+1)^2}$$

$$C = \sqrt{4} = 2$$

$$\text{Given } 2a = 6$$

$$a = 3$$

We know for Ellipse

$$c^2 = a^2 - b^2 \Rightarrow (2)^2 = (3)^2 - b^2$$

$$4 - 9 = -b^2 \Rightarrow b^2 = 5$$

With center $(0, -3)$, required equation of the ellipse is

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$$

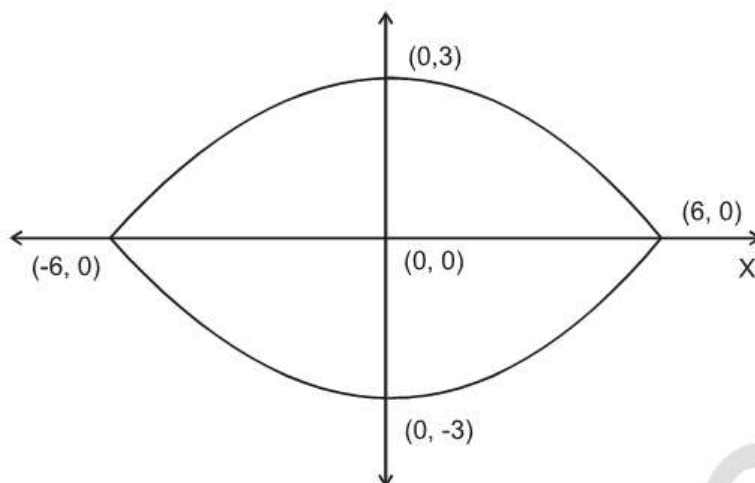
$$\frac{(x-0)^2}{5} + \frac{(y+3)^2}{(3)^2} = 1$$

$$\frac{x^2}{5} + \frac{(y+3)^2}{9} = 1$$



(iii) **Foci $(-3\sqrt{3}, 0)$ & Vertices $(\pm 6, 0)$**
(Lahore Board 2009)

Solution:



$$(\pm ae, 0) = (\pm 3\sqrt{3}, 0) \quad \& \quad (\pm a, 0) = (\pm 6, 0)$$

$$ae = 3\sqrt{3} \quad a = 6$$

$$c = 3\sqrt{3} \quad a = 6$$

We know that $c^2 = a^2 - b^2$

$$\begin{aligned} (3\sqrt{3})^2 &= (6)^2 - b^2 \\ 27 &= 36 - b^2 \quad \Rightarrow \quad b^2 = 9 \end{aligned}$$

Required equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{x^2}{36} + \frac{y^2}{9} = 1$$

Covertices are $= (0, \pm b)$

$$= (0, \pm 3)$$

(iv) **Vertices $(-1, 1), (5, 1)$ foci $(4, 1)$ & $(0, 1)$**

Solution:

$$2a = |VV'| \quad 2c = |FF'|$$

$$2a = \sqrt{(5+1)^2 + (1-1)^2} \quad 2c = \sqrt{(0+4)^2 + (1-1)^2}$$

$$2a = \sqrt{36} \quad 2c = \sqrt{16}$$

$$2a = 6 \quad 2c = 4$$

$$\boxed{a = 3}$$

$$\boxed{c = 2}$$

We know that $c^2 = a^2 - b^2$

$$(2)^2 = (3)^2 - b^2$$



$$4 = 9 - b^2$$

$$\boxed{b^2 = 5}$$

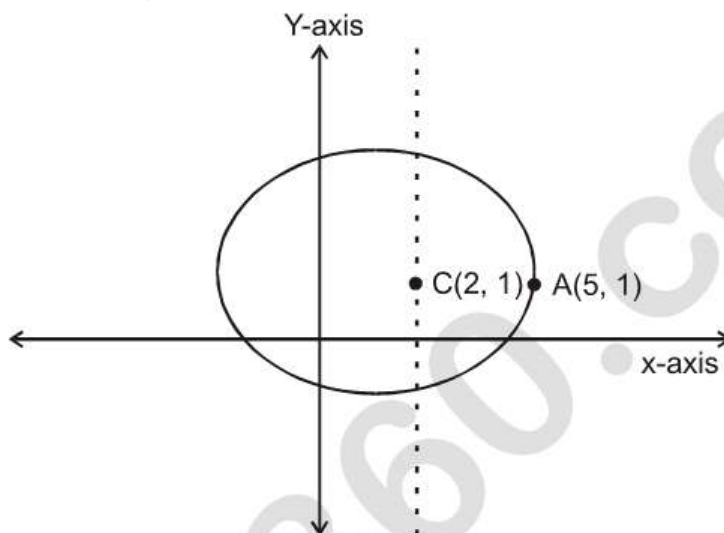
Now center of ellipse = (h, k) = mid point of foci

$$= \left(\frac{4+0}{2}, \frac{1+1}{2} \right)$$

with this center required equation of ellipse

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

i.e; $\frac{(x-2)^2}{9} + \frac{(y-1)^2}{5} = 1$



(v) Foci $(\pm\sqrt{5}, 0)$ & passing through $\left(\frac{3}{2}, \sqrt{3}\right)$

Solution:

Given $(\pm ae, 0) = (\pm\sqrt{5}, 0)$

$c = \sqrt{5}$ we know that $c^2 = a^2 - b^2$

$$(\sqrt{5})^2 = a^2 - b^2$$

$$5 = a^2 - b^2$$

$$\Rightarrow a^2 = 5 + b^2 \quad \dots\dots (1)$$

Since Ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

and it is passing through point $\left(\frac{3}{2}, \sqrt{3}\right)$ therefore it becomes $\frac{9}{4a^2} + \frac{3}{b^2} = 1 \quad \dots\dots(2)$

from (1) we have $a^2 = 5 + b^2$ Put in (2)



$$\frac{9}{4(5+b^2)} + \frac{3}{b^2} = 1$$

$$\frac{9b^2 + 12(5+b^2)}{4b^2(5+b^2)} = 1$$

$$9b^2 + 60 + 12b^2 = 20b^2 + 4b^4$$

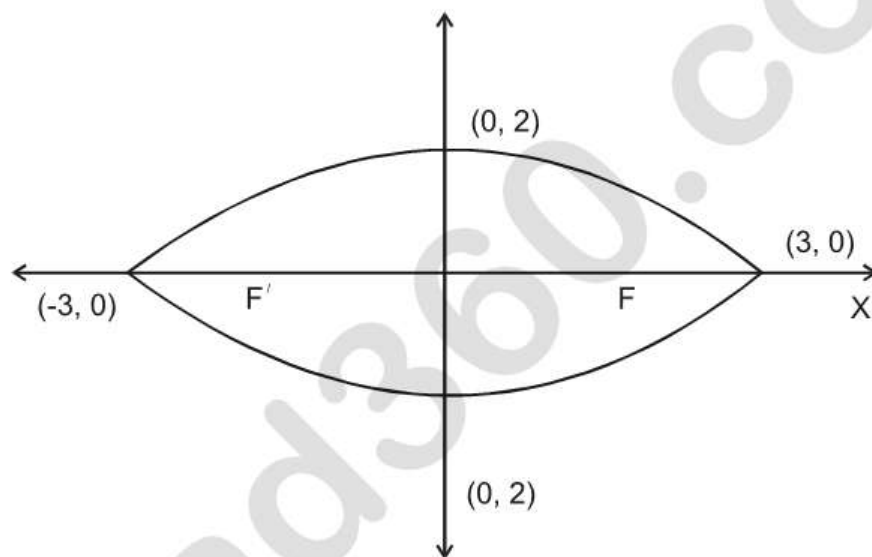
$$4b^4 - b^2 - 60 = 0$$

$$b^2 = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(4)(-60)}}{2(4)}$$

$$b^2 = \frac{1 \pm \sqrt{961}}{8}$$

$$= \frac{1 \pm 31}{8}$$

$$b^2 = \frac{1+31}{8}, \quad \frac{1-31}{8}$$



$$b^2 = \frac{32}{8}, \quad b^2 = \frac{-30}{8} \quad (\text{solution not possible})$$

$$b^2 = 4$$

Required equation of Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\text{i.e; } \frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$\text{Vertices } (\pm a, 0) = (\pm 3, 0)$$

$$\text{Co-vertices } (0, \pm b) = (0, \pm 2)$$



(vi) Vertices $(0, \pm 5)$, eccentricity $= \frac{3}{5}$

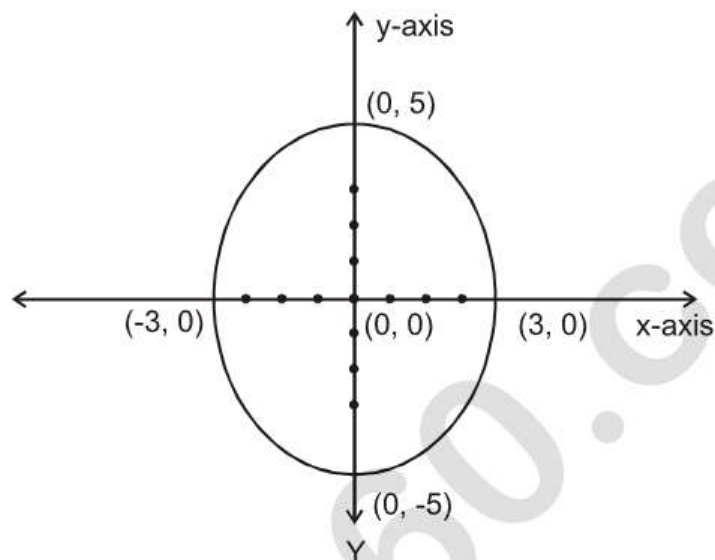
Solution:

$$\text{Vertices } (0, \pm a) = (0, \pm 5)$$

$$a = 5 \quad e = \frac{3}{5}$$

$$ae = 5 \times \frac{3}{5}$$

$$\Rightarrow c = 3$$



We know that

$$c^2 = a^2 - b^2$$

$$(3)^2 = (5)^2 - b^2$$

$$9 = 25 - b^2$$

$$b^2 = 16$$

Required equation of Ellipse is

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$\frac{x^2}{16} + \frac{y^2}{25} = 1$$

Co-vertices

$$(\pm b, 0) = (\pm 4, 0)$$

$$\text{Foci} = (0, \pm c) = (0, \pm 3)$$



(vii) Centre (0, 0) focus (0, -3), vertex (0, 4)
(Lahore Board 2011)

Solution:

$$\begin{aligned}(0, -c) &= (0, -3) & a &= 4 \\ c &= 3\end{aligned}$$

We know that $e^2 = \frac{a^2 - b^2}{a^2}$

$$(ae)^2 = a^2 - b^2$$

$$c^2 = a^2 - b^2$$

$$(3)^2 = 16 - b^2$$

$$9 = 16 - b^2$$

$$b^2 = 16 - 9 \Rightarrow b^2 = 7$$

Required equation of ellipse is $\frac{y^2}{a^2} + \frac{x^2}{b^2} = 1$

$$\Rightarrow \frac{y^2}{16} + \frac{x^2}{7} = 1$$

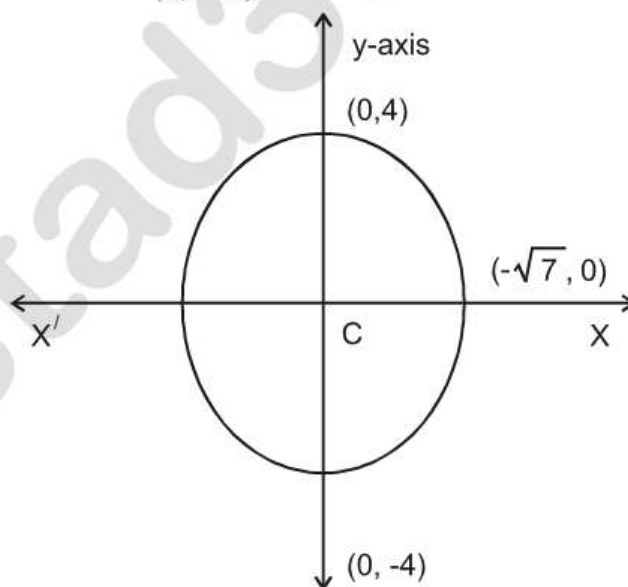
Its vertices are $(0, \pm a) = (0, \pm 4)$

Covertices $(\pm b, 0) = (\pm \sqrt{7}, 0)$

Coordinates of foci $= (0, \pm ae)$

$$= (0, \pm 4 \frac{3}{4})$$

$$= (0, \pm 3)$$



- (viii) Centre (2, 2) major axis parallel to y-axis and of length 8 units, minor axis parallel to x-axis and of length 6 units.

Solution:

$$\text{Center (2, 2) also } 2a = 8 \Rightarrow a = 4$$

$$2b = 6 \Rightarrow b = 3$$

$$\text{Equation of ellipse is } \frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$$

$$\Rightarrow \frac{(y-2)^2}{16} + \frac{(x-2)^2}{9} = 1$$

$$\text{Its vertices are } (0, \pm a) = (0, \pm 2)$$

$$\text{and } e^2 = \frac{a^2 - b^2}{a^2} = \frac{16 - 9}{16} = \frac{7}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$$

$$\text{Coordinates of foci} = (\pm ae, 0) = (\pm \frac{4\sqrt{7}}{4}, 0) = (\pm \sqrt{7}, 0)$$

$$(y-2, x-2) = (\pm \sqrt{7}, 0)$$

$$y = 2 \pm \sqrt{7}, \quad x = 2$$

$$\text{Thus foci are } (2, 2 + \sqrt{7}) \text{ \& } (2, 2 - \sqrt{7})$$

$$\text{For vertices, we have } x-2 = 0 \quad y-2 = \pm a$$

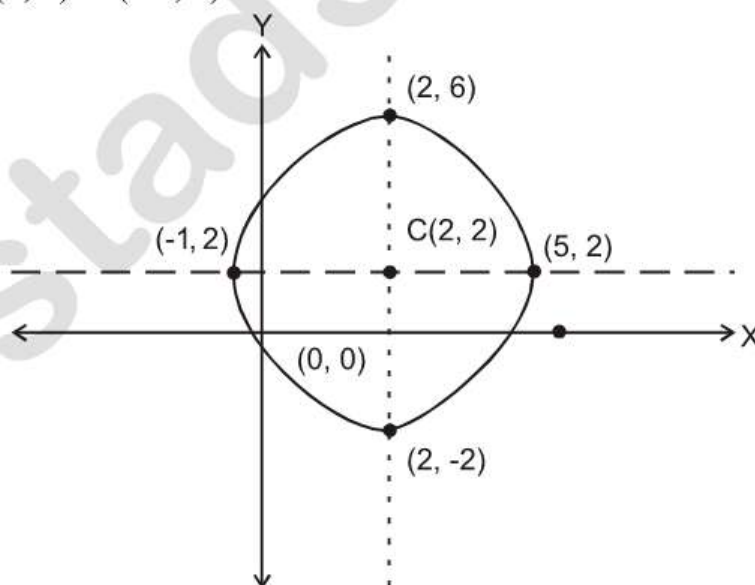
$$x = 2 \quad y = 2 \pm 4 \Rightarrow 6, -2$$

$$\text{So vertices are } (2, 6), (2, -2)$$

$$\text{Next we have } x-2 = \pm b, \quad y-2 = 0$$

$$x = 2 \pm 3, \quad y = 2$$

$$\text{So covertices are } (5, 2) \text{ \& } (-1, 2)$$



- (ix) Center (0, 0) symmetric with respect to both the axes and passing through the points (2, 3) and (6, 1).

Solution:

We know that equation of ellipse with center (0, 0) is given by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Since it passes through the points (2, 3) & (6, 1)

$$\frac{4}{a^2} + \frac{9}{b^2} = 1 \quad (I) \quad \frac{36}{a^2} + \frac{1}{b^2} = 1 \quad (II)$$

Subtracting

$$\begin{array}{rcl} 4b^2 + 9a^2 & = & a^2b^2 \\ -36b^2 \pm a^2 & = & -a^2b^2 \\ \hline -32b^2 + 8a^2 & = & 0 \\ 8a^2 & = & 32b^2 \\ a^2 & = & 4b^2 \quad \text{Put in (I)} \\ \frac{4}{4b^2} + \frac{9}{b^2} & = & 1 \\ \frac{1+9}{b^2} & = & 1 \end{array}$$

$$\boxed{10 = b^2} \quad \boxed{a^2 = 40}$$

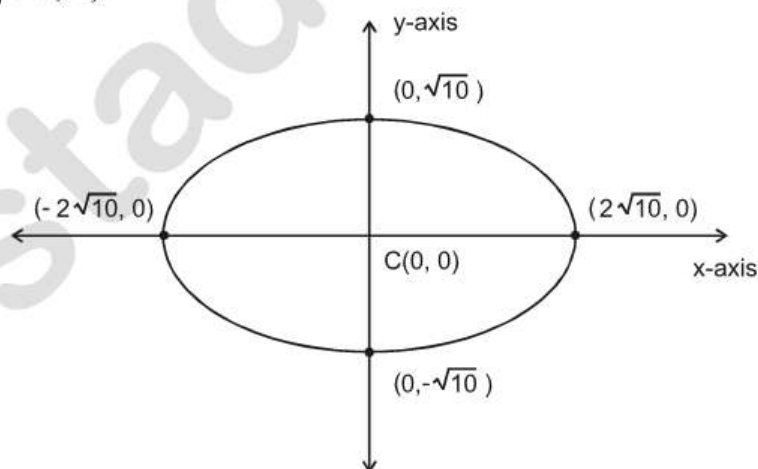
Required equation of ellipse is

$$\frac{x^2}{40} + \frac{y^2}{10} = 1$$

Vertices are $(\pm a, 0) = (\pm \sqrt{40}, 0) = (\pm 2\sqrt{10}, 0)$

Covertices $(0, \pm b) = (0, \pm \sqrt{10})$

Foci $(\pm \sqrt{30}, 0)$



(x) **Center (0, 0) major axis horizontal, the points (3, 1) (4, 0) lie on the graph.**

Solution:

We know that equation of Ellipse with center (0, 0) is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Since it passes through the points (3, 1) & (4, 0)

For (3, 1)

$$\frac{9}{a^2} + \frac{1}{b^2} = 1 \quad (1)$$

For (4, 0)

$$\frac{16}{a^2} + \frac{0}{b^2} = 1 \quad (2)$$

$$\boxed{a^2 = 16} \text{ Put in (1)}$$

$$\frac{9}{16} + \frac{1}{b^2} = 1$$

$$\frac{1}{b^2} = 1 - \frac{9}{16} = \frac{7}{16}$$

$$\boxed{b^2 = \frac{16}{7}}$$

Required equation of ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

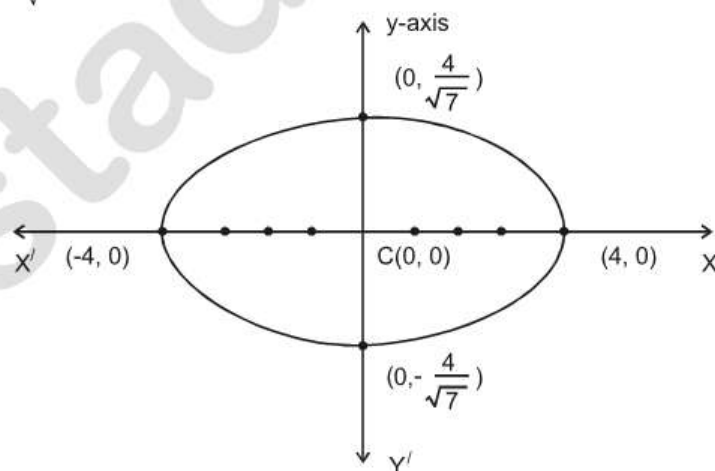
$$\frac{x^2}{16} + \frac{y^2}{16} = 1$$

$$\frac{x^2}{16} + \frac{7y^2}{16} = 1$$

Vertices $(\pm a, 0) = (\pm 4, 0)$

Covertices $(0, \pm b) = (0, \pm \frac{4}{\sqrt{7}})$

Foci $(\pm 4\sqrt{\frac{6}{7}}, 0)$



Q.2: Find the center, foci, eccentricity, vertices and directrix of the ellipse whose equation is given.

(i) $x^2 + 4y^2 = 16$ (Lahore Board 2009 (Supply))

Solution:

$$x^2 + 4y^2 = 16$$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

Here $a^2 = 16$ $b^2 = 4$

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{16 - 4}{16} = \frac{12}{16} = \frac{3}{4}$$

$$e = \frac{\sqrt{3}}{2}$$

Foci are $= (\pm c, 0) = (\pm 2\sqrt{3}, 0)$

Vertices are $= (\pm a, 0) = (\pm 4, 0)$

Directrix are $x = \pm \frac{a}{e} \Rightarrow x = \pm \frac{4}{\frac{\sqrt{3}}{2}} = \pm \frac{8}{\sqrt{3}}$

Clearly center of ellipse is (0, 0)

(ii) $9x^2 + y^2 = 18$

Solution:

$$9x^2 + y^2 = 18$$

$$\frac{x^2}{2} + \frac{y^2}{18} = 1$$

$$a^2 = 18, \quad b^2 = 2$$

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{18 - 2}{18} = \frac{16}{18} = \frac{8}{9} \Rightarrow e = \frac{2\sqrt{2}}{3}$$

foci are $= (0, \pm c) = (0, \pm 4)$

vertices are $= (\pm a, 0) = (\pm 3\sqrt{2}, 0)$

directrix are $= y = \pm \frac{a}{e} = \pm \frac{3\sqrt{2}}{\frac{2\sqrt{2}}{3}} = \pm \frac{9}{2}$

Clearly center is (0, 0)



(iii) $25x^2 + 9y^2 = 225$

Solution:

$$\frac{25x^2}{225} + \frac{9y^2}{225} = 1$$

$$\frac{x^2}{9} + \frac{y^2}{25} = 1 \quad (I)$$

$$a^2 = 25 \quad \& \quad b^2 = 9$$

Eccentricity

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{25 - 9}{25} = \frac{16}{25} \Rightarrow e = \frac{4}{5}$$

$$\text{foci} = (0, \pm c) = (0, \pm 4)$$

$$\text{vertices} = (0, \pm a) = (0, \pm 5)$$

$$\text{Center} = (0, 0)$$

$$\text{Directrix } y = \pm \frac{a}{e} = \pm \frac{5}{\frac{4}{5}} = \pm \frac{25}{4}$$

(iv) $\frac{(2x-1)^2}{4} + \frac{(y+2)^2}{16} = 1$

Solution:

$$\text{Let } 2x - 1 = X, \quad y + 2 = Y$$

Given equation becomes

$$\frac{X^2}{4} + \frac{Y^2}{16} = 1 \quad \text{Here } a^2 = 16, \quad b^2 = 4$$

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{16 - 4}{16} = \frac{12}{16} = \frac{3}{4}$$

Eccentricity

$$e = \frac{\sqrt{3}}{2}$$

Center:- For center put $X = 0, \quad Y = 0$

$$2x - 1 = 0, \quad y + 2 = 0$$

$$x = \frac{1}{2}, \quad y = -2$$

Required center is $(\frac{1}{2}, -2)$

$$\text{Foci } (0, \pm ae) = \left(0, \pm 4 \left(\frac{\sqrt{3}}{2}\right)\right)$$



$$\begin{aligned}
 (X, Y) &= (0, \pm 2\sqrt{3}) \\
 (2x-1, y+2) &= (0, \pm 2\sqrt{3}) \\
 2x-1 &= 0 & y+2 &= \pm 2\sqrt{3} \\
 x &= \frac{1}{2} & y &= -2 \pm 2\sqrt{3}
 \end{aligned}$$

Required foci are $(\frac{1}{2}, -2 \pm 2\sqrt{3})$

$$\begin{aligned}
 \text{Vertices } (0, \pm a) &= (0, \pm 4) \\
 (X, Y) &= (0, \pm 4) \\
 (2x-1, y+2) &= (0, \pm 4) \\
 2x-1 &= 0 & y+2 &= \pm 4 \\
 x &= \frac{1}{2} & y &= -2 \pm 4 \\
 & & y &= -6, 2
 \end{aligned}$$

Vertices are $(\frac{1}{2}, -6), (\frac{1}{2}, 2)$

$$\begin{aligned}
 \text{Directrices } Y &= \pm \frac{a}{e} \\
 y+2 &= \pm \frac{4}{\frac{2}{\sqrt{3}}} \\
 y+2 &= \pm \frac{8}{\sqrt{3}} \Rightarrow y = -2 \pm \frac{8}{\sqrt{3}}
 \end{aligned}$$

(v) $x^2 + 16x + 4y^2 - 16y + 76 = 0$

Solution:

$$\begin{aligned}
 x^2 + 16x + 4(y^2 - 4y) &= -76 \\
 (x^2 + 16x + 64) + 4(y^2 - 4y + 4) &= -76 + 64 + 16 \\
 (x+8)^2 + 4(y-2)^2 &= 4 \\
 \frac{(x+8)^2}{4} + \frac{4(y-2)^2}{4} &= \frac{4}{4} \\
 \frac{(x+8)^2}{4} + (y-2)^2 &= 1
 \end{aligned}$$

Let

$$x+8 = X, \quad y-2 = Y$$



$$\frac{X^2}{4} + \frac{Y^2}{1} = 1 \quad (\text{an ellipse})$$

$$a^2 = 4, \quad b^2 = 1 \quad e^2 = \frac{a^2 - b^2}{a^2} = \frac{4 - 1}{4} = \frac{3}{4}$$

$$\Rightarrow e = \frac{\sqrt{3}}{2}$$

For the center Put $X = 0, \quad Y = 0$

$$x + 8 = 0, \quad y - 2 = 0$$

$$x = -8, \quad y = 2$$

Required centre $(-8, 2)$

Foci $= (\pm ae, 0)$

$$(X, Y) = (\pm 2 \left(\frac{\sqrt{3}}{2} \right), 0)$$

$$(x + 8, y - 2) = (\pm \sqrt{3}, 0)$$

$$x + 8 = \pm \sqrt{3} \quad y - 2 = 0$$

$$x = -8 \pm \sqrt{3} \quad y = 2$$

Required foci are $(-8 \pm \sqrt{3}, 2)$

Vertices are $= (\pm a, 0)$

$$(X, Y) = (\pm a, 0)$$

$$(x + 8, y - 2) = (\pm 2, 0)$$

$$x + 8 = \pm 2, \quad y - 2 = 0$$

$$x = -8 \pm 2 \quad y = 2$$

Required vertices are $(-6, 2)$ & $(-10, 2)$

$$\text{Directrix } X = \pm \frac{a}{e}$$

$$x + 8 = \pm \frac{\frac{2}{\sqrt{3}}}{\frac{\sqrt{3}}{2}} = \pm \frac{4}{\sqrt{3}}$$

$$x = -8 \pm \frac{4}{\sqrt{3}}$$

$$(vi) \quad 25x^2 + 4y^2 - 250x - 16y + 541 = 0$$

Solution:

$$25x^2 - 250x + 4y^2 + 16y = -541$$

$$25(x^2 - 10x) + 4(y^2 - 4y) = -541$$

$$25(x^2 - 10x + 25) + 4(y^2 - 4y + 4) = -541 + 625 + 16$$



$$25(x-5)^2 + 4(y-2)^2 = 100$$

Dividing both sides by 100

$$\frac{(x-5)^2}{4} + \frac{(y-2)^2}{25} = 1$$

Let

$$x-5 = X, \quad y-2 = Y \quad \text{above equation}$$

Becomes

$$\frac{X^2}{4} + \frac{Y^2}{25} = 1 \quad (\text{an Ellipse})$$

$$a^2 = 25, \quad b^2 = 4$$

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{25 - 4}{25} = \frac{21}{25}$$

$$e = \frac{\sqrt{21}}{5}$$

For Center Put $X = 0, Y = 0$

$$(x-5, y-2) = (0, 0)$$

$$x-5 = 0, \quad y-2 = 0$$

$$x = 5, \quad y = 2$$

Center is (5, 2)

$$\text{Foci} = (0, \pm ac) = (0, \pm 5 \frac{\sqrt{21}}{5})$$

$$(X, Y) = (0, \pm \sqrt{21})$$

$$(x-5, y-2) = (0, \pm \sqrt{21})$$

$$x-5 = 0, \quad y-2 = \pm \sqrt{21}$$

$$x = 5, \quad y = 2 \pm \sqrt{21}$$

Foci are $(5, 2 \pm \sqrt{21})$

$$\text{Vertices } (0, \pm a) = (0, \pm 5)$$

$$(X, Y) = (0, \pm 5)$$

$$(x-5, y-2) = (0, \pm 5)$$

$$x-5 = 0, \quad y-2 = \pm 5$$

$$x = 5, \quad y = 2 \pm 5$$

$$y = 7, -3$$

Vertices are (5, 7) & (5, -3)



Directrix are

$$Y = \pm \frac{a}{e}$$

$$y - 2 = \pm \frac{5}{\frac{\sqrt{21}}{5}}$$

$$y = 2 \pm \frac{25}{\sqrt{21}}$$

Q.3: Let a be a +ve number and $0 < c < a$. Let $F(-c, 0)$ & $F'(c, 0)$ be two given points. Prove that the locus of points $p(x, y)$ such that $|PF| + |PF'| = 2a$ is an ellipse.

Solution:

Given $P(x, y)$, $F(-c, 0)$, $F'(c, 0)$

$$\therefore |PF| + |PF'| = 2a$$

$$\sqrt{(x+c)^2 + (y-0)^2} + \sqrt{(x-c)^2 + (y-0)^2} = 2a$$

$$\sqrt{(x+c)^2 + y^2} = 2a - \sqrt{(x-c)^2 + y^2}$$

Squaring on both sides

$$[\sqrt{(x+c)^2 + y^2}]^2 = [2a - \sqrt{(x-c)^2 + y^2}]^2$$

$$(x+c)^2 + y^2 = 4a^2 + (x-c)^2 + y^2 - 4a\sqrt{(x-c)^2 + y^2}$$

$$x^2 + c^2 + 2cx + y^2 = 4a^2 + x^2 + c^2 - 2cx + y^2 - 4a\sqrt{(x-c)^2 + y^2}$$

$$4cx - 4a^2 = -4a\sqrt{(x-c)^2 + y^2}$$

(Dividing by 4)

$$(cx - a^2) = -a\sqrt{(x-c)^2 + y^2}$$

Again Squaring

$$(cx - a^2)^2 = [-a\sqrt{(x-c)^2 + y^2}]^2$$

$$c^2x^2 + a^4 - 2cxa^2 = a^2(x^2 + c^2 - 2cx + y^2)$$

$$c^2x^2 + a^4 - 2cxa^2 = a^2x^2 + a^2c^2 - 2cxa^2 + a^2y^2$$

$$x^2(c^2 - a^2) - a^2y^2 = a^2(c^2 - a^2)$$

Dividing throughout by $c^2 - a^2$

$$x^2 - \frac{a^2y^2}{c^2 - a^2} = a^2 \quad (1)$$

We know that for ellipse $c^2 = a^2 - b^2 \Rightarrow c^2 - a^2 = -b^2$ put in (1)

$$x^2 - \frac{a^2y^2}{-b^2} = a^2$$



$$x^2 + \frac{a^2 y^2}{b^2} = a^2$$

(Dividing throughout by a^2)

$$\frac{x^2}{a^2} + \frac{a^2 y^2}{a^2 b^2} = \frac{a^2}{a^2}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{which is an ellipse}$$

Q.4: Use problem 3 to find equation of the ellipse as locus of points $P(x, y)$ such that the sum of the distances from P to the points $(0, 0)$ & $(1, 1)$ is 2.

Solution:

Given $P(x, y)$, $F(0, 0)$ & $F'(1, 1)$ Also given that $2a = 2$

For ellipse we know that

$$|PF| + |PF'| = 2a$$

$$\sqrt{(x-0)^2 + (y-0)^2} + \sqrt{(x-1)^2 + (y-1)^2} = 2$$

$$\sqrt{x^2 + y^2} + \sqrt{(x-1)^2 + (y-1)^2} = 2$$

$$\sqrt{(x-1)^2 + (y-1)^2} = 2 - \sqrt{x^2 + y^2} \quad \text{Squaring}$$

$$(x-1)^2 + (y-1)^2 = 4 + (x^2 + y^2) - 4\sqrt{x^2 + y^2}$$

$$x^2 + 1 - 2x + y^2 + 1 - 2y = 4 + x^2 + y^2 - 4\sqrt{x^2 + y^2}$$

$$-2x - 2y - 2 = -4\sqrt{x^2 + y^2}$$

$$x + y + 1 = 2\sqrt{x^2 + y^2} \quad (\text{Dividing both sides by 2})$$

Again squaring

$$(x + y + 1)^2 = (2\sqrt{x^2 + y^2})^2$$

$$x^2 + y^2 + 1 + 2xy + 2y + 2x = 4(x^2 + y^2)$$

$$4x^2 + 4y^2 - x^2 - y^2 - 1 - 2xy - 2y - 2x = 0$$

$$3x^2 + 3y^2 - 2xy - 2x - 2y - 1 = 0 \quad \text{required ellipse}$$

Q.5: Prove that latusrectum of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$

Solution:

The given ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (1)$$

From the figure, the points $A(c, h)$ & $B(c, -h)$ lies on ellipse (1) therefore.

For $A(c, h)$ equation (1) becomes

$$\frac{c^2}{a^2} + \frac{h^2}{b^2} = 1$$



$$\frac{h^2}{b^2} = 1 - \frac{c^2}{a^2}$$

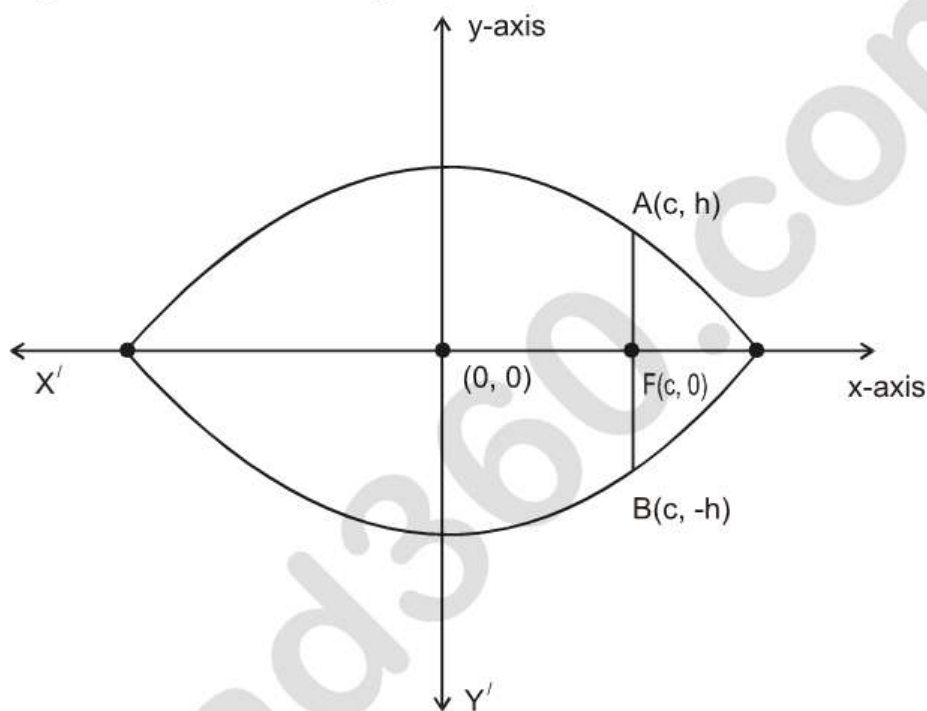
$$\frac{h^2}{b^2} = \frac{a^2 - c^2}{a^2} \quad (2)$$

For ellipse, we know that

$$\begin{aligned} c^2 &= a^2 - b^2 \\ \Rightarrow b^2 &= a^2 - c^2 \end{aligned} \quad \text{Put in (2)}$$

$$\frac{h^2}{b^2} = \frac{b^2}{a^2}$$

$$h^2 = \frac{b^4}{a^2} \Rightarrow h = \pm \frac{b^2}{a}$$



Points A & B becomes

$$A\left(c, \frac{b^2}{a}\right) \text{ and } B\left(c, -\frac{b^2}{a}\right)$$

Length of Latus rectum AB

$$\begin{aligned} &= \sqrt{(c - c)^2 + \left(\frac{b^2}{a} - \left(-\frac{b^2}{a}\right)\right)^2} \\ &= \sqrt{\left(\frac{2b^2}{a}\right)^2} = \sqrt{\frac{4b^4}{a^2}} \\ &= \frac{2b^2}{a} \quad \text{Hence proved.} \end{aligned}$$



Q.6: The major axis of an ellipse in standard form lies along the x-axis and has length $4\sqrt{2}$. The distance between the foci equals the length of minor axis. Write an equation of the ellipse.

Solution:

Since the major axis of the ellipse lies along x-axis so its equation is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots\dots (1)$$

By the given condition $2a = 4\sqrt{2}$

$$\Rightarrow a = 2\sqrt{2}$$

We know that

Distance between foci = length of minor axis

$$2c = 2b$$

$$\Rightarrow c = b$$

Since for ellipse $c^2 = a^2 - b^2$
 $b^2 = a^2 - b^2$

$$2b^2 = a^2 \Rightarrow b^2 = \frac{a^2}{2}$$

$$\Rightarrow b^2 = \frac{(2\sqrt{2})^2}{2} = \frac{8}{2} = 4$$

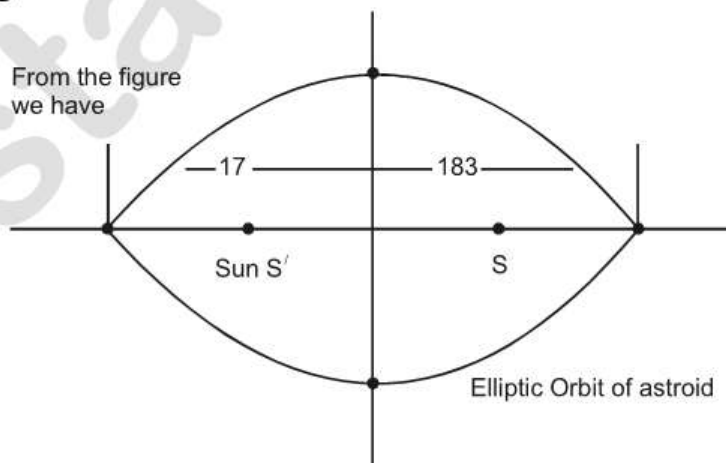
Putting values of a & b in (1)

$$\frac{x^2}{8} + \frac{y^2}{4} = 1 \quad \text{Required equation of ellipse}$$

Q.7: An astroid has elliptic orbit with the sun at one focus. Its distance from the sun ranges from 17 million miles to 183 million miles. Write an equation of the orbit of the astroid.

Solution: (Lahore Board 2004)

From the figure we have



Greatest distance of astroid from the sun is $a + c = 183$ (1)

And least distance of astroid from the sun is

$$a - c = 17 \quad (2)$$

Adding (1) & (2)

$$a + c = 183$$

$$\underline{a - c = 17}$$

$$2a = 200$$

$$a = 100 \quad \text{Put in (1)}$$

$$100 + c = 183$$

$$c = 83$$

For ellipse, we know that

$$c^2 = a^2 - b^2 \Rightarrow b^2 = a^2 - c^2$$

$$\Rightarrow b^2 = (100)^2 - (83)^2$$

$$b^2 = 3111$$

Required equation of the orbit of astroid is

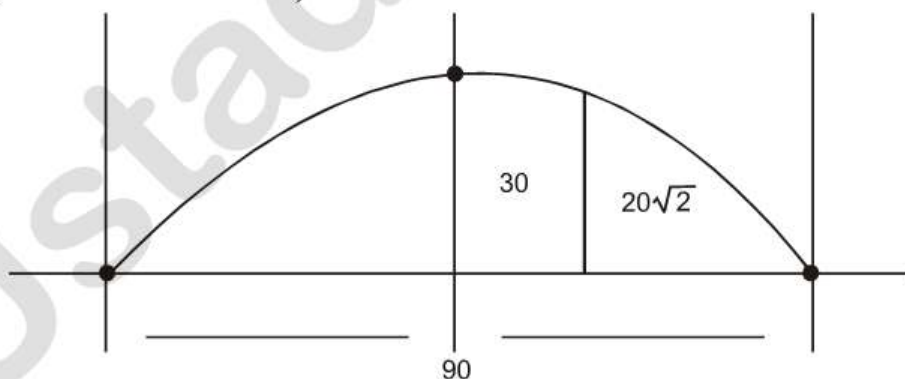
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{(100)^2} + \frac{y^2}{3111} = 1$$

$$\frac{x^2}{10000} + \frac{y^2}{3111} = 1 \quad \text{Ans.}$$

Q.8: An arch in the shape of semi ellipse is 90m wide at the base and 30 m high at the center. At what distance from the center is the arch $20\sqrt{2}$ m high?

Solution: (Lahore Board 2004)



Let the equation of the semi-elliptic are be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (1)

Since the arch is 90m wide at the base



$$\text{so } 2a = 90 \Rightarrow a = 45$$

Since arch is 30 m high at the center, so

$$b = 30, \quad \text{putting } a \text{ \& } b \text{ in (1)}$$

$$\frac{x^2}{(45)^2} + \frac{y^2}{(30)^2} = 1 \quad \dots\dots\dots (2)$$

Let d be the required distance from the center when arch is $20\sqrt{2}$ m high.

$$\text{Putting } x = d, \quad y = 20\sqrt{2} \text{ in (2)}$$

$$\frac{d^2}{(45)^2} + \frac{(20\sqrt{2})^2}{(30)^2} = 1$$

$$\frac{d^2}{2025} + \frac{800}{900} = 1$$

$$\frac{d^2}{2025} = 1 - \frac{800}{900}$$

$$d^2 = \frac{100}{900} \times 2025$$

$$d^2 = \frac{(45)^2}{(3)^2}$$

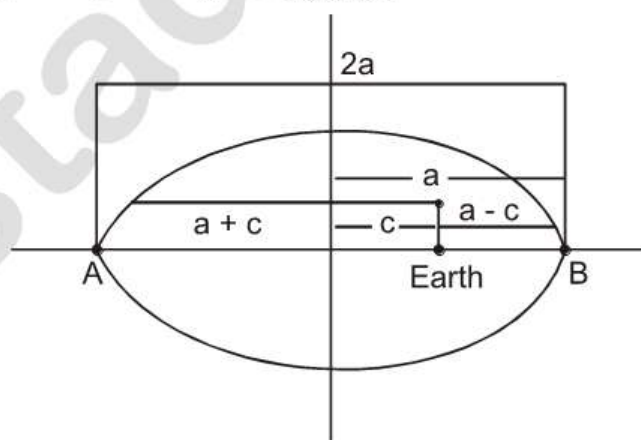
$$d = \frac{45}{3} = 15 \text{ meter}$$

Q.9: The moon orbits the earth in an elliptic path with earth at one focus. The major and minor axes of the orbit are 768, 806 km & 767, 746 km respectively. Find the greatest and least distances in astronomy called apogee & perigee) of the moon from the earth.

Solution:

By the given conditions

$$2a = 768806 \Rightarrow a = 384403$$



$$2b = 767746 \Rightarrow b = 383874$$

$$\text{Since } \sqrt{c^2} = \sqrt{a^2 - b^2}$$



$$c = \sqrt{(384403)^2 - (383874)^2}$$

$$c = 20179 \text{ km}$$

The greatest distance

$$\begin{aligned} a + c &= 384403 + 20179 \\ &= 404582 \text{ km} \end{aligned}$$

The least distance

$$\begin{aligned} a - c &= 384403 - 20179 \\ &= 364224 \text{ km} \end{aligned}$$

Hyperbola

Let $e > 1$ and F be a fixed point and L be line not containing F . Also let $P(x, y)$ be any point in the plane and $|PM|$ be the perpendicular distance of P from L . The set of all the points $P(x, y)$ such that $\frac{|PF|}{|PM|} = e > 1$ is called hyperbola.

Key points about Hyperbola

(1) Equation (Standard form)

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

(2) Eccentricity $e^2 = \frac{a^2+b^2}{a^2}$

$$(ae)^2 = a^2 + b^2 \Rightarrow c^2 = a^2 + b^2$$

(3) Foci $(\pm ae, 0)$

(4) Directrix $x = \pm \frac{a}{e}$

(5) Center $(0, 0)$

(6) Vertices $(\pm a, 0)$

(7) Covertices $(0, \pm b)$

Length of latus rectum is $\frac{2b^2}{a}$.

If center is not $(0, 0)$ then equations become

(1) (Lahore Board 2009)

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

(2) Eccentricity $e^2 = \frac{a^2+b^2}{a^2}$

$$(ae)^2 = a^2 + b^2$$

$$c^2 = a^2 + b^2$$

(3) Foci $(0, \pm ae)$

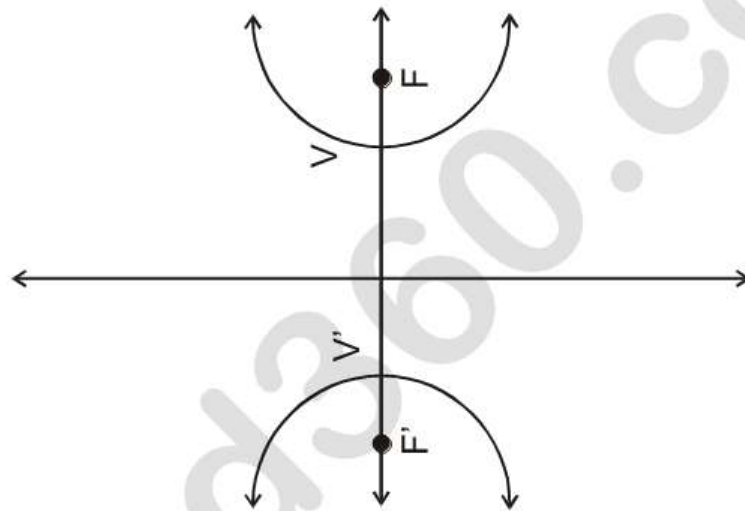
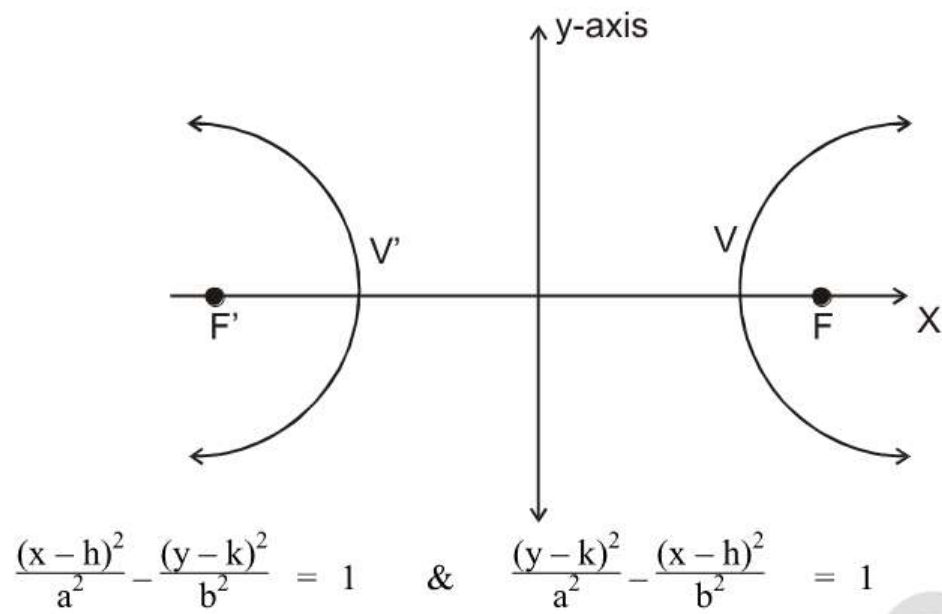
(4) Directrix $y = \pm \frac{a}{e}$

(5) Center $(0, 0)$

(6) Vertices $(0, \pm a)$

(7) Covertices $(\pm b, 0)$





EXERCISE 6.6

Q.1: Find an equation of the hyperbola with given data. Sketch the graph of each.

(i) Center $(0, 0)$ Focus $(6, 0)$, Vertex $(4, 0)$

Solution:

$$\text{Given } (ae, 0) = (6, 0) \quad , \quad (a, 0) = (4, 0)$$

$$(c, 0) = (6, 0) \quad a = 4$$

$$c = 6$$



For hyperbola

$$c^2 = a^2 + b^2$$

$$(6)^2 = (4)^2 + b^2$$

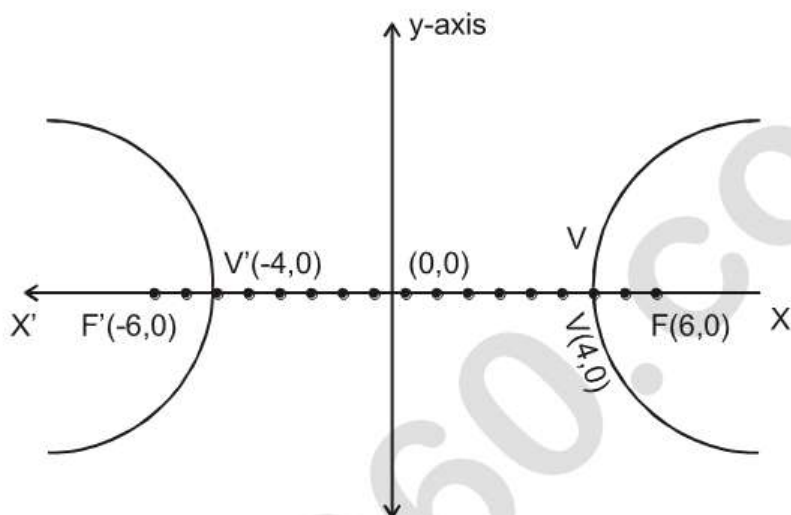
$$36 - 16 = b^2$$

$$20 = b^2$$

Required equation of hyperbola is

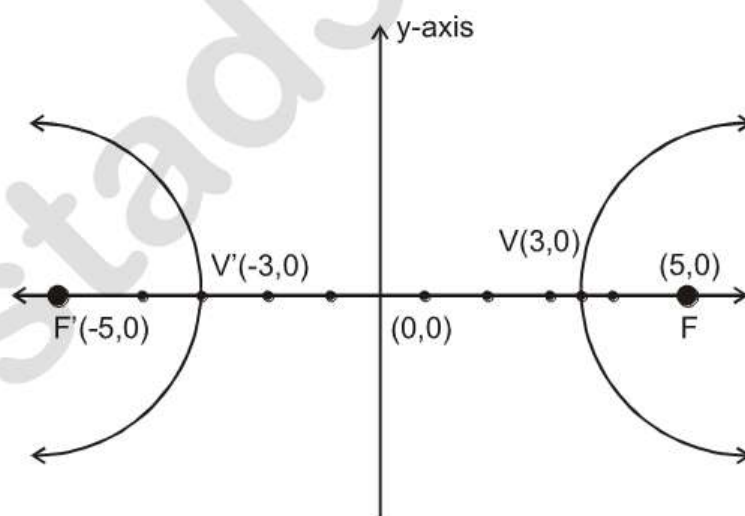
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{16} - \frac{y^2}{20} = 1$$



(ii) Foci $(\pm 5, 0)$, vertex $(3, 0)$

Solution:



Foci $(\pm 5, 0)$ Vertex $(3, 0)$

$$(\pm c, 0) = (\pm 5, 0) \quad (a, 0) = (3, 0)$$



$$\boxed{c = 5}$$

$$\boxed{a = 3}$$

For hyperbola

$$c^2 = a^2 + b^2$$

$$(5)^2 = (3)^2 + b^2$$

$$25 = 9 + b^2$$

$$16 = 9 + b^2$$

$$16 = b^2$$

Required equation of hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1 \quad \text{Ans.}$$

(iii) **Foci $(2 \pm 5\sqrt{2}, -7)$, Length of the transverse axis 10.**

Solution:

$$(\pm ae, 0) = (2 \pm 5\sqrt{2}, -7)$$

$$\text{Foci are } (2 + 5\sqrt{2}, -7) \text{ \& } F' (2 - 5\sqrt{2}, -7)$$

Also given $2a = 10$

$$a = 5$$

Center = mid point of foci

$$= \left(\frac{2 + 5\sqrt{2} + 2 - 5\sqrt{2}}{2}, \frac{-7 - 7}{2} \right)$$

$$\text{Center} = (2, -7)$$

We know that

$$c = \text{distance between center and focus}$$

$$= \sqrt{(2 - 2 - 5\sqrt{2})^2 + (-7 + 7)^2} = \sqrt{50}$$

For hyperbola

$$c^2 = a^2 + b^2$$

$$(\sqrt{50})^2 = (5)^2 + b^2$$

$$50 - 25 = b^2$$

$$25 = b^2$$

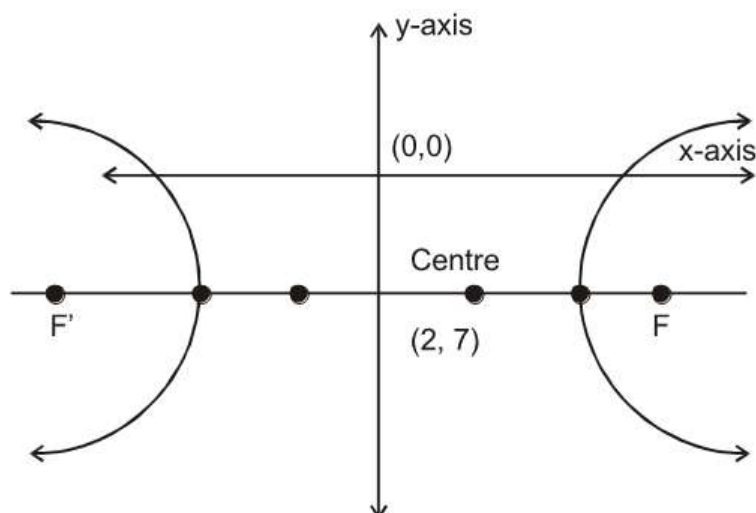
$$b^2 = 25$$

with center $(2, -7)$ equation of hyperbola is

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-2)^2}{25} - \frac{(y+7)^2}{25} = 1 \quad \text{Ans}$$





(iv) Foci $(0, \pm 6)$, $e = 2$ (Lahore Board 2011)

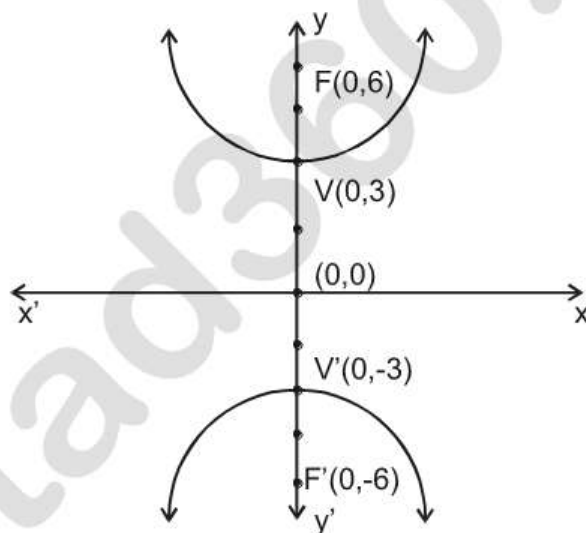
Solution:

$$\text{Foci } (0, \pm c) = (0, \pm 6)$$

$$c = 6, e = 2$$

$$ae = c$$

$$a = \frac{c}{e} = \frac{6}{2} = \boxed{3 = a}$$



For hyperbola

$$c^2 = a^2 + b^2$$

$$(6)^2 = (3)^2 + b^2$$

$$36 - 9 = b^2$$

$$b^2 = 27$$

Equation of hyperbola is



$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$\frac{y^2}{9} - \frac{x^2}{27} = 1$$

(v) **Foci (0, ± 9) directrix y = ± 4**

Solution:

$$\text{Foci } (0, \pm ae) = (0, \pm 9) \quad , \quad y = \pm \frac{a}{e}$$

$$ae = 9 \quad , \quad \pm 4 = \pm \frac{a}{e}$$

$$\Rightarrow c = 9 \quad , \quad e = \frac{a}{4}$$

$$\Rightarrow a \left(\frac{a}{4} \right) = 9$$

$$a^2 = 36$$

For hyperbola

$$c^2 = a^2 + b^2$$

$$(9)^2 = 36 + b^2$$

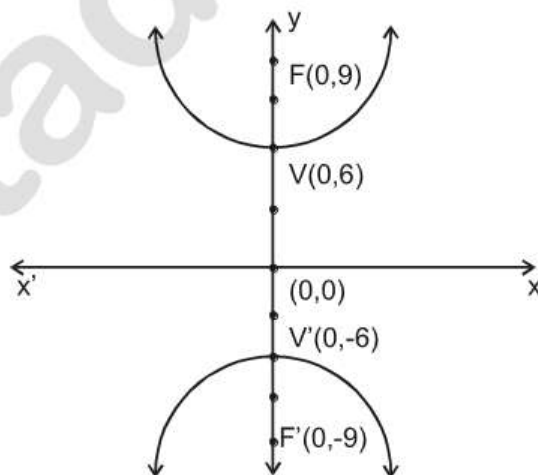
$$81 - 36 = b^2$$

$$b^2 = 45$$

Required equation of hyperbola is

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$\frac{y^2}{36} - \frac{x^2}{45} = 1$$



(vi) Centre (2, 2) horizontal transverse axis of length 6 & eccentricity $e = 2$.

Solution:

Centre (2, 2) & $2a = 6 \Rightarrow \boxed{a = 3}$ & $e = 2$

$$c = ae = 3(2) = 6$$

For hyperbola

$$c^2 = a^2 + b^2$$

$$(6)^2 = (3)^2 + b^2$$

$$36 - 9 = b^2$$

$$b^2 = 27$$

with centre (2, 2) required equation of hyperbola is

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-2)^2}{9} - \frac{(y-2)^2}{27} = 1$$

$$\text{foci are } (\pm ae, 0) = (\pm 6, 0)$$

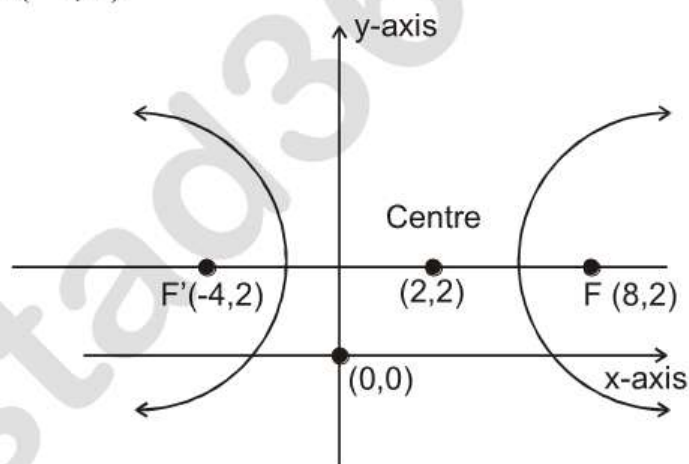
$$(x-2, y-2) = (\pm 6, 0)$$

$$x-2 = \pm 6 \quad y-2 = 0$$

$$x = \pm 6 + 2 \quad y = 2$$

$$x = 8, -4 \quad y = 2$$

$$\therefore F(8, 2) \text{ \& } (-4, 2)$$



(vii) Vertices (2, ± 3) and (0, 5) lies on the curve.

Solution:

Center of the hyperbola = mid point of vertices

$$= \left(\frac{2+2}{2}, \frac{3-3}{2} \right)$$

$$\text{Center} = (2, 0)$$



Since $V = (2, 3)$ & $V' = (2, -3)$

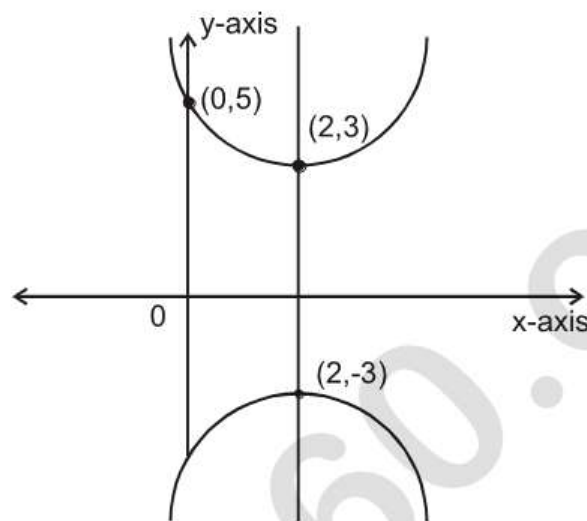
$$2a = |VV'| = \sqrt{(2-2)^2 + (3+3)^2} = 6$$

$$\boxed{a = 3}$$

With center $(2, 0)$ equation of hyperbola is

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$\frac{(y-0)^2}{9} - \frac{(x-2)^2}{b^2} = 1 \quad (1)$$



Since $(0, 5)$ lies on (1)

$$\frac{25}{9} - \frac{4}{b^2} = 1$$

$$\frac{25}{9} - 1 = \frac{4}{b^2}$$

$$\frac{16}{9} = \frac{4}{b^2}$$

$$b^2 = \frac{9}{4} \quad (1) \text{ becomes } \frac{y^2}{9} - \frac{(x-2)^2}{\frac{9}{4}} = 1 \quad \text{Ans.}$$

(viii) Foci $(5, -2)$ $(5, 4)$ & One vertex $(5, 3)$

Solution:

Center = mid point of foci

$$= \left(\frac{5+5}{2}, \frac{-2+4}{2} \right)$$

$$\text{Center} = (5, 1)$$



$$\begin{aligned}
 c &= \text{distance between center \& focus} \\
 &= \sqrt{(5-5)^2 + (1-4)^2} \\
 &= \sqrt{9} \\
 &= 3
 \end{aligned}$$

$$\boxed{c = 3}$$

$$\begin{aligned}
 a &= \text{distance between vertex \& center} \\
 &= \sqrt{(5-5)^2 + (3-1)^2} \\
 &= \sqrt{4} \\
 &= 2
 \end{aligned}$$

$$\boxed{a = 2}$$

For hyperbola

$$\begin{aligned}
 c^2 &= a^2 + b^2 \\
 (3)^2 &= (2)^2 + b^2 \\
 9 - 4 &= b^2
 \end{aligned}$$

$$\boxed{b^2 = 5}$$

with center (5, 1) required equation of hyperbola is

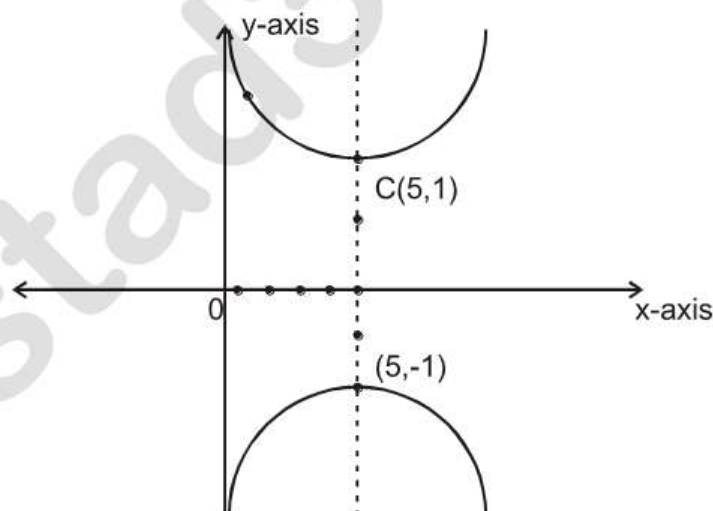
$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$\frac{(y-1)^2}{4} - \frac{(x-5)^2}{5} = 1$$

The second vertex is

$$(5, 1-2)$$

$$(5, -1)$$



Q.2: Find the center, foci, eccentricity, vertices and equations of directrices of each of the following. (i) $x^2 - y^2 = 9$

Solution:

$$\frac{x^2}{9} - \frac{y^2}{9} = 1$$

$$\Rightarrow a^2 = 9, \quad b^2 = 9$$

$$c^2 = a^2 + b^2$$

$$c^2 = 9 + 9 = 18$$

$$c^2 = 18 \Rightarrow c = 3\sqrt{2}$$

$$e^2 = \frac{a^2 + b^2}{a^2} = \frac{18}{9} = 2$$

$$e = \sqrt{2}$$

$$\text{foci} = (\pm c, 0) = (\pm 3\sqrt{2}, 0)$$

$$\text{Vertices} = (\pm a, 0) = (\pm 3, 0)$$

$$\text{Directrix } x = \pm \frac{a}{e}$$

$$\text{Directrix } x = \pm \frac{3}{\sqrt{2}}$$

Center is (0, 0)

(ii) $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Solution:

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$

$$a^2 = 4$$

$$b^2 = 9$$

$$\therefore c^2 = a^2 + b^2 = 4 + 9 = 13 \Rightarrow c = \sqrt{13}$$

$$\therefore ae = c \Rightarrow e = \frac{c}{a}$$

$$\text{eccentricity } e = \frac{\sqrt{13}}{2}$$

$$\text{foci} = (\pm c, 0) = (\pm \sqrt{13}, 0)$$

$$\text{Vertices} = (\pm a, 0) = (\pm 2, 0)$$

$$\text{Directrix } x = \pm \frac{a}{e}$$



$$x = \pm \frac{\frac{2}{\sqrt{13}}}{2}$$

$$x = \pm \frac{4}{\sqrt{13}}$$

Center is (0, 0)

$$(iii) \quad \frac{y^2}{16} - \frac{x^2}{9} = 1$$

Solution:

$$\frac{y^2}{16} - \frac{x^2}{9} = 1$$

$$a^2 = 16, \quad b^2 = 9$$

$$\therefore c^2 = a^2 + b^2$$

$$= 16 + 9$$

$$c^2 = 25$$

$$\text{eccentricity } ae = c \Rightarrow e = \frac{c}{a} \Rightarrow \frac{5}{4}$$

$$\boxed{e = \frac{5}{4}}$$

$$\text{foci} = (0, \pm c)$$

$$= (0, \pm 5)$$

$$\text{Vertices} = (0, \pm a) = (0, \pm 4)$$

$$\text{Directrix } y = \pm \frac{a}{e}$$

$$\text{Directrix } y = \pm \frac{4}{\frac{5}{4}} = \pm \frac{16}{5}$$

Center is (0, 0)

$$(iv) \quad \frac{y^2}{4} - x^2 = 1$$

Solution:

$$\frac{y^2}{4} - \frac{x^2}{1} = 1$$

Center = (0, 0)

$$a^2 = 4, \quad b^2 = 1$$



$$c^2 = a^2 + b^2$$

$$c^2 = 4 + 1$$

$$c^2 = 5$$

$$\text{eccentricity } ae = c \Rightarrow e = \frac{c}{a} = \boxed{\frac{\sqrt{5}}{2} = e}$$

$$\text{foci} = (0, \pm c) = (0, \pm \sqrt{5})$$

$$\text{Vertices} = (0, \pm a) = (0, \pm 2)$$

$$\text{Directrix } y = \pm \frac{a}{e}$$

$$\begin{aligned} \text{Directrix } y &= \pm \frac{2}{\frac{\sqrt{5}}{2}} \\ &= \pm \frac{4}{\sqrt{5}} \end{aligned}$$

$$(v) \quad \frac{(x-1)^2}{2} - \frac{(y-1)^2}{9} = 1$$

Solution:

$$\text{Let } x-1 = X, \quad y-1 = Y$$

$$\frac{X^2}{2} - \frac{Y^2}{9} = 1$$

$$a^2 = 2, \quad b^2 = 9 \quad c^2 = a^2 + b^2 = 2 + 9 = 11$$

$$ae = c, \quad e = \frac{c}{a} = \frac{\sqrt{11}}{\sqrt{2}}$$

$$\text{Eccentricity } e = \sqrt{\frac{11}{2}}$$

$$\text{Coordinates of foci } (\pm c, 0)$$

$$(X, Y) = (\pm \sqrt{11}, 0)$$

$$(x-1, y-1) = (\pm \sqrt{11}, 0)$$

$$x-1 = \pm \sqrt{11} \quad y-1 = 0$$

$$x = 1 \pm \sqrt{11} \quad y = 1$$

$$\text{foci } (1 \pm \sqrt{11}, 1)$$

$$\text{For center Put } X = 0, \quad Y = 0$$

$$x-1 = 0, \quad y-1 = 0$$

$$\Rightarrow x = 1, \quad y = 1$$

$$\boxed{\text{cenere} = (1, 1)}$$



$$\begin{aligned}
 \text{Vertices } (X, Y) &= (\pm\sqrt{2}, 0) \\
 (x-1, y-1) &= (\pm\sqrt{2}, 0) \\
 x-1 &= \pm\sqrt{2} = y-1 = 0 \\
 x &= 1 \pm \sqrt{2} \quad y = 1
 \end{aligned}$$

$$\text{Vertices are } \boxed{(1 + \sqrt{2}, 1), (1 - \sqrt{2}, 1)}$$

Equation of directrix

$$\begin{aligned}
 X &= \pm \frac{a}{e} \\
 x-1 &= \pm \frac{\frac{\sqrt{2}}{\sqrt{11}}}{\frac{\sqrt{2}}{\sqrt{11}}} \\
 x-1 &= \pm \frac{2}{\sqrt{11}} \\
 \boxed{x = 1 \pm \frac{2}{\sqrt{11}}}
 \end{aligned}$$

$$(vi) \quad \frac{(y+2)^2}{9} - \frac{(x-2)^2}{16} = 1$$

Solution:

$$\text{Let } y+2 = Y, \quad x-2 = X$$

Given equation becomes

$$\frac{Y^2}{9} - \frac{X^2}{16} = 1$$

$$a^2 = 9, \quad b^2 = 16 \quad c^2 = a^2 + b^2 = 9 + 16 = 25$$

$$\Rightarrow c = \pm 5$$

$$ae = c$$

$$\text{Eccentricity } \Rightarrow e = \frac{c}{a} = \frac{5}{3}$$

$$\text{Foci } (X, Y) = (0, \pm c)$$

$$(x-2, y+2) = (0, \pm 5)$$

$$x-2 = 0, \quad y+2 = \pm 5$$

$$x = 2, \quad y = 2 \pm 5$$

$$y = -7, 3$$

$$\text{Foci } (2, -7), (2, 3)$$



For the center put $X = 0$ $Y = 0$

$$\begin{aligned} x - 2 &= 0 & , & \quad y + 2 = 0 \\ x &= 2 & , & \quad y = -2 \end{aligned}$$

Center $(2, -2)$

$$\begin{aligned} \text{Vertices } (X, Y) &= (0, \pm a) \\ (x - 2, y + 2) &= (0, \pm 3) \\ x - 2 &= 0 & \quad y + 2 = \pm 3 \\ x &= 2 & \quad y = -2 \pm 3 \\ & & \quad y = -5, 1 \end{aligned}$$

Vertices

$$(2, -5) \text{ \& } (2, 1)$$

Equations of directrices are

$$\begin{aligned} Y &= \pm \frac{a}{e} \\ y + 2 &= \pm \frac{\frac{3}{5}}{\frac{3}{5}} \end{aligned}$$

$$y = -2 \pm \frac{9}{5} \quad \text{Ans.}$$

(vii) $9x^2 - 12x - y^2 - 2y + 2 = 0$

Solution:

$$9x^2 - 12x - y^2 - 2y = -2$$

$$9\left(x^2 - \frac{12}{9}x\right) - 1(y^2 + 2y) = -2$$

$$9\left(x^2 - \frac{4}{3}x + \frac{4}{9}\right) - 1(y^2 + 2y + 1) = -2 + 4 - 1$$

$$9\left(x - \frac{2}{3}\right)^2 - 1(y + 1)^2 = 1$$

$$\frac{\left(x - \frac{2}{3}\right)^2}{\frac{1}{9}} - \frac{(y + 1)^2}{1} = 1 \quad (1)$$

Let $x - \frac{2}{3} = X$, $y + 1 = Y$ Equation (1) becomes

$$\frac{X^2}{\frac{1}{9}} - \frac{Y^2}{1} = 1 \quad \text{which represents hyperbola}$$



$$a^2 = \frac{1}{9} \quad , \quad b^2 = 1$$

$$e^2 = \frac{a^2 + b^2}{a^2} = \frac{\frac{1}{9} + 1}{\frac{1}{9}} = \frac{\frac{10}{9}}{\frac{1}{9}} = 10$$

$$e = \sqrt{10} \quad c^2 = \frac{10}{9} \Rightarrow c = \frac{\sqrt{10}}{3}$$

Coordinates of foci (X, Y) = ($\pm C$, 0)

$$(x - \frac{2}{3}, y + 1) = (\pm \frac{\sqrt{10}}{3}, 0)$$

$$x - \frac{2}{3} = \pm \frac{\sqrt{10}}{3} \quad y + 1 = 0$$

$$x = \frac{2}{3} \pm \frac{\sqrt{10}}{3} \quad y = -1$$

$$x = \frac{2 \pm \sqrt{10}}{3} \quad , \quad y = -1$$

Foci ($\frac{2 \pm \sqrt{10}}{3}$, -1)

For center put X = 0, Y = 0

$$x - \frac{2}{3} = 0 \quad , \quad y + 1 = 0$$

$$x = \frac{2}{3} \quad y = -1 \quad \text{Center } (\frac{2}{3}, -1)$$

Vertices ($\pm a$, 0)

$$(X + Y) = (\pm \frac{1}{3}, 0)$$

$$(x - \frac{2}{3}, y + 1) = (\pm \frac{1}{3}, 0)$$

$$x - \frac{2}{3} = \pm \frac{1}{3} \quad , \quad y + 1 = 0$$

$$x = \frac{2}{3} \pm \frac{1}{3} \quad y = -1$$

$$\text{Vertices } (1, -1) \quad , \quad (\frac{1}{3}, -1)$$

Equation of directrices are

$$X = \pm \frac{a}{e}$$

$$x - \frac{2}{3} = \pm \frac{1}{3} \div \sqrt{10}$$

$$x = \frac{2}{3} \pm \frac{1}{3\sqrt{10}}$$



(viii) $4y^2 + 12y - x^2 + 4x + 1 = 0$

Solution:

$$4(y^2 + 3y) - 1(x^2 - 4x) = -1$$

$$4(y^2 + 3y + \frac{9}{4}) - 1(x^2 - 4x + 4) = -1 + 9 - 4$$

$$4(y + \frac{3}{2})^2 - 1(x - 2)^2 = 4$$

$$\frac{(y + \frac{3}{2})^2}{1} - \frac{(x - 2)^2}{4} = 1 \quad (1)$$

Let $y + \frac{3}{2} = Y$, $x - 2 = X$

Equation (1) becomes

$$\frac{Y^2}{1} - \frac{X^2}{4} = 1 \quad (\text{Represents a hyperbola})$$

$$a^2 = 1, \quad b^2 = 4$$

$$\therefore c^2 = a^2 + b^2 = 1 + 4 = 5 \Rightarrow c = \sqrt{5}$$

Eccentricity $ae = c \Rightarrow e = \frac{c}{a} = \frac{\sqrt{5}}{1} = \boxed{\sqrt{5}} = e$

Foci $(X, Y) = (0, \pm c)$

$$(x - 2, y + \frac{3}{2}) = (0, \pm \sqrt{5})$$

$$x - 2 = 0 \quad y + \frac{3}{2} = \pm \sqrt{5}$$

$$x = 2 \quad y = \frac{-3}{2} \pm \sqrt{5}$$

Foci $(2, \frac{-3}{2} \pm \sqrt{5})$

For center Put $X = 0, Y = 0$

$$x - 2 = 0, \quad y + \frac{3}{2} = 0$$

$$x = 2 \quad y = \frac{-3}{2}$$

Center $(2, \frac{-3}{2})$

Vertices $(X, Y) = (0, \pm a)$



$$(x-2, y+\frac{3}{2}) = (0, \pm 1)$$

$$x-2 = 0, \quad y + \frac{3}{2} = \pm 1$$

$$x = 2, \quad y = \frac{-3}{2} \pm 1$$

$$y = \frac{-1}{2}, \frac{-5}{2}$$

Vertices are $(2, \frac{-1}{2})$, $(2, \frac{-5}{2})$

Equation of directrices are

$$Y = \pm \frac{a}{e}$$

$$y + \frac{3}{2} = \pm \frac{1}{\sqrt{5}}$$

$$y = \frac{-3}{2} \pm \frac{1}{\sqrt{5}} \quad \text{Ans}$$

(ix) $x^2 - y^2 + 8x - 2y - 10 = 0$

Solution:

$$x^2 + 8x - y^2 - 2y = 10$$

$$(x^2 + 8x + 16) - (y^2 + 2y + 1) = 10 + 16 - 1$$

$$(x+4)^2 - (y+1)^2 = 25$$

$$\frac{(x+4)^2}{25} - \frac{(y+1)^2}{25} = 1 \quad \text{(i)}$$

Let $x+4 = X$, $y+1 = Y$

(i) Becomes $\frac{X^2}{25} - \frac{Y^2}{25} = 1$ (which represents hyperbola)

$$a^2 = 25, \quad b^2 = 25 \quad c^2 = a^2 + b^2 = 50 \Rightarrow c = \sqrt{50}$$

$$ae = c$$

Eccentricity $e = \frac{c}{a} = \frac{\sqrt{50}}{5} = \frac{5\sqrt{2}}{5} = \sqrt{2}$

$$\text{Foci} = (\pm c, 0)$$

$$(X, Y) = (\pm 5\sqrt{2}, 0)$$

$$(x+4, y+1) = (\pm 5\sqrt{2}, 0)$$

$$x+4 = \pm 5\sqrt{2}, \quad y+1 = 0$$

$$x = -4 \pm 5\sqrt{2} \quad y = -1 \quad \text{foci } (-4 \pm 5\sqrt{2}, -1)$$



For center put $X = 0, Y = 0$

$$x + 4 = 0, \quad y + 1 = 0 \Rightarrow x = -4, \quad y = -1$$

Center $(-4, -1)$

Vertices $(X, Y) = (\pm a, 0)$

$$(x + 4, y + 1) = (\pm 5, 0)$$

$$x + 4 = \pm 5 \quad y + 1 = 0$$

$$x = -4 \pm 5 \quad y = -1$$

$$= -9, 1$$

Vertices $(-9, -1) \quad \& \quad (1, -1)$

Equations of directrices

$$X = \pm \frac{a}{e}$$

$$x + 4 = \pm \frac{5}{\sqrt{2}}$$

$$x = -4 \pm \frac{5}{\sqrt{2}} \quad \text{Ans.}$$

$$(x) \quad 9x^2 - y^2 - 36x - 6y + 18 = 0$$

Solution:

$$9x^2 - 36x - y^2 - 6y = -18$$

$$9(x^2 - 4x) - 1(y^2 + 6y) = -18$$

$$9(x^2 - 4x + 4) - 1(y^2 + 6y + 9) = -18 + 36 - 9$$

$$9(x - 2)^2 - 1(y + 3)^2 = 9$$

$$\frac{(x - 2)^2}{1} - \frac{(y + 3)^2}{1} = 1 \quad (i)$$

Let $x - 2 = X, \quad y + 3 = Y$ Equation (i) becomes

$$\frac{X^2}{1} - \frac{Y^2}{9} = 1$$

$$a^2 = 1, \quad b^2 = 9$$

$$\therefore c^2 = a^2 + b^2 \Rightarrow c^2 = 1 + 9 = 10 \quad \boxed{c^2 = 10}$$

$$\therefore ae = c$$

$$\text{Eccentricity} \quad \therefore e = \frac{c}{a} = \frac{\sqrt{10}}{1} \quad \boxed{e = \sqrt{10}}$$

$$\text{Foci} \quad (\pm c, 0) = (\pm \sqrt{10}, 0)$$

$$(X, Y) = \pm \sqrt{10}, 0$$

$$(x - 2, y + 3) = (\pm \sqrt{10}, 0)$$



$$x - 2 = \pm\sqrt{10} \quad , \quad y + 3 = 0$$

$$x = 2 \pm \sqrt{10} \quad , \quad y = -3$$

Foci $(2 \pm \sqrt{10}, -3)$

For center Put $X = 0 \quad , \quad Y = 0$

$$x - 2 = 0 \quad y + 3 = 0$$

$$x = 2 \quad , \quad y = -3 \quad \boxed{\text{center } (2, -3)}$$

Vertices $(X, Y) = (\pm a, 0)$

$$(x - 2, y + 3) = (\pm 1, 0)$$

$$x - 2 = \pm 1 \quad y + 3 = 0$$

$$x = 2 \pm 1 \quad y = -3$$

$$x = 3, 1 \quad , \quad y = -3 \quad \boxed{\text{Vertices } (3, -3), (1, -3)}$$

Equations of directrices

$$X = \pm \frac{a}{e}$$

$$x - 2 = \pm \frac{1}{\sqrt{10}}$$

$$x = 2 \pm \frac{1}{\sqrt{10}} \quad \text{Ans}$$

Q.3: Let $0 < a < c$ and $F(-c, 0)$, $F'(c, 0)$ be two fixed points. Show that the set of points $P(x, y)$ such that $|PF| - |PF'| = \pm 2a$ is the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{c^2 - a^2} = 1$$

Solution:

Given $F(-c, 0)$, $F'(c, 0)$ and $p(x, y)$

$$\text{Also } |PF| - |PF'| = 2a$$

$$\sqrt{(x + c)^2 + (y - 0)^2} - \sqrt{(x - c)^2 + (y - 0)^2} = 2a$$

$$\sqrt{(x + c)^2 + y^2} = 2a + \sqrt{(x - c)^2 + y^2}$$

Squaring on both sides

$$(\sqrt{(x + c)^2 + y^2})^2 = (2a + \sqrt{(x - c)^2 + y^2})^2$$

$$x^2 + c^2 + 2cx + y^2 = 4a^2 + x^2 + c^2 - 2cx + y^2 + 4a\sqrt{(x - c)^2 + y^2}$$

$$4cx - 4a^2 = 4a^2\sqrt{x^2 + c^2 - 2cx + y^2}$$

$$4(cx - a^2) = 4a^2\sqrt{x^2 + c^2 - 2cx + y^2}$$

Again Squaring

$$c^2x^2 + a^4 - 2cxa^2 = a^2(x^2 + c^2 - 2cx + y^2)$$



$$c^2x^2 + a^4 - 2cxa^2 - a^2x^2 - a^2c^2 + 2a^2cx - a^2y^2 = 0$$

$$x^2(c^2 - a^2) - a^2y^2 = a^2(c^2 - a^2)$$

Dividing throughout by $c^2 - a^2$

$$\frac{x^2(c^2 - a^2)}{c^2 - a^2} - \frac{a^2y^2}{c^2 - a^2} = \frac{a^2(c^2 - a^2)}{c^2 - a^2}$$

$$x^2 - \frac{a^2y^2}{c^2 - a^2} = a^2$$

Dividing throughout by a^2 $\frac{x^2}{a^2} - \frac{y^2}{c^2 - a^2} = 1$ Hence proved.

Q.4: Find an equation of the hyperbola with foci $(-5, -5)$ & $(5, 5)$ vertices $(-3\sqrt{2}, -3\sqrt{2})$ and $(3\sqrt{2}, 3\sqrt{2})$.

Solution:

Since $2a =$ distance between two vertices

$$2a = \sqrt{(3\sqrt{2} + 3\sqrt{2})^2 + (3\sqrt{2} + 3\sqrt{2})^2}$$

$$2a = \sqrt{(6\sqrt{2})^2 + (6\sqrt{2})^2} = \sqrt{72 + 72} = \sqrt{144} = 12$$

$$2a = 12 \quad \boxed{a = 6}$$

Also given $F(-5, -5)$ & $F'(5, 5)$. Let $P(x, y)$ be any point on the hyperbola such that $|PF| - |PF'| = 2a$

$$\sqrt{(x+5)^2 + (y+5)^2} - \sqrt{(x-5)^2 + (y-5)^2} = 12$$

Squaring $\sqrt{(x+5)^2 + (y+5)^2} = 12 + \sqrt{(x-5)^2 + (y-5)^2}$

$$x^2 + 25 + 10x + y^2 + 25 + 10y = x^2 + 25 - 10x + y^2 + 25 - 10y + 144 + 24\sqrt{(x-5)^2 + (y-5)^2}$$

$$20x + 20y - 144 = 24\sqrt{(x-5)^2 + (y-5)^2}$$

$$4(5x + 5y - 36) = 4[6\sqrt{(x-5)^2 + (y-5)^2}]$$

Again squaring on both sides

$$(5x + 5y - 36)^2 = 36(x^2 + 25 - 10x + y^2 + 25 - 10y)$$

$$25x^2 + 5y^2 + 1296 + 50xy - 360y - 360x = 36x^2 + 900 - 360x + 36y^2 + 900 - 360y$$

$$25x^2 + 25y^2 + 1296 + 50xy - 360y - 360x - 36x^2 - 900 + 360x - 36y^2 - 900 + 360y = 0$$

$$11x^2 - 50xy - 11y^2 + 504 = 0 \quad \text{Ans}$$

Q.5: For any point on hyperbola the difference of its distances from the points $(2, 2)$ & $(10, 2)$ is 6. Find an equation of hyperbola.

Solution:

Given $2a = 6$ $F = (2, 2)$ & $F' = (10, 2)$

Let $P(x, y)$ be any point on the hyperbola, such that

$$|PF| - |PF'| = 2a$$



$$\begin{aligned}\sqrt{(x-2)^2 + (y-2)^2} - \sqrt{(x-10)^2 + (y-2)^2} &= 6 \\ \sqrt{(x-2)^2 + (y-2)^2} &= 6 + \sqrt{(x-10)^2 + (y-2)^2}\end{aligned}$$

Squaring

$$\begin{aligned}x^2 + 4 - 4x + y^2 + 4 - 4y &= 36 + x^2 + 100 - 20x + y^2 + 4 - 4y + 12 \\ \sqrt{(x-10)^2 + (y-2)^2} \\ 16x - 132 &= 12\sqrt{(x-10)^2 + (y-2)^2} \\ 4(4x - 33) &= 12\sqrt{(x-10)^2 + (y-2)^2} \\ 4x - 33 &= 3\sqrt{(x-10)^2 + (y-2)^2}\end{aligned}$$

Again squaring

$$\begin{aligned}16x^2 + 1089 - 264x &= 9(x^2 + 100 - 20x + y^2 + 4 - 4y) \\ 16x^2 + 1089 - 264x - 9x^2 + 900 + 180x - 9y^2 - 36 + 36y &= 0 \\ 7x^2 - 9y^2 - 84x + 36y + 153 &= 0 \quad \text{Ans}\end{aligned}$$

Q.6: Two listening posts hear the sound of an enemy gun. The difference in time is one second. If the listening posts are 1400 feet apart, write an equation of the hyperbola passing through the position of the enemy gun.

Solution:

Since the difference in time is one second, so let the listening posts F & F' near the second of the enemy gun after t, t-1 seconds respectively. If P is position of enemy then

$$|PF| - |PF'| = 2a \quad \dots\dots (1)$$

Since posts are 1400 feet apart, so

$$2c = 1400 \Rightarrow c = 700$$

since distance = (Velocity) (time)

So $|PF| = (1080)t$ & $|PF'| = (1080)(t-1)$

Putting these values in (1)

$$(1080)t - (1080)(t-1) = 2a$$

$$1080t - 1080t + 1080 = 2a$$

$$a = \frac{1080}{2} = 540$$

For hyperbola, we know that $c^2 = a^2 + b^2$

$$(700)^2 = (540)^2 + b^2$$

$$b^2 = 490000 - 291600$$

$$b^2 = 198400$$

Required equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\frac{x^2}{291600} - \frac{y^2}{198400} = 1 \quad \text{Ans}$$



Tangents to Conics

(I) Parabola

Equations of Tangent in different forms

(i) Point form:

The equation of the tangent to the parabola

$$y^2 = 4ax \text{ at the point } (x, y) \text{ is } yy_1 = 2a(x + x_1)$$

(ii) Slope form:

The equation of the tangent to the parabola

$$y^2 = 4ax \text{ in the term of slope 'm' is}$$

$$y = mx + \frac{a}{m} \quad (\because c = \frac{a}{m})$$

Note:

The equation of tangent at (x_1, y_1) can also be obtained by replacing x^2 by xx_1 , y^2 by yy_1 , x by $\frac{1}{2}(x + x_1)$, y by $\frac{1}{2}(y + y_1)$ and xy by $\frac{xy_1 + yx_1}{2}$

(II) Ellipse

Equations of tangents in different forms.

(i) Point form:

Equations of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at point (x_1, y_1) is

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

(ii) Slope form:

The equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in terms of slope m is

$$y = mx \pm \sqrt{a^2 m^2 + b^2} \quad (\because c = \pm \sqrt{a^2 m^2 + b^2})$$

(III) Hyperbola

Equations of tangent in different forms

(i) Point form:

The equation of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at point (x_1, y_1) is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

(ii) Slope form:

The equation of tangent to hyperbola in terms of 'm' is

$$y = mx \pm \sqrt{a^2 m^2 - b^2} \quad (c = \pm \sqrt{a^2 m^2 - b^2})$$



(IV) Circle

Equations of tangent in different forms

(i) Point form:Equation of tangent to the circle at (x_1, y_1) is $xx_1 + yy_1 = a^2$ **(ii) Slope form:**Equation of tangent in terms of slope 'm' is $y = mx \pm a\sqrt{1 + m^2}$

$$(\because c^2 = a^2 (1 + m^2))$$

Equations of Normal**(i) Parabola $y^2 = 4ax$ is at (x_1, y_1)**

$$y - y_1 = \frac{-y_1}{2a}(x - x_1)$$

(ii) Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (x_1, y_1) is

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$$

(iii) Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at (x_1, y_1) is

$$\frac{xa^2}{x_1} + \frac{yb^2}{y_1} = a^2 + b^2$$



EXERCISE 6.7

Q.1: Find equations of tangent and normal to each of the following at the indicated point.

(i) $y^2 = 4ax$ at $(at^2, 2at)$

Solution:

Equation of tangent at $(at^2, 2at)$ is

$$yy_1 = 2a(x + x_1)$$

$$y(2at) = 2a(x + at^2)$$

$$2ayt = 2ax + 2a^2t^2$$

$$2ayt = 2a(x + at^2)$$

$$yt = x + at^2$$

And equation of normal at $(at^2, 2at)$ is

$$y - y_1 = \frac{-y_1}{2a}(x - x_1)$$

$$y - 2at = \frac{-2at}{2a}(x - at^2)$$



$$y - 2at = -tx + at^3$$

$$tx + y - 2at - at^3 = 0$$

(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $(a \cos \theta, b \sin \theta)$

Solution:

Equation of tangent $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

$$\frac{x(a \cos \theta)}{a^2} + \frac{y(b \sin \theta)}{b^2} = 1$$

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

and equation of normal

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$$

$$\frac{a^2x}{a \cos \theta} - \frac{b^2y}{b \sin \theta} = a^2 - b^2$$

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2 \quad ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2$$

(iii) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $(a \sec \theta, b \tan \theta)$

Solution:

Equation of tangent

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

$$\frac{x a \sec \theta}{a^2} - \frac{y b \tan \theta}{b^2} = 1$$

$$\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$$

And equation of normal

$$\frac{xa^2}{x_1} + \frac{yb^2}{y_1} = a^2 + b^2$$

$$\frac{xa^2}{a \sec \theta} + \frac{yb^2}{b \tan \theta} = a^2 + b^2$$

$$\frac{xa}{\sec \theta} + \frac{yb}{\tan \theta} = a^2 + b^2$$

$$\text{OR} \quad x a \cos \theta + y b \cot \theta = a^2 + b^2$$



Q.2: Write equation of the tangent to the given conic at the indicated point

(i) $3x^2 = -16y$ at the points whose ordinate is -3

Solution:

$$3x^2 = -16y \quad \dots\dots (1)$$

Put $y = -3$ in

$$3x^2 = -16(-3)$$

$$3x^2 = 48 \quad \Rightarrow \quad x^2 = 16$$

$$\Rightarrow x = \pm 4$$

Hence points are

$(4, -3)$ & $(-4, -3)$

Now diff. (1) w.r.t 'x'

$$6x = -16 \frac{dy}{dx}$$

$$\frac{6x}{-16} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{-3}{8} x$$

$$m = \text{Slope} = \frac{dy}{dx} \Big|_{(4, -3)} = \frac{-3}{8} (4) = \frac{-3}{2}$$

$$\text{Also } m = \frac{dy}{dx} \Big|_{(-4, -3)} = \frac{-3}{8} (4) = \frac{3}{2}$$

$\therefore$ Equation of tangent at $(4, -3)$ is

$$y - y_1 = m(x - x_1)$$

$$y + 3 = \frac{-3}{2} (x - 4)$$

$$2y + 6 = -3x + 12$$

$$3x + 2y = 6$$

$$3x + 2y - 6 = 0$$

Equation of tangent at $(-4, -3)$ is

$$y - y_1 = m(x - x_1)$$

$$y + 3 = \frac{+3}{2} (x + 4)$$

$$2y + 6 = 3x + 12$$

$$3x - 2y = -6$$

$$3x - 2y + 6 = 0$$

(ii) $3x^2 - 7y^2 = 20$ at points where $y = -1$.

Solution:

$$3x^2 - 7y^2 = 20 \quad \dots\dots (1)$$

Put $y = -1$ in (1)

$$3x^2 - 7(-1)^2 = 20$$

$$3x^2 = 20 + 7$$

$$3x^2 = 27 \quad \Rightarrow \quad x^2 = 9 \quad \Rightarrow \quad x = \pm 3$$

Thus the required points on the conic are $(3, -1)$ & $(-3, -1)$



Now diff (1) w.r.t. 'x' we have

$$6x - 14y \frac{dy}{dx} = 0$$

$$14 \frac{dy}{dx} = 6x$$

$$\frac{dy}{dx} = \frac{6x}{14y} = \frac{3x}{7y}$$

Now $m = \text{Slope} = \left. \frac{dy}{dx} \right|_{(3, -1)} = \frac{9}{-7}$ Also $m = \left. \frac{dy}{dx} \right|_{(-3, -1)} = \frac{9}{7}$

Therefore equation of tangent at (3, -1) is

$$y - y_1 = m(x - x_1)$$

$$y + 1 = \frac{-9}{7} (x - 3)$$

$$7y + 7 = -9x + 27$$

$$9x + 7y = 20$$

$$9x + 7y - 20 = 0 \quad \text{Ans}$$

Equation of tangent at (-3, -1)

$$y - y_1 = m(x - x_1)$$

$$y + 1 = \frac{9}{7} (x + 3)$$

$$7y + 7 = 9x + 27$$

$$9x - 7y = -20$$

$$9x - 7y + 20 = 0 \quad \text{Ans}$$

(iii) $3x^2 - 7y^2 + 2x - y - 48 = 0$, at point where $x = 4$

Solution:

$$3x^2 - 7y^2 + 2x - y - 48 = 0 \quad \dots (1)$$

Put $x = 4$ in (1)

$$3(4)^2 - 7y^2 + 2(4) - y - 48 = 0$$

$$48 - 7y^2 + 8 - y - 48 = 0$$

$$\Rightarrow -7y^2 - y + 8 = 0 \quad \Rightarrow 7y^2 + y - 8 = 0$$

$$7y^2 + 8y - 7y - 8 = 0$$

$$y(7y + 8) - 1(7y + 8) = 0$$

$$(7y + 8)(y - 1) = 0$$

Either

$$7y + 8 = 0, \quad y - 1 = 0$$

$$y = \frac{-8}{7}, \quad y = 1$$

Therefore, required points on the conic are $(4, \frac{-8}{7})$ & $(4, 1)$

Now diff. (1) w.r.t. 'x' $6x - 14y \frac{dy}{dx} + 2 - \frac{dy}{dx} = 0$

$$(-14y - 1) \frac{dy}{dx} = -6x - 2$$



$$\frac{dy}{dx} = \frac{6x+2}{14y+1}$$

$$m = \frac{dy}{dx} \Big|_{(4,1)} = \frac{6(4)+2}{14(1)+1} = \frac{26}{15} \quad \text{Also } m = \frac{dy}{dx} \Big|_{(4, -\frac{8}{7})} = \frac{6(4)+2}{14(-\frac{8}{7})+1} = \frac{26}{-15}$$

Equation of tangent at (4, 1) is

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{26}{15}(x - 4)$$

$$15y - 15 = 26x - 104$$

$$26x - 15y - 89 = 0 \quad \text{Ans}$$

Equation of tangent at $(4, -\frac{8}{7})$ is

$$y - y_1 = m(x - x_1)$$

$$y + \frac{8}{7} = \frac{-26}{15}(x - 4)$$

$$105y - 120 = -182x + 728$$

$$182x + 105y - 608 = 0 \quad \text{Ans}$$

Q.3: Find equations of the tangents to each of the following through the given point

(i) $x^2 + y^2 = 25$, through (7, -1)

Solution:

$$x^2 + y^2 = 25 \Rightarrow r = 5$$

We know that condition of tangency for the circle is

$$c^2 = r^2(1 + m^2)$$

$$c^2 = 25(1 + m^2)$$

$$\Rightarrow c = \pm 5\sqrt{1 + m^2}$$

Let the required equation of tangent be

$$y = mx + c \quad \dots (1) \quad \text{Putting value of } C \text{ in } (1)$$

$$y = mx \pm 5\sqrt{1 + m^2} \quad \dots (2)$$

Since tangent line passes through point (7, -1), therefore

$$-1 = 7m \pm 5\sqrt{1 + m^2}$$

$$\pm 5\sqrt{1 + m^2} = 7m + 1 \quad \text{Squaring}$$

$$25(1 + m^2) = (7m + 1)^2$$

$$25 + 25m^2 = 49m^2 + 1 + 14m$$

$$-24m^2 - 14m + 24 = 0$$

$$12m^2 + 7m - 12 = 0$$

$$12m^2 + 16m - 9m - 12 = 0$$

$$4m(3m + 4) - 3(3m + 4) = 0$$

$$(3m + 4)(4m - 3) = 0$$

$$m = \frac{-4}{3} \quad m = \frac{3}{4}$$



with $m = \frac{-4}{3}$ (2) becomes

$$\begin{aligned} y &= -\frac{4}{3}x \pm 5\sqrt{1 + \frac{16}{9}} \\ &= -\frac{4}{3}x \pm 5\frac{5}{3} \end{aligned}$$

$$3y = -4x \pm 25$$

$$4x + 3y \pm 25 = 0$$

(ii) $y^2 = 12x$ through (1, 4)

Solution:

$$y^2 = 12x$$

As standard form is

$$y^2 = 4ax$$

$$4a = 12 \Rightarrow a = 3$$

Let $y = mx + c$ (1) be the required equation of tangent. For Parabola we know that condition of tangency is $c = \frac{a}{m} = \frac{3}{m}$ put in (1)

$$y = mx + \frac{3}{m} \text{ (2)}$$

Since tangent line passes through point (1, 4)

So (2) becomes

$$4 = m + \frac{3}{m} \Rightarrow 4m = m^2 + 3$$

$$m^2 - 4m + 3 = 0$$

$$(m - 1)(m - 3) = 0$$

$$m = 1, \quad m = 3 \quad \text{Put in (1)}$$

$$y = x + 3 \quad \& \quad y = 3x + \frac{3}{m}$$

$$x - y + 3 = 0 \quad y = 3x + \frac{3}{3}$$

$$y = 3x + 1$$

$$3x - y + 1 = 0 \quad \text{Ans}$$

(iii) $x^2 - 2y^2 = 2$ through (1, -2)

Solution:

$$x^2 - 2y^2 = 2$$

$$\frac{x^2}{2} - \frac{y^2}{1} = 1$$

$$\Rightarrow a^2 = 2, \quad b^2 = 1$$

with $m = \frac{3}{4}$ (2) becomes

$$\begin{aligned} y &= \frac{3x}{4} \pm 5\sqrt{1 + \frac{9}{16}} \\ &= \frac{3x}{4} \pm \frac{25}{4} \end{aligned}$$

$$4y = 3x \pm 25$$

$$3x - 4y \pm 25 = 0$$



For hyperbola, we know that condition of tangent is

$$c^2 = a^2 m^2 - b^2$$

$$\Rightarrow c^2 = 2m^2 - 1 \quad \Rightarrow c = \pm \sqrt{2m^2 - 1}$$

Let $y = mx + c$ be tangent to the given hyperbola then $y = mx \pm \sqrt{2m^2 - 1}$ (1)

Since (1) passes through $(1, -2)$ (1) becomes

$$-2 = m \pm \sqrt{2m^2 - 1}$$

$$-2 - m = \pm \sqrt{2m^2 - 1} \quad \text{Squaring}$$

$$4 + m^2 + 4m = 2m^2 - 1$$

$$2m^2 - 1 - m^2 - 4m - 4 = 0$$

$$m^2 - 4m - 5 = 0$$

$$\Rightarrow (m - 5)(m + 1) = 0$$

$$\Rightarrow m = 5, \quad m = -1$$

Putting values of m in (1) we get

$$y = 5x \pm \sqrt{2(25) - 1},$$

$$y = -x \pm \sqrt{2 - 1}$$

$$y = 5x \pm \sqrt{49},$$

$$= -x \pm 1$$

$$y = 5x \pm 7,$$

$$y + x \pm 1 = 0$$

$$5x - y \pm 7 = 0 \quad \text{Ans}$$

Q.4: Find equations of normal to the Parabola $y^2 = 8x$, which are parallel to the line $2x + 3y = 10$.

Solution:

$$y^2 = 8x \quad \dots\dots (1)$$

$$2x + 3y = 10 \quad \dots\dots (2)$$

Diff. (1) w.r.t. 'x'

$$2y \frac{dy}{dx} = 8$$

$$\frac{dy}{dx} = \frac{8}{2y} = \frac{4}{y}$$

$$m_1 = \frac{dy}{dx} = \frac{4}{y}$$

$$m_1 = \text{Slope of normal} = \frac{-y}{4}$$

$m_2 = \text{Slope of line}$

$$= \frac{-\text{coeff of } x}{\text{coeff of } y}$$

$$= -\frac{2}{3}$$

Since normal and given line are Parallel

$$m_1 = m_2$$

$$\frac{-y}{4} = \frac{-2}{3} \quad \Rightarrow y = \frac{8}{3}$$

Put in (1)

$$\left(\frac{8}{3}\right)^2 = 8x$$



$$\frac{64}{9 \times 8} = x \quad \Rightarrow \quad x = \frac{8}{9}$$

Required point $(\frac{8}{9}, \frac{8}{3})$

with $y = \frac{8}{3}$, m_1 become

$$m_1 = -\frac{8}{3} \times \frac{1}{4} = -\frac{2}{3}$$

Required equation of normal at $(\frac{8}{9}, \frac{8}{3})$ is

$$y - y_1 = m(x - x_1)$$

$$y - \frac{8}{3} = -\frac{2}{3}(x - \frac{8}{9})$$

$$3y - 8 = -2(\frac{9x - 8}{9})$$

$$27y - 72 = -18x + 16$$

$$18x + 27y - 88 = 0$$

Q.5: Find equations of tangents to the ellipse $\frac{x^2}{4} + y^2 = 1$, which are parallel to the line $2x - 4y + 5 = 0$.

Solution:

$$\frac{x^2}{4} + \frac{y^2}{1} = 1$$

$$2x - 4y + 5 = 0$$

$$\Rightarrow a^2 = 4, \quad b^2 = 1$$

$$m = \frac{-\text{coeff of } x}{\text{coeff of } y} = \frac{-2}{-4} = \frac{1}{2}$$

We know that condition of tangent for ellipse is

$$c^2 = a^2 m^2 + b^2$$

$$c^2 = 4m^2 + 1$$

$$c = \pm \sqrt{4m^2 + 1}$$

Since tangent is parallel to line $2x - 4y + 5 = 0$

$$\therefore \text{Slope is also } m = \frac{1}{2}$$

$$c = \pm \sqrt{4 \cdot \frac{1}{4} + 1} = \pm \sqrt{2}$$

Let the equation of required tangent by

$$y = mx + c$$



$$y = \frac{1}{2}x \pm \sqrt{2}$$

$$2y = x \pm 2\sqrt{2}$$

$$x - 2y \pm 2\sqrt{2} = 0 \quad \text{Ans}$$

Q.6: Find equations of the tangents to the conics $9x^2 - 4y^2 = 36$ Parallel to $5x - 2y + 7 = 0$.

Solution:

$$9x^2 - 4y^2 = 36$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1 \quad (\text{Dividing by 36})$$

$$\Rightarrow a^2 = 4, \quad b^2 = 9$$

$$5x - 2y + 7 = 0$$

$$m = \text{slope of line} = \frac{5}{2}$$

For hyperbola, we know that

$$c^2 = a^2m^2 - b^2$$

$$c^2 = 4m^2 - 9$$

Since tangent and given line are parallel so their slopes are same. Thus $m = \frac{5}{2}$

$$c^2 = 4\left(\frac{25}{4}\right) - 9 \quad c^2 = 16 \quad \Rightarrow c = \pm 4$$

Let $y = mx + c$ be the required equation of the tangent then $y = \frac{5}{2}x \pm 4$

$$2y = 5x \pm 8$$

$$5x - 2y \pm 8 = 0 \quad \text{Ans.}$$

Q.7: Find equations of common tangents to the given conics.

(i) $x^2 = 80y$ & $x^2 + y^2 = 81$

Solution:

$$x^2 = 80y \quad \dots (1) \quad x^2 + y^2 = 81 \quad \dots (2)$$

Let $y = mx + c$ (3) be the required common tangent. Let a be radius of circle then (2) becomes $a^2 = 81$ Put in (1)

$$x^2 = 80(mx + c)$$

$$x^2 - 80mx - 80c = 0$$

For equal roots, we know that $\text{Disc} = 0$

$$b^2 - 4ac = 0$$

$$(-80m)^2 - 4(1)(-80c) = 0$$

$$80(80m^2 + 4c) = 0$$



$$80m^2 + 4c = 0 \quad c = -20m^2$$

Condition of tangency for circle is $c^2 = a^2(1 + m^2)$ (4)

$$(-20m^2)^2 = 81(1 + m^2)$$

$$400m^4 = 81 + 81m^2$$

$$400m^4 - 81m^2 - 81 = 0$$

By Quadratic Formula

$$m^2 = \frac{-(-81) \pm \sqrt{(-81)^2 - 4(400)(-81)}}{2(400)}$$

$$= \frac{81 \pm \sqrt{136161}}{800} = \frac{9}{16}$$

$$m = \pm \frac{3}{4}$$

$$\therefore c = -20\left(\frac{9}{16}\right) = -\frac{45}{4}$$

Putting values of m & c in $y = mx + c$

$$y = \pm \frac{3}{4}x - \frac{45}{4}$$

$$4y = \pm 3x - 45$$

$$\pm 3x - 4y - 45 = 0 \quad \text{Ans.}$$

(ii) $y^2 = 16x$ & $x^2 = 2y$

Solution:

$$y^2 = 16x \quad \dots (1) \quad x^2 = 2y \quad \dots (2)$$

$$y^2 = 4ax$$

$$4a = 16$$

$$\boxed{a = 4}$$

We know that condition of tangency for Parabola is $c = \frac{a}{m}$

$$c = \frac{4}{m}$$

Let $y = mx + c \dots\dots (3)$ be required tangent

then $y = mx + \frac{4}{m}$ Putting value of y in (2)

$$x^2 = 2\left(mx + \frac{4}{m}\right) \Rightarrow mx^2 = 2m^2x + 8$$

$$mx^2 - 2m^2x - 8 = 0$$

For equal roots, we know that $\text{Disc} = 0$



$$\text{i.e; } b^2 - 4ac = 0$$

$$(-2m^2)^2 - 4(m)(-8) = 0$$

$$4m^4 + 32m = 0$$

$$4m(m^3 + 8) = 0$$

$$m = 0, \quad m^3 = -8, \quad m = -2$$

Equation of tangent is

$$y = mx + c$$

$$y = -2x + \frac{4}{-2}$$

$$y = -2x - 2$$

$$\boxed{2x + y + 2 = 0}$$

Ans.

Q.8: Find the points of intersection of the given conics.

$$(i) \quad \frac{x^2}{18} + \frac{y^2}{8} = 1 \quad \& \quad \frac{x^2}{3} - \frac{y^2}{3} = 1$$

Solution:

$$\frac{x^2}{18} + \frac{y^2}{8} = 1 \quad \& \quad \frac{x^2}{3} - \frac{y^2}{3} = 1$$

$$8x^2 + 18y^2 = 144 \quad x^2 - y^2 = 3 \quad \dots\dots (2)$$

$$4x^2 + 9y^2 = 72 \quad \dots (1) \quad (\text{Dividing by 2})$$

Multiplying Eq. (2) by 9 & add in (1)

$$9x^2 - 9y^2 = 27$$

$$4x^2 + 9y^2 = 72$$

$$13x^2 = 99$$

$$x^2 = \frac{99}{13} \quad \Rightarrow \quad x = \pm \sqrt{\frac{99}{13}}$$

Put in (2)

$$\frac{99}{13} - y^2 = 3$$

$$\frac{99}{13} - 3 = y^2$$

$$\frac{99 - 39}{13} = y^2$$

$$y^2 = \frac{60}{13} \quad \Rightarrow \quad y = \pm \sqrt{\frac{60}{13}}$$



Points of intersection are $\left(\pm \sqrt{\frac{99}{13}}, \pm \sqrt{\frac{60}{13}} \right)$ Ans.

(ii) $x^2 + y^2 = 8$ & $x^2 - y^2 = 1$

Solution:

$$x^2 + y^2 = 8 \quad \dots (1) \quad x^2 - y^2 = 1 \quad \dots (2)$$

Adding (1) & (2)

$$x^2 + y^2 = 8$$

$$x^2 - y^2 = 1$$

$$2x^2 = 9 \Rightarrow x^2 = \frac{9}{2} \Rightarrow x = \pm \frac{3}{\sqrt{2}}$$

Put in (1) $\frac{9}{2} + y^2 = 8$

$$y^2 = 8 - \frac{9}{2}$$

$$y^2 = \frac{16 - 9}{2} = \frac{7}{2}$$

$$y = \pm \sqrt{\frac{7}{2}}$$

Hence points of intersection are $\left(\pm \frac{3}{\sqrt{2}}, \pm \sqrt{\frac{7}{2}} \right)$ Ans

(iii) $3x^2 - 4y^2 = 12$ & $3y^2 - 2x^2 = 7$

Solution:

$$3x^2 - 4y^2 = 12 \quad \dots (1)$$

$$3y^2 - 2x^2 = 7 \quad \dots (2)$$

Multiplying equation (1) by (2) & (2) by 3 and adding

$$6x^2 - 8y^2 = 24$$

$$-6x^2 + 9y^2 = 21$$

$$y^2 = 45 \Rightarrow y = \pm \sqrt{45}$$

Put in (2)

$$-2x^2 + 3(45) = 7$$

$$-2x^2 + 135 = 7$$

$$135 - 7 = 2x^2$$

$$128 = 2x^2$$

$$x^2 = 64 \Rightarrow x = \pm 8$$

Hence points of intersection are

$$(\pm 8, \pm \sqrt{45}) \text{ Ans.}$$



(iv) $3x^2 + 5y^2 = 60$ and $9x^2 + y^2 = 124$

Solution:

$$3x^2 + 5y^2 = 60 \quad \dots\dots (1) \quad \quad 9x^2 + y^2 = 124 \quad \dots\dots (2)$$

Multiplying (1) by (3) & Subtracting from (2)

$$\begin{array}{r} 9x^2 + y^2 = 124 \\ -9x^2 + 15y^2 = -180 \\ \hline -14y^2 = -56 \\ y^2 = 4 \end{array}$$

$$\Rightarrow y = \pm 2$$

Put in (1)

$$9x^2 + 4 = 124$$

$$9x^2 = 120$$

$$x^2 = \frac{120}{9} = \frac{40}{3} \quad x = \pm \sqrt{\frac{40}{3}}$$

Hence points of intersection are $\left(\pm \sqrt{\frac{40}{3}} \pm 2 \right)$



EXERCISE 6.8

Q.1: Find an equation of each of the following with respect to new parallel axes obtained by shifting the origin to the indicated point.

Remember



Solution:

(i) $x^2 + 16y - 16 = 0$ (1) $O'(0, 1) \Rightarrow h = 0, k = 1$

We know that equations of transformation are

$$x = X + h, \quad y = Y + k$$

$$x = X + 0, \quad y = Y + 1 \quad \text{Put in (1)}$$

$$X^2 + 16(Y + 1) - 16 = 0$$

$$X^2 + 16Y + 16 - 16 = 0$$

$$X^2 + 16Y = 0 \quad \text{Ans}$$

(ii) $4x^2 + y^2 + 16x - 10y + 37 = 0$ $O'(-2, 5)$

Solution:

$$4x^2 + y^2 + 16x - 10y + 37 = 0 \quad \text{..... (i), } O'(-2, 5) \Rightarrow h = -2, k = 5$$

We know that equations of transformation are



$$x = x + h, \quad y = Y + k$$

$$x = x - 2, \quad y = Y + 5$$

Put in (1)

$$4(X-2)^2 + (Y+5)^2 + 16(X-2) - 10(Y+5) + 37 = 0$$

$$4(X^2 + 4 - 4X) + Y^2 + 25 + 10Y + 16X - 32 - 10Y - 50 + 37 = 0$$

$$4X^2 + 16 - 16X + Y^2 + 16X - 20 = 0$$

$$4X^2 + Y^2 - 4 = 0 \quad \text{Ans}$$

$$(iii) \quad 9x^2 + 4y^2 + 18x - 16y - 11 = 0 \quad O'(-1, 2)$$

Solution:

$$9x^2 + 4y^2 + 18x - 16y - 11 = 0 \quad \dots\dots\dots (1), \quad O'(-1, 2) \Rightarrow h = -1, k = 2$$

We know that equations of transformation are

$$x = X + h, \quad y = Y + k$$

$$x = X - 1, \quad y = Y + 2$$

Put in (1)

$$9(X-1)^2 + 4(Y+2)^2 + 18(X-1) - 16(Y+2) - 11 = 0$$

$$9(X^2 + 1 - 2x) + 4(Y^2 + 4 + 4Y) + 18X - 18 - 16Y - 32 - 11 = 0$$

$$9X^2 + 9 - 18X + 4Y^2 + 16 + 16Y + 18X - 18 - 16Y - 32 - 11 = 0$$

$$9X^2 + 4Y^2 - 36 = 0 \quad \text{Ans}$$

$$(iv) \quad x^2 - y^2 + 4x + 8y - 11 = 0 \quad O'(-2, 4)$$

Solution:

$$x^2 - y^2 + 4x + 8y - 11 = 0 \quad \dots (1), \quad O'(-2, 4) \Rightarrow h = -2, k = 4$$

We know that equations of transformations of transformation

$$x = X + h, \quad y = Y + k$$

$$x = X - 2, \quad y = Y + 4$$

Put in (1)

$$(X-2)^2 - (Y+4)^2 + 4(X-2) + 8(Y+4) - 11 = 0$$

$$X^2 + 4 - 4X - Y^2 - 16 - 8Y + 4X - 8 + 8Y + 32 - 11 = 0$$

$$X^2 - Y^2 + 1 = 0 \quad \text{Ans.}$$

$$(v) \quad 9x^2 - 4y^2 + 36x + 8y - 4 = 0 \quad O'(-2, 1)$$

Solution:

$$9x^2 - 4y^2 + 36x + 8y - 4 = 0 \quad \dots (1) \quad O'(-2, 1) \Rightarrow h = -2, k = 1$$

We know that equations of transformation are

$$x = X + h, \quad y = Y + k$$

$$x = X - 2, \quad y = Y + 1$$

Put in (1)

$$9(X-2)^2 - 4(Y+1)^2 + 36(X-2) + 8(Y+1) - 4 = 0$$



$$\begin{aligned}
 9(X^2 + 4 - 4X) - 4(Y^2 + 1 + 2Y) + 36X - 72 + 8Y + 8 - 4 &= 0 \\
 9X^2 + 36 - 36X - 4Y^2 - 4 - 8Y + 36x - 72 + 8Y + 4 &= 0 \\
 9X^2 - 4Y^2 - 36 &= 0 \quad \text{Ans.}
 \end{aligned}$$

Q.2: Find coordinates of the new origin so that first-degree terms are removed from the transformed equation of each of the following. Also find the transformed equation.

(i) $3x^2 - 2y^2 + 24x + 12y + 24 = 0$

Solution:

$$3x^2 - 2y^2 + 24x + 12y + 24 = 0 \quad \dots (1)$$

Let the coordinates of the new origin be (h, k)

Equations of transformed axes are $x = X + h$, $y = Y + K$ Put in (i)

$$3(X + h)^2 - 2(Y + K)^2 + 24(X + h) + 12(Y + k) + 24 = 0$$

$$3(X^2 + h^2 + 2Xh) - 2(Y^2 + k^2 + 2Yk) + 24X + 24h + 12Y + 12k + 24 = 0$$

$$3X^2 + 3h^2 + 6Xh - 2Y^2 - 2k^2 - 4Yk + 24X + 24h + 12Y + 12k + 24 = 0$$

$$3X^2 - 2Y^2 + X(6h + 24) - Y(4k - 12) + 3h^2 - 2k^2 + 24h + 12k + 24 = 0 \quad \dots (2)$$

Now, we remove first-degree terms from the transformed equation

$$6h + 24 = 0, \quad 4k - 12 = 0$$

$$\Rightarrow h = -4, \quad k = 3$$

Thus new origin $O' (h, k) = O' (-4, 3)$

Put $h = -4$, $k = 3$ in (2)

$$3X^2 - 2Y^2 + 0 - 0 + 3(16) - 2(9) + 24(-4) + 12(3) + 24 = 0$$

$$3X^2 - 2Y^2 + 48 - 18 - 96 + 36 + 24 = 0$$

$$3X^2 - 2Y^2 - 6 = 0$$

(ii) $25x^2 + 9y^2 + 50x - 36y - 164 = 0$

Solution:

$$25x^2 + 9y^2 + 50x - 36y - 164 = 0 \quad \dots (1)$$

Let the coordinates of the new origin be (h, k) equation of transformed axes are $x = X + h$, $y = Y + k$

Put in (1)

$$25(X + h)^2 + 9(Y + k)^2 + 50(X + h) - 36(Y + k) - 164 = 0$$

$$25(X^2 + h^2 + 2Xh) + 9(Y^2 + k^2 + 2Yk) + 50X + 50h - 36Y - 36k - 164 = 0$$

$$25X^2 + 25h^2 + 50Xh + 9Y^2 + 9k^2 + 18Yk + 50X + 50h - 36Y - 36k - 164 = 0$$

$$25X^2 + 9Y^2 + (50h + 50)X + (18k - 36)Y + 25h^2$$

$$+ 9k^2 + 50h - 36k - 164 = 0 \quad \dots (2)$$

Now we remove first - degree terms, from the transformed equation

$$50h + 50 = 0$$

$$18k - 36 = 0$$



$$50h = -50$$

$$\boxed{h = -1}$$

$$18k = 36$$

$$\boxed{k = 2}$$

New origin is $O'(h, k) = O'(-1, 2)$

Put $h = -1$, $K = 2$ in (2)

$$25X^2 + 9Y^2 + 0 + 0 + 25(1) + 9(4) + 50(-1) - 36(2) - 164 = 0$$

$$25X^2 + 9Y^2 + 25 + 36 - 50 - 72 - 164 = 0$$

$$25X^2 + 9Y^2 - 225 = 0$$

(iii) $x^2 - y^2 - 6x + 2y + 7 = 0$

Solution:

$$x^2 - y^2 - 6x + 2y + 7 = 0 \quad (1)$$

Let the coordinates of the new origin O' be (h, k) . The equations of transformation are $x = X + h$, $y = Y + k$.

Put in (1)

$$(X + h)^2 - (Y + k)^2 - 6(X + h) + 2(Y + k) + 7 = 0$$

$$X^2 + h^2 + 2Xh - Y^2 - k^2 - 2Yk - 6X - 6h + 2Y + 2k + 7 = 0$$

$$X^2 - Y^2 + X(2h - 6) - Y(2k - 2) + h^2 - k^2 - 6h + 2k + 7 = 0 \quad \dots\dots\dots (2)$$

Now we remove first degrees terms from the transformed equation

$$2h - 6 = 0 \quad 2k - 2 = 0$$

$$h = \frac{6}{2} \quad 2k = 2$$

$$h = 3 \quad k = 1$$

New origin is $O'(h, k) = O'(3, 1)$

Put $h = 3$, $k = 1$ in (2)

$$X^2 - Y^2 + 0 - 0 + 9 - 1 - 18 + 2 + 7 = 0$$

$$X^2 - Y^2 - 1 = 0$$

Q.3: In each of the following, find an equation referred to the new axes obtained by rotation of axes about the origin through the given angle.

(i) $xy = 1 \dots (1)$, $\theta = 45^\circ$

Solution:

We know that equations of rotation are

$$x = X \cos \theta - Y \sin \theta \Rightarrow x = X \cos 45^\circ - Y \sin 45^\circ \Rightarrow x = \frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}}$$

$$y = X \sin \theta + Y \cos \theta \Rightarrow y = X \sin 45^\circ + Y \cos 45^\circ \Rightarrow y = \frac{X}{\sqrt{2}} + \frac{Y}{\sqrt{2}}$$

Putting these values in (1) we get



$$\left(\frac{X-y}{\sqrt{2}}\right)\left(\frac{X+y}{\sqrt{2}}\right) = 1$$

$$\frac{X^2 - Y^2}{2} = 1 \quad \Rightarrow \quad X^2 - Y^2 = 2 \quad \text{Ans.}$$

(ii) $7x^2 - 8xy + y^2 - 9 = 0$, $\theta = \arctan 2$

Solution:

$$7x^2 - 8xy + y^2 - 9 = 0 \quad (1) \quad \theta = \tan^{-1} 2$$

$$\Rightarrow \tan \theta = 2 \quad \Rightarrow \quad \cot \theta = \frac{1}{2}$$

$$\sec \theta = \sqrt{1 + \tan^2 \theta} \quad , \quad \operatorname{cosec} \theta = \sqrt{1 + \cot^2 \theta}$$

$$= \sqrt{1 + 4} \quad , \quad = \sqrt{1 + \frac{1}{4}}$$

$$\sec \theta = \sqrt{5} \quad , \quad \operatorname{cosec} \theta = \frac{\sqrt{5}}{2}$$

$$\cos \theta = \frac{1}{\sqrt{5}} \quad , \quad \sin \theta = \frac{2}{\sqrt{5}}$$

$$x = X \cos \theta - Y \sin \theta \quad , \quad y = X \sin \theta + Y \cos \theta$$

$$= X \frac{1}{\sqrt{5}} - Y \frac{2}{\sqrt{5}} \quad , \quad y = X \frac{2}{\sqrt{5}} + Y \frac{1}{\sqrt{5}}$$

$$x = \frac{X - 2Y}{\sqrt{5}} \quad , \quad y = \frac{2X + Y}{\sqrt{5}}$$

Putting these values in (1)

$$7 \left(\frac{X - 2Y}{\sqrt{5}} \right)^2 - 8 \left(\frac{X - 2Y}{\sqrt{5}} \right) \left(\frac{2X + Y}{\sqrt{5}} \right) + \left(\frac{2X + Y}{\sqrt{5}} \right)^2 - 9 = 0$$

$$7 \left(\frac{X^2 + 4Y^2 - 4XY}{5} \right) - 8 \left(\frac{2X^2 + XY - 4XY - 2Y^2}{5} \right) + \frac{4X^2 + Y^2 + 4XY}{5} - 9 = 0$$

$$7X^2 - 28XY + 28Y^2 - 16X^2 + 24XY + 16Y^2 + 4X^2 + 4XY + Y^2 - 45 = 0$$

$$-5X^2 + 45Y^2 - 45 = 0$$

$$-5(X^2 - 9Y^2 + 9) = 0$$

$$X^2 - 9Y^2 + 9 = 0$$



$$(iii) \quad 9x^2 + 12xy + 4y^2 - x - y = 0 \quad \theta = \arctan \frac{2}{3}$$

Solution:

$$9x^2 + 12xy + 4y^2 - x - y = 0 \quad (1)$$

$$\theta = \tan^{-1} \frac{2}{3}$$

$$\tan \theta = \frac{2}{3} \Rightarrow \cot \theta = \frac{3}{2}$$

$$\begin{aligned} \sec \theta &= \sqrt{1 + \tan^2 \theta} \\ &= \sqrt{1 + \frac{4}{9}} = \frac{\sqrt{13}}{3} \end{aligned}$$

$$\sec \theta = \frac{\sqrt{13}}{3} \Rightarrow \cos \theta = \frac{3}{\sqrt{13}}$$

$$\sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{9}{13}} = \sqrt{\frac{13-9}{13}} = \sqrt{\frac{4}{13}} = \frac{2}{\sqrt{13}}$$

$$x = X \cos \theta - Y \sin \theta \Rightarrow x = X \frac{3}{\sqrt{13}} - Y \frac{2}{\sqrt{13}} = \frac{3X - 2Y}{\sqrt{13}}$$

$$y = X \sin \theta + Y \cos \theta \Rightarrow y = X \frac{2}{\sqrt{13}} + Y \frac{3}{\sqrt{13}} \quad y = \frac{2X + 3Y}{\sqrt{13}}$$

Putting the values in (1)

$$\begin{aligned} &9 \left(\frac{3X - 2Y}{\sqrt{13}} \right)^2 + 12 \left(\frac{3X - 2Y}{\sqrt{13}} \right) \left(\frac{2X + 3Y}{\sqrt{13}} \right) + 4 \left(\frac{2X + 3Y}{\sqrt{13}} \right)^2 - \left(\frac{3X - 2Y}{\sqrt{13}} \right) \\ &- \left(\frac{2X + 3Y}{\sqrt{13}} \right) = 0 \end{aligned}$$

$$\begin{aligned} &9 \frac{(9X^2 + 4Y^2 - 12XY)}{13} + \frac{12}{13} (6X^2 + 9XY - 4XY - 6Y^2) + \frac{4}{13} (4X^2 + 9Y^2 + \\ &12XY) - \frac{3X - 2Y}{\sqrt{13}} - \frac{2X + 3Y}{\sqrt{13}} = 0 \end{aligned}$$

$$\begin{aligned} &81X^2 + 36Y^2 - 108XY + 72X^2 + 108XY - 48XY - 72Y^2 + 16X^2 + 36Y^2 + \\ &48XY - 3\sqrt{13}X + 2\sqrt{13}Y - 2\sqrt{13}X - 3\sqrt{13}Y = 0 \end{aligned}$$

$$169X^2 - 5\sqrt{13}X - \sqrt{13}Y = 0$$

Dividing throughout by $\sqrt{13}$

$$13\sqrt{13}X^2 - 5X - Y = 0 \quad \text{Ans}$$



Q.4: Find the measure of angle through which the axes be rotated, so that the product term XY is removed from transformed equation. Also find transformation.

$$2x^2 + 6xy + 10y^2 - 11 = 0 \quad (1)$$

Solution:

$$2x^2 + 6xy + 10y^2 - 11 = 0 \quad \dots\dots (1)$$

The transformed equations for rotated axes are

$$x = X \cos \theta - Y \sin \theta, \quad y = X \sin \theta + Y \cos \theta \quad \text{Put in (1)}$$

$$2(X \cos \theta - Y \sin \theta)^2 + 6(X \cos \theta - Y \sin \theta)(X \sin \theta + Y \cos \theta) + 10(X \sin \theta + Y \cos \theta)^2 - 11 = 0$$

$$2X^2 \cos^2 \theta - 2Y^2 \sin^2 \theta - 4XY \sin \theta \cos \theta + 6(X^2 \cos \theta \sin \theta + XY \cos^2 \theta$$

$$- XY \sin^2 \theta - Y^2 \sin \theta \cos \theta) + 10X^2 \sin^2 \theta + 10Y^2 \cos^2 \theta$$

$$+ 20XY \sin \theta \cos \theta - 11 = 0$$

$$2X^2 \cos^2 \theta + 10X^2 \sin^2 \theta + 2Y^2 \sin^2 \theta + 10Y^2 \cos^2 \theta + XY (6 \cos^2 \theta - 6 \sin^2 \theta + 20 \sin \theta \cos \theta - 4 \cos \theta \sin \theta) + 6X^2 \cos \theta \sin \theta - 6Y^2 \sin^2 \theta = 0$$

$$X^2(2\cos^2 \theta + 10\sin^2 \theta + 6 \sin \theta \cos \theta) + XY (6 \cos^2 \theta - 6 \sin^2 \theta + 20 \sin \theta \cos \theta - 4 \cos \theta \sin \theta) + Y^2 (2 \sin^2 \theta - 6 \sin \theta \cos \theta + 10 \cos^2 \theta) - 11 = 0 \quad \dots\dots (2)$$

To remove XY term, we put

$$6\cos^2 \theta - 6\sin^2 \theta + 20 \sin \theta \cos \theta - 4 \sin \theta \cos \theta = 0$$

$$3\cos^2 \theta - 3\sin^2 \theta + 8 \sin \theta \cos \theta = 0$$

$$3 - 3 \tan^2 \theta - 8 \tan \theta = 0 \quad (\text{Dividing throughout by } \cos^2 \theta)$$

$$3 \tan^2 \theta + 8 \tan \theta - 3 = 0$$

$$\tan \theta = \frac{-8 \pm \sqrt{(8)^2 - 4(3)(-3)}}{2(3)} = \frac{8 \pm \sqrt{64 + 36}}{6}$$

$$= \frac{8 \pm \sqrt{100}}{6} = \frac{8 \pm 10}{6}$$

Either

$$\tan \theta = \frac{8+10}{6} = \frac{18}{6} \quad \text{or} \quad \tan \theta = \frac{8-10}{6} = \frac{-2}{6}$$

$$\tan \theta = 3, \quad \tan \theta = \frac{-1}{3}$$

As θ is taken in 1st quadrant so $\tan \theta = 3$ is only admissible value

$$\sec \theta = \sqrt{1 + \tan^2 \theta} = \sqrt{1 + 9} = \sqrt{10}$$

$$\cos \theta = \frac{1}{\sqrt{10}}$$



$$\begin{aligned}\sin \theta &= \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{1}{10}} = \sqrt{\frac{10-1}{10}} \\ &= \sqrt{\frac{9}{10}} = \frac{3}{\sqrt{10}}\end{aligned}$$

Put in (2)

$$X^2 \left[2 \left(\frac{1}{\sqrt{10}} \right)^2 + 6 \frac{1}{\sqrt{10}} \frac{3}{\sqrt{10}} + 10 \left(\frac{3}{\sqrt{10}} \right)^2 \right] + XY(0) + Y^2$$

$$\left[2 \left(\frac{3}{\sqrt{10}} \right)^2 - 6 \frac{3}{\sqrt{10}} \frac{1}{\sqrt{10}} + 10 \left(\frac{1}{\sqrt{10}} \right)^2 \right] - 11 = 0$$

$$X^2 \left[\frac{2}{10} + \frac{18}{10} + \frac{90}{10} \right] + Y^2 \left[\frac{18}{10} - \frac{18}{10} + \frac{10}{10} \right] - 11 = 0$$

$$X^2 \left[\frac{2+18+90}{10} \right] + Y^2 \left[\frac{18-18+10}{10} \right] - 11 = 0$$

$$11X^2 + Y^2 - 11 = 0 \quad \text{Ans.}$$

(ii) $xy + 4x - 3y - 10 = 0$ (1)

We know that for rotation at axes

$$x = X \cos \theta - Y \sin \theta, \quad y = X \sin \theta + Y \cos \theta \quad \text{Put in (1)}$$

$$(X \cos \theta - Y \sin \theta)(X \sin \theta + Y \cos \theta) + 4(X \cos \theta - Y \sin \theta) - 3(X \sin \theta + Y \cos \theta) - 10 = 0$$

$$X^2 \cos \theta \sin \theta + XY \cos^2 \theta - XY \sin^2 \theta - Y^2 \sin \theta \cos \theta + 4X \cos \theta - 4Y \sin \theta - 3X \sin \theta - 3Y \cos \theta - 10 = 0$$

$$X^2 \cos \theta \sin \theta - Y^2 \sin \theta \cos \theta + XY (\cos^2 \theta - \sin^2 \theta) + X(4 \cos \theta - 3 \sin \theta) + Y(-4 \sin \theta - 3 \cos \theta) - 10 = 0 \quad (2)$$

To remove XY terms we put

$$\cos^2 \theta - \sin^2 \theta = 0 \Rightarrow \cos 2\theta = 0$$

$$\Rightarrow 2\theta = \cos^{-1}(0) = 90^\circ$$

$$\Rightarrow \theta = 45^\circ \quad \text{Put in (2)}$$

$$X^2 \cos 45^\circ \sin 45^\circ - Y^2 \sin 45^\circ \cos 45^\circ + XY(0) + X(4 \cos 45^\circ - 3 \sin 45^\circ) + Y(-4 \sin 45^\circ - 3 \cos 45^\circ) - 10 = 0$$

$$X^2 \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} - Y^2 \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} + X \left(4 \times \frac{1}{\sqrt{2}} - 3 \times \frac{1}{\sqrt{2}} \right) + Y \left(-4 \frac{1}{\sqrt{2}} - 3 \frac{1}{\sqrt{2}} \right) - 10 = 0$$

$$\frac{1}{2} X^2 - \frac{1}{2} Y^2 + X \left(\frac{1}{\sqrt{2}} \right) + Y \left(\frac{-7}{\sqrt{2}} \right) - 10 = 0$$

$$X^2 - Y^2 + \frac{2}{\sqrt{2}} X - \frac{14Y}{\sqrt{2}} - 20 = 0$$



$$X^2 - Y^2 + \sqrt{2} X - 7\sqrt{2} - 20 = 0 \quad \text{Ans.}$$

(iii) $5x^2 - 6xy + 5y^2 - 8 = 0$

Solution:

$$5x^2 - 6xy + 5y^2 - 8 = 0 \quad (1)$$

We know that for rotation of axes

$$x = X \cos \theta - Y \sin \theta, \quad y = X \sin \theta + Y \cos \theta \quad \text{Put in (1)}$$

$$5(X \cos \theta - Y \sin \theta)^2 - 6(X \cos \theta - Y \sin \theta)(X \sin \theta + Y \cos \theta) + 5(X \sin \theta + Y \cos \theta)^2 - 8 = 0$$

$$5(X^2 \cos^2 \theta + Y^2 \sin^2 \theta - 2XY \sin \theta \cos \theta) - 6(X^2 \cos \theta \sin \theta + XY \cos^2 \theta - XY \sin^2 \theta - Y^2 \sin \theta \cos \theta) + 5(X^2 \sin^2 \theta + Y^2 \cos^2 \theta + 2XY \sin \theta \cos \theta) - 8 = 0$$

$$5X^2 \cos^2 \theta + 5Y^2 \sin^2 \theta - 10XY \sin \theta \cos \theta - 6X^2 \cos \theta \sin \theta - 6XY \cos^2 \theta + 6XY \sin^2 \theta + 6Y^2 \sin \theta \cos \theta + 5X^2 \sin^2 \theta + 5Y^2 \cos^2 \theta + 10XY \sin \theta \cos \theta - 8 = 0$$

$$X^2(5\cos^2 \theta - 6\cos \theta \sin \theta + 5\sin^2 \theta) + XY(-10\cos \theta \sin \theta - 6\cos^2 \theta + 6\sin^2 \theta + 10\sin \theta \cos \theta) + Y^2(5\sin^2 \theta + 6\sin \theta \cos \theta + 5\cos^2 \theta) - 8 = 0 \quad \dots (2)$$

To remove XY terms put (2)

$$-6\cos^2 \theta + 6\sin^2 \theta = 0$$

$$\cos^2 \theta = \sin^2 \theta \Rightarrow \tan^2 \theta = 1$$

$$\tan \theta = 1 \quad (\theta \text{ is taken as } 1^{\text{st}} \text{ Quadrant})$$

$$\theta = 45^\circ \quad \text{Put in (2)}$$

$$X^2(5(\cos 45^\circ)^2 - 6\cos 45^\circ \sin 45^\circ + 5(\sin 45^\circ)^2) + 0 + Y^2(5(\sin 45^\circ)^2 + 6\sin 45^\circ \cos 45^\circ + 5(\cos 45^\circ)^2) - 8 = 0$$

$$X^2 \left(5 + \left(\frac{1}{\sqrt{2}} \right)^2 - 6 \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} + 5 \left(\frac{1}{\sqrt{2}} \right)^2 \right) + Y^2 \left(5 + \left(\frac{1}{\sqrt{2}} \right)^2 + 6 \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} + 5 \left(\frac{1}{\sqrt{2}} \right)^2 \right)$$

$$X^2 \left(\frac{5}{2} - \frac{6}{2} + \frac{5}{2} \right) + Y^2 \left(\frac{5}{2} + \frac{6}{2} + \frac{5}{2} \right) - 8 = 0$$

$$2X^2 + 8Y^2 - 8 = 0$$

$$\Rightarrow X^2 + 4Y^2 - 4 = 0 \quad \text{Ans}$$



EXERCISE 6.9

Q.1: By rotation of axes, eliminates the xy -term in each of the following equations. Identify the conic & find its elements.

(i) $4x^2 - 4xy + y^2 - 6 = 0$

Solution:

$$4x^2 - 4xy + y^2 - 6 = 0 \quad (1)$$

$$\text{Here } a = 4 \quad b = 1 \quad 2h = -4$$



$$\tan 2\theta = \frac{2h}{a-b}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{-4}{4-1} = \frac{-4}{3}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{-4}{3}$$

$$6 \tan \theta = -4 + 4 \tan^2 \theta$$

$$4 \tan^2 \theta - 6 \tan \theta - 4 = 0 \Rightarrow 2 \tan^2 \theta - 3 \tan \theta - 2 = 0$$

$$2 \tan^2 \theta - 4 \tan \theta + \tan \theta - 2 = 0$$

$$2 \tan \theta (\tan \theta - 2) + 1 (\tan \theta - 2) = 0$$

$$(\tan \theta - 2) (2 \tan \theta + 1) = 0$$

$$\tan \theta = 2, \tan \theta = \frac{-1}{2} \text{ (is not admissible)}$$

$$\sec \theta = \sqrt{1 + \tan^2 \theta} = \sqrt{1 + 4} = \sqrt{5}$$

$$\cos \theta = \frac{1}{\sqrt{5}} \quad \sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{1}{5}}$$

$$= \sqrt{\frac{5-1}{5}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$$

Since equations of transformation are

$$\begin{aligned} x &= X \cos \theta - Y \sin \theta \\ y &= X \sin \theta + Y \cos \theta \end{aligned} \quad \text{putting values of } \sin \theta \text{ \& } \cos \theta \text{ in above equations}$$

$$\begin{aligned} x &= X \frac{1}{\sqrt{5}} - Y \frac{2}{\sqrt{5}} = \frac{X - 2Y}{\sqrt{5}} \\ x &= X \frac{2}{\sqrt{5}} - Y \frac{1}{\sqrt{5}} = \frac{2X + Y}{\sqrt{5}} \end{aligned} \quad (2)$$

Using (2) in (1)

$$4 \left(\frac{X - 2Y}{\sqrt{5}} \right)^2 - 4 \left(\frac{X - 2Y}{\sqrt{5}} \right) \left(\frac{2X + Y}{\sqrt{5}} \right) + \left(\frac{2X + Y}{\sqrt{5}} \right)^2 - 6 = 0$$

$$4 \left(\frac{X^2 + 4Y^2 - 4XY}{5} \right) - 4 \left(\frac{2X^2 + XY - 4XY - 2Y^2}{5} \right) + \left(\frac{4X^2 + Y^2 + 4XY}{5} \right) - 6 = 0$$

$$4X^2 + 16Y^2 - 16XY - 8X^2 - 4XY + 16XY + 8Y^2 + 4X^2 + Y^2 + 4XY - 30 = 0$$

$$25Y^2 - 30 = 0$$

$$25Y^2 = 30$$

$$Y^2 = \frac{30}{25} \Rightarrow Y^2 = \frac{6}{5}$$



$$\Rightarrow Y = \pm \sqrt{\frac{6}{5}} \rightarrow (3)$$

$$\text{From (2) we have } X - 2Y = \sqrt{5}x \dots\dots (4) \quad , \quad 2X + Y = \sqrt{5}y \dots\dots (5)$$

Multiply eq. (4) by 2 & subtract from 5

$$\begin{array}{rcl} 2X + Y & = & \sqrt{5}y \\ -2X + 4Y & = & -2\sqrt{5}x \\ \hline 5Y & = & \sqrt{5}y - 2\sqrt{5}x \\ 5Y & = & \sqrt{5}(y - 2x) \\ Y & = & -\frac{\sqrt{5}}{5}(2x - y) \rightarrow (6) \end{array}$$

Using (3) in 6

$$\pm \sqrt{\frac{6}{5}} = -\frac{\sqrt{5}}{5}(2x - y)$$

$$\pm \frac{\sqrt{6}}{\sqrt{5}} = -\frac{1}{\sqrt{5}}(2x - y)$$

$$2x - y \pm \sqrt{6} = 0$$

$$2x - y + \sqrt{6} = 0 \quad , \quad 2x - y - \sqrt{6} = 0 \quad \text{Ans.}$$

$$(ii) \quad x^2 - 2xy + y^2 - 8x - 8y = 0 \quad (1)$$

Solution:

$$\text{Here } a = 1, \quad 2h = -2, \quad b = 1$$

$$\tan 2\theta = \frac{2h}{a-b} = \frac{-2}{1-1} = \frac{-2}{0} = \infty$$

$$\tan 2\theta = \infty$$

$$2\theta = \tan^{-1} \infty = 90^\circ$$

$$\theta = 45^\circ$$

Equations of transformation are

$$x = X \cos \theta - Y \sin \theta \Rightarrow x = X \cos 45^\circ - Y \sin 45^\circ$$

$$x = \frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}} = \frac{X - Y}{\sqrt{2}}$$

$$y = X \sin \theta + Y \cos \theta \Rightarrow y = X \sin 45^\circ + Y \cos 45^\circ$$

$$y = \frac{X}{\sqrt{2}} + \frac{Y}{\sqrt{2}} = \frac{X + Y}{\sqrt{2}}$$

Using (2) in (1)



$$\begin{aligned} & \left(\frac{X-Y}{\sqrt{2}}\right)^2 - 2\left(\frac{X-Y}{\sqrt{2}}\right)\left(\frac{X+Y}{\sqrt{2}}\right) + \left(\frac{X+Y}{\sqrt{2}}\right)^2 - 8\left(\frac{X-Y}{\sqrt{2}}\right) - 8\left(\frac{X+Y}{\sqrt{2}}\right) = 0 \\ & \frac{X^2 + Y^2 - 2XY}{2} - \frac{2}{2}(X^2 - Y^2) + \frac{X^2 + Y^2 + 2XY}{2} - \frac{8}{\sqrt{2}}(X - Y) - \frac{8}{\sqrt{2}}(X + Y) = 0 \\ & X^2 + Y^2 - 2XY - 2(X^2 - Y^2) + X^2 + Y^2 + 2XY - 8\sqrt{2}(X - Y) - 8\sqrt{2}(X + Y) = 0 \\ & X^2 + Y^2 - 2XY - 2X^2 + 2Y^2 + X^2 + Y^2 + 2XY - 8\sqrt{2}X + 8\sqrt{2}Y - 8\sqrt{2}X - 8\sqrt{2}Y = 0 \end{aligned}$$

$$4Y^2 - 16\sqrt{2}X = 0$$

$$Y^2 - 4\sqrt{2}X = 0 \quad \text{which represents a Parabola}$$

To solve (2) for values of X & Y

$$x = \frac{X-Y}{\sqrt{2}} \Rightarrow \sqrt{2}x = X-Y \rightarrow (3)$$

$$y = \frac{X+Y}{\sqrt{2}} \Rightarrow \sqrt{2}y = X+Y \rightarrow (4)$$

Adding (3) & (4)

$$\sqrt{2}(x+y) = 2X$$

$$X = \frac{\sqrt{2}}{2}(x+y) \Rightarrow X = \frac{1}{\sqrt{2}}(x+y)$$

Subtracting (3) & (4)

$$\sqrt{2}x = X - Y$$

$$-\sqrt{2}y = -X + Y$$

$$\sqrt{2}(x-y) = -2Y$$

$$Y = \frac{\sqrt{2}}{2}(x-y)$$

$$Y = \frac{-1}{\sqrt{2}}(x-y)$$

For vertex of Parabola, put $X = 0$, $Y = 0$

$$0 = \frac{1}{\sqrt{2}}(x+y) \quad , \quad 0 = \frac{-1}{\sqrt{2}}(x-y)$$

$$\Rightarrow x+y = 0$$

$$\Rightarrow x-y = 0$$

$$x+y = 0$$

$$x+y = 0$$

Adding $x-y = 0$

Subtract $-x \mp y = 0$

$$2x = 0$$

$$\Rightarrow \boxed{x = 0}$$

$$2y = 0$$

$$\Rightarrow \boxed{y = 0}$$

$$\boxed{\text{Vertex} = (0, 0)}$$



Axis of Parabola is $Y = 0$

$$\frac{-1}{\sqrt{2}} (x - y) = 0$$

$$x - y = 0 \Rightarrow \boxed{x = y}$$

Since $Y^2 = 4\sqrt{2} X$

As standard form is

$$y^2 = 4aX$$

$$4a = 4\sqrt{2}$$

$$a = \sqrt{2}$$

Focus of Parabola is $(0, 0)$

$$(X, Y) = (\sqrt{2}, 0)$$

$$\left[\frac{1}{\sqrt{2}} (x + y), \frac{1}{\sqrt{2}} (x - y) \right] = (\sqrt{2}, 0)$$

$$\frac{1}{\sqrt{2}} (x + y) = \sqrt{2}, \quad \frac{1}{\sqrt{2}} (x - y) = 0$$

$$x + y = 2$$

$$x - y = 0$$

Solving above equations

$$x + y = 2$$

$$x - y = 2$$

$$2x = 2$$

$$\boxed{x = 1}$$

$$1 - y = 0$$

$$\boxed{y = 1}$$

$$\boxed{\text{Focus } (1, 1)}$$

Ans

(iii) $x^2 + 2xy + y^2 + 2\sqrt{2}x - 2\sqrt{2}y + 2 = 0$

Solution:

$$x^2 + 2xy + y^2 + 2\sqrt{2}x - 2\sqrt{2}y + 2 = 0 \quad \dots\dots (1)$$

$$a = 1 \quad \& \quad b = 1 \quad \Rightarrow \quad a = b, \quad 2h = 2$$

$$\tan 2\theta = \frac{2h}{a - b}$$

$$\tan 2\theta = \frac{2}{0} = \infty$$

$$2\theta = 90^\circ \Rightarrow \theta = 45^\circ$$

Now equations of transformation are

$$x = X \cos \theta - Y \sin \theta = X \cos 45^\circ - Y \sin 45^\circ$$



$$x = \frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}} = \frac{X-Y}{\sqrt{2}}$$

$$y = X \sin \theta + Y \cos \theta$$

$$y = X \sin 45^\circ + Y \cos 45^\circ$$

$$y = \frac{X}{\sqrt{2}} + \frac{Y}{\sqrt{2}} = \frac{X+Y}{\sqrt{2}}$$

$$x = \frac{X-Y}{\sqrt{2}}, \quad y = \frac{X+Y}{\sqrt{2}} \quad] \rightarrow (2)$$

Using (2) in (1)

$$\left(\frac{X-Y}{\sqrt{2}}\right)^2 + 2\left(\frac{X-Y}{\sqrt{2}}\right)\left(\frac{X+Y}{\sqrt{2}}\right) + \left(\frac{X+Y}{\sqrt{2}}\right)^2 + 2\sqrt{2}\left(\frac{X-Y}{\sqrt{2}}\right) - 2\sqrt{2}\left(\frac{X+Y}{\sqrt{2}}\right) + 2 = 0$$

$$\frac{X^2 + Y^2 - 2XY}{2} + \frac{2}{2}(X^2 - Y^2) + \frac{X^2 + Y^2 + 2XY}{2} + 2(X-Y) - 2(X+Y) + 2 = 0$$

$$X^2 + Y^2 - 2XY + 2X^2 - 2Y^2 + X^2 + Y^2 + 2XY + 4X - 4Y - 4X - 4Y + 4 = 0$$

$$4X^2 - 8Y + 4 = 0 \Rightarrow X^2 - 2Y + 1 = 0$$

$$X^2 = 2Y - 1 \Rightarrow X^2 = 2(Y-1) \rightarrow (3)$$

Which represents a Parabola

Now, solve (2) for X & Y

	$X + Y = \sqrt{2} y$		$X - Y = \sqrt{2} y$
Adding	$X - Y = \sqrt{2} x$	Subtracting	$-X \pm Y = -\sqrt{2} y$
	$2X = \sqrt{2} (x + y)$		$-2Y = \sqrt{2} (x - y)$
			$Y = \frac{-1}{\sqrt{2}} (x - y)$

$$X = \frac{\sqrt{2}}{2} (x + y) \Rightarrow X = \frac{1}{\sqrt{2}} (x + y)$$

For vertex of Parabola

$$\text{Put } X = 0, \quad Y = 0$$

$$\frac{1}{\sqrt{2}} (x + y) = 0 \quad \quad \quad -\frac{1}{\sqrt{2}} (x - y) = 0$$

$$x + y = 0$$

$$x - y = 0$$



$$\begin{array}{rcl} x + y & = & 0 \\ x - y & = & 0 \\ \hline 2x & = & 0 \end{array}$$

$$\Rightarrow \boxed{x = 0}$$

$$\begin{array}{rcl} x + y & = & 0 \\ -x + y & = & -0 \\ \hline 2y & = & 0 \end{array}$$

$$\Rightarrow \boxed{y = 0}$$

$$\boxed{\text{Vertex} = (0, 0)}$$

Axis of Parabola

$$X = 0$$

$$\frac{1}{\sqrt{2}}(x + y) = 0 \Rightarrow \boxed{x + y = 0}$$

Focus $(X, Y) = (0, a)$

$$\therefore X^2 = 2\left(Y - \frac{1}{2}\right)$$

$$x^2 = 4ay$$

$$\Rightarrow 4a = 2 \quad \boxed{a = \frac{1}{2}}$$

$$\left(\frac{1}{\sqrt{2}}(x + y), \frac{-1}{\sqrt{2}}(x - y) \right) = \left(0, -\frac{1}{2} \right)$$

$$\frac{1}{\sqrt{2}}(x + y) = 0$$

$$\frac{-1}{\sqrt{2}}(x - y) = \frac{1}{2}$$

$$(x + y) = 0$$

$$(x - y) = -\sqrt{2}$$

$$x + y = 0$$

$$x - y = -\sqrt{2}$$

$$\frac{2x}{2} = -\sqrt{2}$$

$$\Rightarrow \boxed{x = \frac{-1}{\sqrt{2}}}$$

$$x - y = -\sqrt{2}$$

$$\frac{-1}{\sqrt{2}} - y = -\sqrt{2}$$

$$\frac{-1 - \sqrt{2}y}{\sqrt{2}} = -2$$

$$-1 - \sqrt{2} = -2$$

$$-1 + 2 = \sqrt{2} y$$

$$1 = \sqrt{2} y$$

$$\Rightarrow y = \frac{1}{\sqrt{2}}$$

$$\text{Focus} = \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

Ans.



(iv) $x^2 + xy + y^2 - 4 = 0$

Solution:

Here $a = 1$, $b = 1$, $2h = 1$

If $a = b$ $\theta = 45^\circ$ (always)

New equations of transformation are

$$\begin{aligned} x &= X \cos \theta - Y \sin \theta = X \cos 45^\circ - Y \sin 45^\circ = \frac{X}{\sqrt{2}} - \frac{Y}{\sqrt{2}} \\ y &= X \sin \theta + Y \cos \theta = Y \sin 45^\circ + Y \cos 45^\circ = \frac{X}{\sqrt{2}} + \frac{Y}{\sqrt{2}} \end{aligned} \quad (2)$$

Putting in (1)

$$\begin{aligned} \left(\frac{X-Y}{\sqrt{2}}\right)^2 + \left(\frac{X-Y}{\sqrt{2}}\right)\left(\frac{X+Y}{\sqrt{2}}\right) + \left(\frac{X+Y}{\sqrt{2}}\right)^2 - 4 &= 0 \\ \frac{X^2 + Y^2 - 2XY}{2} + \frac{X^2 - Y^2}{2} + \frac{X^2 + Y^2 + 2XY}{2} - 4 &= 0 \\ X^2 + Y^2 - 2XY + X^2 - Y^2 + X^2 + Y^2 + 2XY - 8 &= 0 \\ 3X^2 + Y^2 &= 8 \\ \frac{3X^2}{8} + \frac{Y^2}{8} &= 1 \Rightarrow \frac{X^2}{\frac{8}{3}} + \frac{Y^2}{8} = 1 \quad (\text{which represents an ellipse}) \end{aligned}$$

Solve (2) for X & Y

$$\begin{array}{rcl} \sqrt{2}x & = & X - Y \\ \text{Adding } \sqrt{2}y & = & X + Y \\ \hline \sqrt{2}(x+y) & = & 2X \\ X & = & \frac{1}{\sqrt{2}}(x+y) \end{array} \quad \begin{array}{rcl} \sqrt{2}x & = & X - Y \\ -\sqrt{2}y & = & -X + Y \\ \hline \sqrt{2}(x-y) & = & -2Y \\ Y & = & -\frac{1}{\sqrt{2}}(x-y) \end{array}$$

For center put $X = 0$, $Y = 0$

$$\begin{array}{rcl} \frac{1}{\sqrt{2}}(x+y) & = & 0 \\ (x+y) & = & 0 \\ x+y & = & 0 \\ \text{Adding } x-y & = & 0 \\ \hline 2x & = & 0 \Rightarrow \boxed{x = 0} \end{array} \quad \begin{array}{rcl} -\frac{1}{\sqrt{2}}(x-y) & = & 0 \\ x-y & = & 0 \\ x+y & = & 0 \\ -x+y & = & -0 \\ \hline 2y & = & 0 \Rightarrow \boxed{y = 0} \end{array}$$



Required center (0, 0)

From (3) we have $a^2 = 8$ (length of major axis)

$$b^2 = \frac{8}{3} \quad (\text{Length of minor axis})$$

$$\therefore e^2 = \frac{a^2 - b^2}{a^2} = \frac{8 - \frac{8}{3}}{8} = \frac{24 - 8}{3} \times \frac{1}{8} = \frac{16}{3} \times \frac{1}{8} = \frac{2}{3}$$

$$\Rightarrow e = \sqrt{\frac{2}{3}}$$

Coordinates of foci

$$= (0, \pm ae)$$

$$(X, Y) = (0, \pm 2\sqrt{2} \sqrt{\frac{2}{3}})$$

$$\left(\frac{1}{\sqrt{2}}(x+y), \frac{-1}{\sqrt{2}}(x-y) \right) = \left(0, \pm \frac{4}{\sqrt{3}} \right)$$

$$\frac{1}{\sqrt{2}}(x+y) = 0, \quad \frac{-1}{\sqrt{2}}(x-y) = \pm \frac{4}{\sqrt{3}}$$

$$\Rightarrow x+y = 0 \quad x-y = \pm \frac{4\sqrt{2}}{\sqrt{3}}$$

Now $x+y = 0$

Adding $x-y = \pm \frac{4\sqrt{2}}{\sqrt{3}}$

$$2x = \pm \frac{4\sqrt{2}}{\sqrt{3}} \Rightarrow \boxed{x = \pm \frac{2\sqrt{2}}{\sqrt{3}}} \quad \boxed{y = \pm \frac{2\sqrt{2}}{\sqrt{3}}}$$

Coordinates of foci are $\left(\pm \frac{2\sqrt{2}}{\sqrt{3}}, \pm \frac{2\sqrt{2}}{\sqrt{3}} \right)$ Ans.

Vertices of ellipse are $(0, \pm a)$

$$(X, Y) = (0, \pm 2\sqrt{2})$$

$$\frac{1}{\sqrt{2}}(x-y) = 0 \quad \frac{-1}{\sqrt{2}}(x+y) = \pm 2\sqrt{2}$$

$$\Rightarrow x-y = 0 \quad x+y = \pm 4$$



$$x - y = 0$$

$$x + y = \pm 4$$

$$2x = \pm 4 \Rightarrow \boxed{x = \pm 2}$$

$$y = \pm 2$$

Required vertices are $(\pm 2, \pm 2)$

For equation of major axis Put $Y = 0$

$$\frac{-1}{\sqrt{2}}(x + y) = 0 \Rightarrow x + y = 0$$

For equation of minor axis Put $X = 0$

$$\frac{1}{\sqrt{2}}(x - y) = 0 \Rightarrow x - y = 0$$

(v) $7x^2 - 6\sqrt{3}xy + 13y^2 - 16 = 0$

Solution:

$$7x^2 - 6\sqrt{3}xy + 13y^2 - 16 = 0 \quad \dots (1)$$

$$a = 7, \quad 2h = -6\sqrt{3}, \quad b = 13$$

$$\tan 2\theta = \frac{2h}{a - b}$$

$$\tan 2\theta = \frac{-6\sqrt{3}}{7 - 13} = \frac{-6\sqrt{3}}{-6}$$

$$\tan 2\theta = \sqrt{3} \Rightarrow 2\theta = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$\theta = 30^\circ$$

Now equations of transformation are

$$\left. \begin{aligned} x &= X \cos 30^\circ - Y \sin 30^\circ = X \frac{\sqrt{3}}{2} - Y \frac{1}{2} = \frac{\sqrt{3}X - Y}{2} \\ y &= X \sin 30^\circ + Y \cos 30^\circ = X \frac{1}{2} + Y \frac{\sqrt{3}}{2} = \frac{X + \sqrt{3}Y}{2} \end{aligned} \right] \dots (2)$$

Putting these values in (1)

$$7 \left(\frac{\sqrt{3}X - Y}{2} \right)^2 - 6\sqrt{3} \left(\frac{\sqrt{3}X - Y}{2} \right) \left(\frac{X + \sqrt{3}Y}{2} \right) + 13 \left(\frac{X + \sqrt{3}Y}{2} \right)^2 - 16 = 0$$

$$7 \left(\frac{3X^2 + Y^2 - 2\sqrt{3}XY}{4} \right) - \frac{3\sqrt{3}}{2} (\sqrt{3}X^2 + 3XY - XY - \sqrt{3}Y^2) + \frac{13}{4} (X^2 + 3Y^2 + 2\sqrt{3}XY) - 16 = 0$$

$$21X^2 + 7Y^2 - 14\sqrt{3}XY - 6\sqrt{3}(\sqrt{3}X^2 + 2XY - \sqrt{3}Y^2) + 13X^2 + 39Y^2 + 26\sqrt{3}XY - 64 = 0$$



$$21X^2 + 7Y^2 - 14\sqrt{3}XY - 18X^2 - 12\sqrt{3}XY + 18Y^2 + 13X^2 + 39Y^2 + 26\sqrt{3}XY - 64 = 0$$

$$16X^2 + 64Y^2 - 64 = 0$$

$$16X^2 + 64Y^2 = 64$$

$$\frac{X^2}{4} + \frac{Y^2}{1} = 1 \rightarrow (3) \text{ which represents an ellipse}$$

Now for values of X & Y solving (2)

$$\begin{array}{rcl} \sqrt{3}X - Y & = & 2x \\ -\sqrt{3}X + 3Y & = & -2\sqrt{3}y \\ \hline -4Y & = & 2x - 2\sqrt{3}y \\ -4Y & = & 2(x - \sqrt{3}y) \\ Y & = & \frac{-1}{2}(x - \sqrt{3}y) \dots (4) \end{array}$$

$$\begin{array}{rcl} 3X - \sqrt{3}Y & = & 2\sqrt{3}x \\ X + \sqrt{3}Y & = & 2y \\ \hline 4X & = & 2\sqrt{3}x + 2y \\ X & = & \frac{1}{2}(\sqrt{3}x + y) \dots (5) \end{array}$$

From (3), we have

$$a^2 = 4, \quad b^2 = 1 \quad \therefore \quad e^2 = \frac{a^2 - b^2}{a^2} = \frac{4 - 1}{4} = \frac{3}{4}$$

$$\Rightarrow e = \frac{\sqrt{3}}{2}$$

Coordinates of foci = $(\pm ae, 0)$

$$(X, Y) = (\pm\sqrt{3}, 0)$$

$$\left[\frac{1}{2}(\sqrt{3}x + y), \frac{-1}{2}(x - \sqrt{3}y) \right] = (\pm\sqrt{3}, 0)$$

$$\frac{1}{2}(\sqrt{3}x + y) = \pm\sqrt{3} \quad \frac{-1}{2}(x - \sqrt{3}y) = 0$$

$$\sqrt{3}x + y = \pm 2\sqrt{3} \quad (i) \quad x - \sqrt{3}y = 0 \quad (ii)$$

Multiplying (ii) by $\sqrt{3}$ & subtracting from (i)

$$\begin{array}{rcl} \sqrt{3}x + y & = & 2\sqrt{3} \\ -\sqrt{3}x + 3y & = & 0 \\ \hline 4y & = & 2\sqrt{3} \\ y & = & \frac{\sqrt{3}}{2} \end{array}$$

$$\begin{array}{rcl} \sqrt{3}x + y & = & -2\sqrt{3} \\ -\sqrt{3}x + 3y & = & 0 \\ \hline 4y & = & -2\sqrt{3} \\ y & = & \frac{-1}{2}\sqrt{3} \end{array}$$

$$x - \sqrt{3} \left(\frac{\sqrt{3}}{2} \right) = 0 \quad \therefore \quad x - \sqrt{3} \left(-\frac{1}{2}\sqrt{3} \right) = 0$$



$$x - \frac{3}{2} = 0 \quad x + \frac{3}{2} = 0$$

$$x = \frac{3}{2} \quad x = -\frac{3}{2}$$

Coordinates of foci

$$\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right) \quad \left(-\frac{3}{2}, -\frac{\sqrt{3}}{2}\right)$$

Vertices

$$(\pm a, 0)$$

$$(X, Y) = (\pm 2, 0)$$

$$\frac{1}{2}(\sqrt{3}x + y) = \pm 2, \quad \frac{1}{2}(\sqrt{3}y - x) = 0$$

$$\sqrt{3}x + y = \pm 4 \quad \sqrt{3}y - x = 0$$

Solving above equations

$$\sqrt{3}x + y = 4 \quad \text{(iii)}$$

$$\sqrt{3}x + y = -4 \quad \text{(iv)}$$

$$-x + \sqrt{3}y = 0 \quad -x + \sqrt{3}y = 0 \quad \text{(v)}$$

Multiply equation (v) by $\sqrt{3}$ & adding in (iii) and (iv)

$$\begin{array}{rcl} \sqrt{3}x + y & = & 4 \\ -\sqrt{3}x + 3y & = & 0 \\ \hline 4y & = & 4 \end{array} \quad \begin{array}{rcl} \sqrt{3}x + y & = & -4 \\ -\sqrt{3}x + 3y & = & 0 \\ \hline 4y & = & -4 \end{array}$$

$$4y = 4$$

$$\boxed{y = 1}$$

$$\therefore -x + \sqrt{3}(1) = 0$$

$$\boxed{x = \sqrt{3}}$$

$$4y = -4$$

$$y = -1$$

$$\therefore -x + \sqrt{3}(-1) = 0$$

$$x = -\sqrt{3}$$

Required vertex $(\sqrt{3}, 1)$ & $(-\sqrt{3}, -1)$

For equation of major axes, for equation of minor axes

$$\text{Put } Y = 0$$

$$\frac{1}{2}(\sqrt{3}y - x) = 0$$

$$\sqrt{3}y - x = 0$$

$$\text{Put } X = 0$$

$$\frac{1}{2}(\sqrt{3}x + y) = 0$$

$$\sqrt{3}x + y = 0$$



$$(vi) \quad 4x^2 - 4xy + 7y^2 + 12x + 6y - 9 = 0 \quad (1)$$

Solution:

$$a = 4, \quad b = 7, \quad 2h = -4$$

$$\begin{aligned} \tan 2\theta &= \frac{2h}{a-b} \\ &= \frac{-4}{4-7} = \frac{-4}{-3} = \frac{4}{3} \end{aligned}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{4}{3} \Rightarrow 6 \tan \theta = 4 - 4 \tan^2 \theta$$

$$\Rightarrow 4 \tan^2 \theta + 6 \tan \theta - 4 = 0$$

$$2 \tan^2 \theta + 3 \tan \theta - 2 = 0$$

$$2 \tan^2 \theta + 4 \tan \theta - \tan \theta - 2 = 0$$

$$2 \tan \theta (\tan \theta + 2) - 1 (\tan \theta + 2) = 0$$

$$(\tan \theta + 2) (2 \tan \theta - 1) = 0$$

$$\tan \theta + 2 = 0, \quad 2 \tan \theta - 1 = 0$$

$$\tan \theta = -2 \quad (\text{neglect it, because } \theta \text{ is taken from } 1^{\text{st}} \text{ Quadrant})$$

And also

$$2 \tan \theta - 1 = 0$$

$$2 \tan \theta = 1 \Rightarrow \tan \theta = \frac{1}{2}$$

$$\cot \theta = 2 \Rightarrow \operatorname{cosec} \theta = \sqrt{1 + \cot^2 \theta} = \sqrt{1 + 4} = \sqrt{5}$$

$$\sin \theta = \frac{1}{\sqrt{5}} \Rightarrow \cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$= \sqrt{1 - \frac{1}{5}} = \sqrt{\frac{5-1}{5}} = \frac{2}{\sqrt{5}}$$

Now equations of transformation are

$$\begin{aligned} x &= X \cos \theta - Y \sin \theta = X \left(\frac{2}{\sqrt{5}} \right) - Y \left(\frac{1}{\sqrt{5}} \right) = \frac{2X - Y}{\sqrt{5}} \\ y &= X \sin \theta + Y \cos \theta = X \frac{1}{\sqrt{5}} + Y \frac{2}{\sqrt{5}} = \frac{X + 2Y}{\sqrt{5}} \end{aligned} \quad \dots (2)$$

Now putting values in (1)

$$4 \left(\frac{2X - Y}{\sqrt{5}} \right)^2 - 4 \left(\frac{2X - Y}{\sqrt{5}} \right) \left(\frac{X + 2Y}{\sqrt{5}} \right) + 7 \left(\frac{X + 2Y}{\sqrt{5}} \right)^2 + 12 \left(\frac{2X - Y}{\sqrt{5}} \right) + 6 \left(\frac{X + 2Y}{\sqrt{5}} \right) - 9 = 0$$

$$4 \left(\frac{4X^2 + Y^2 - 4XY}{5} \right) - \frac{4}{5} (2X^2 + 4XY - XY - 2Y^2) + \frac{7}{15} (X^2 + 4Y^2 + 4XY) + \frac{12}{\sqrt{5}}$$



$$(2X - Y) + \frac{6}{\sqrt{5}} (X + 2Y) - 9 = 0$$

$$4(4X^2 - 4XY + Y^2) - 4(2X^2 + 3XY - 2Y^2) + 7(X^2 + 4Y^2 + 4XY) + 12\sqrt{5}(2X - Y) + 6\sqrt{5}(X + 2Y) - 45 = 0$$

$$16X^2 - 16XY + 4Y^2 - 8X^2 - 12XY + 8Y^2 + 7X^2 + 28Y^2 + 28XY + 24\sqrt{5}X - 12\sqrt{5}Y + 6\sqrt{5}X + 12\sqrt{5}Y - 45 = 0$$

$$15X^2 + 40Y^2 + 30\sqrt{5}X - 45 = 0$$

$$\Rightarrow 3X^2 + 8Y^2 + 6\sqrt{5}X - 9 = 0$$

$$3X^2 + 6\sqrt{5}X + 8Y^2 = 9$$

$$3(X^2 + 2\sqrt{5}X + (\sqrt{5})^2) + 8Y^2 = 9 + 3(\sqrt{5})^2$$

$$\Rightarrow 3(X + \sqrt{5})^2 + 8Y^2 = 24$$

$$\frac{(X + \sqrt{5})^2}{8} + \frac{Y^2}{3} = 1 \quad \text{..... (3) which represents an ellipse}$$

Now, for values of X & Y solving (2)

$$2X - Y = \sqrt{5}x$$

$$2X - Y = \sqrt{5}x$$

$$X + 2Y = \sqrt{5}y$$

$$X + 2Y = \sqrt{5}y$$

Multiply 1st Equation by 2 & multiply equation (2) by 2 and adding

$$4X - 2y = 2\sqrt{5}x$$

$$2X - Y = \sqrt{5}x$$

$$X + 2y = \sqrt{5}y$$

$$-2X + 4Y = -2\sqrt{5}y$$

$$5X = \sqrt{5}(2x + y)$$

$$-5Y = \sqrt{5}(x - 2y)$$

$$\Rightarrow X = \frac{1}{\sqrt{5}}(2x + y) \quad \text{...(4)}$$

$$Y = \frac{-1}{\sqrt{5}}(x - 2y) \rightarrow (5)$$

From (3) $a^2 = 8$ & $b^2 = 3$

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{8 - 3}{8} = \frac{5}{8} \Rightarrow e = \sqrt{\frac{5}{8}}$$

Coordinates of foci = $(\pm ae, 0)$

$$(X + \sqrt{5}, Y) = (\pm\sqrt{5}, 0)$$

$$\Rightarrow \frac{1}{\sqrt{5}}(2x + y) + \sqrt{5} = \pm\sqrt{5}$$

$$2x + y = \pm 5 - 5$$

$$2x + y = 0, \quad 2x + y = -10$$

$$x + 2y = 0, \quad x - 2y = 0$$

Multiplying 1st equation by 2



$$\begin{array}{rcl}
 4x + 2y = 0 & & 2x + y = -10 \\
 -x + 2y = 0 & & -2x + 4y = 0 \\
 \hline
 3x = 0 & \Rightarrow & \boxed{x = 0} \therefore \boxed{y = 0} \\
 & & 5y = -10 \\
 & & \boxed{y = -2} \therefore \boxed{x = -4}
 \end{array}$$

Coordinates of foci are

$$(0, 0) \text{ \& \ } (-4, -2)$$

Vertices

$$(X + \sqrt{5}, Y) = (\pm\sqrt{8}, 0)$$

$$X + \sqrt{5} = \pm\sqrt{8} \quad Y = 0$$

$$\frac{1}{\sqrt{5}}(2x + y) + \sqrt{5} = \sqrt{8}, \quad \frac{1}{\sqrt{5}}(2x + y) + \sqrt{5} = -\sqrt{8} \text{ and } \frac{-1}{\sqrt{5}}(x - 2y) = 0$$

$$2x + y + 5 = \sqrt{40}, \quad 2x + y + 5 = -40 \text{ \& \ } x - 2y = 0 \dots\dots (2)$$

$$\Rightarrow 2x + y = \sqrt{40} - 5, \quad 2x + y = -40 - 5$$

$$-2x + 4y = 0$$

$$5y = \sqrt{40} - 5$$

$$y = \frac{\sqrt{40} - 5}{5} \therefore x - 2\left(\frac{\sqrt{40} - 5}{5}\right) = 0$$

$$x = 2\left(\frac{\sqrt{40} - 5}{5}\right)$$

$$\text{Vertex is } \left(\frac{2\sqrt{40}}{5} - 1, \frac{\sqrt{40}}{5} - 1\right)$$

$$(vii) \quad xy - 4x - 2y = 0$$

Solution:

$$xy - 4x - 2y = 0 \quad (1)$$

$$a = 0, \quad b = 0, \quad 2h = 1$$

Since $a = b$ therefore $\theta = 45^\circ$ (always)

Now equations of transformations are

$$\begin{aligned}
 x &= X \cos \theta - Y \sin \theta = X \cos 45^\circ - Y \sin 45^\circ = \frac{X - Y}{\sqrt{2}} \\
 y &= X \sin \theta + Y \cos \theta = X \sin 45^\circ + Y \cos 45^\circ = \frac{X + Y}{\sqrt{2}}
 \end{aligned} \dots\dots (2)$$



Putting values in (1)

$$\left(\frac{X-Y}{\sqrt{2}}\right)\left(\frac{X+Y}{\sqrt{2}}\right) - 4\left(\frac{X-Y}{\sqrt{2}}\right) - 2\left(\frac{X+Y}{\sqrt{2}}\right) = 0$$

$$\frac{X^2 - Y^2}{2} - \frac{4}{\sqrt{2}}(X-Y) - \frac{2}{\sqrt{2}}(X+Y) = 0$$

$$X^2 - Y^2 - 4\sqrt{2}X + 4\sqrt{2}Y - 2\sqrt{2}X - 2\sqrt{2}Y = 0$$

$$X^2 - 6\sqrt{2}X - Y^2 + 2\sqrt{2}Y = 0$$

$$(X^2 - 6\sqrt{2}X + 18) - (Y^2 - 2\sqrt{2}Y + 2) = 18 - 2$$

$$(X^2 - 3\sqrt{2})^2 - (Y - \sqrt{2})^2 = 16$$

$$\frac{(X - 3\sqrt{2})^2}{16} - \frac{(Y - \sqrt{2})^2}{16} = 1 \quad \dots (3) \text{ which represents hyperbola}$$

Now for the values of X & Y, solving (2)

	$X - Y = \sqrt{2}x$		$X - Y = \sqrt{2}x$
Adding	$X + Y = \sqrt{2}y$	Subtracting	$-X \pm Y = -\sqrt{2}y$
	$2X = \sqrt{2}(x+y)$		$-2Y = \sqrt{2}(x-y)$
	$X = \frac{1}{\sqrt{2}}(x+y)$		$Y = \frac{-1}{\sqrt{2}}(x-y)$

From (3) we have $a^2 = 16$, $b^2 = 16$

$$e^2 = \frac{a^2 + b^2}{a^2} = \frac{16 + 16}{16} = \frac{32}{16} = 2$$

$$\Rightarrow e = \sqrt{2}$$

$$\text{foci } (\pm ae = 0)$$

$$(X - 3\sqrt{2}, Y - \sqrt{2}) = (\pm 4\sqrt{2} = 0)$$

$$X - 3\sqrt{2} = \pm 4\sqrt{2} \quad Y - \sqrt{2} = 0$$

$$X = 3\sqrt{2} \pm 4\sqrt{2} \quad Y = \sqrt{2}$$

$$\frac{1}{\sqrt{2}}(x+y) = 3\sqrt{2} \pm 4\sqrt{2} \Rightarrow x+y = 6 \pm 8 \Rightarrow x+y = 14, \quad x+y = -2$$

$$\frac{x-y=-2}{2x=12}, \quad \frac{x-y=-2}{2x=-4}$$

$$\boxed{x = 6}$$

$$\boxed{x = -2}$$

$$\Rightarrow \begin{array}{l} x = 6 \quad \& \quad x = -2 \\ 6+y = 14, \quad -2+y = -2 \end{array}$$

$$\boxed{y = 8}$$

$$\boxed{y = 0}$$

Required foci (6, 8) & (-2, 0)



Vertices = $(\pm a, 0)$

$$(X - 3\sqrt{2}, Y - \sqrt{2}) = (\pm 4, 0)$$

$$X - 3\sqrt{2} = \pm 4 \quad Y - \sqrt{2} = 0$$

$$X = 3\sqrt{2} \pm 4 \quad \frac{-1}{\sqrt{2}}(x - y) = \sqrt{2}$$

$$\Rightarrow \frac{1}{\sqrt{2}}(x + y) = 3\sqrt{2} \pm 4, \quad x - y = -2 \Rightarrow x + y = 6 \pm 4\sqrt{2}$$

$\begin{array}{rcl} x + y & = & 6 + 4\sqrt{2} \\ x - y & = & -2 \\ \hline 2x & = & 4 + 4\sqrt{2} \\ \Rightarrow x & = & 2 + 2\sqrt{2} \\ \therefore y & = & x + 2 \\ & = & 2 + 2\sqrt{2} + 2 \\ y & = & 4 + 2\sqrt{2} \end{array}$	$\begin{array}{rcl} x + y & = & 6 - 4\sqrt{2} \\ x - y & = & -2 \\ \hline 2x & = & 4 - 4\sqrt{2} \\ \Rightarrow x & = & 2 - 2\sqrt{2} \\ \therefore y & = & x + 2 \\ & = & 2 - 2\sqrt{2} + 2 \\ y & = & 4 - 2\sqrt{2} \end{array}$
--	--

Required vertices $(2 + 2\sqrt{2}, 4 + 2\sqrt{2})$ & $(2 - 2\sqrt{2}, 4 - 2\sqrt{2})$

Next, equation of focal axis

$$Y - \sqrt{2} = 0$$

$$Y = \sqrt{2}$$

$$\frac{-1}{\sqrt{2}}(x - y) = \sqrt{2}$$

$$x - y = -2$$

Equation of conjugate axis $X - 3\sqrt{2} = 0$

$$\Rightarrow X = 3\sqrt{2}$$

$$\frac{1}{\sqrt{2}}(x + y) = 3\sqrt{2}$$

$$x + y = 6$$

$$(viii) \quad x^2 + 4xy - 2y^2 - 6 = 0 \quad (1)$$

Solution:

$$a = 1, \quad 2h = 4 \quad b = -2$$

$$\tan 2\theta = \frac{2h}{a - b} = \frac{4}{3}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{4}{3}$$



$$2 \tan^2 \theta + 3 \tan \theta - 2 = 0$$

$$\tan \theta = \frac{-3 \pm \sqrt{25}}{4} = \frac{1}{2}$$

$$\sec \theta = \sqrt{1 + \tan^2 \theta} = \sqrt{1 + \frac{1}{4}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$$

$$\cos \theta = \frac{2}{\sqrt{5}} \quad \sin \theta = \frac{1}{\sqrt{5}}$$

Equations of transformation are

$$\begin{aligned} x &= X \cos \theta - Y \sin \theta \Rightarrow \frac{2X - Y}{\sqrt{5}} \dots (2) \\ y &= X \sin \theta + Y \cos \theta \Rightarrow \frac{X + 2Y}{\sqrt{5}} \dots (3) \end{aligned}$$

Put in (1) we have

$$\left(\frac{2X - Y}{\sqrt{5}} \right)^2 + 4 \left(\frac{2X - Y}{\sqrt{5}} \right) \left(\frac{X + 2Y}{\sqrt{5}} \right) - 2 \left(\frac{X + 2Y}{\sqrt{5}} \right)^2 - 6 = 0$$

$$\frac{1}{5} (4X^2 + Y^2 - 4XY) + \frac{4}{5} (2X^2 + 4XY - XY - 2Y^2) - \frac{2}{5} (X^2 + 4Y^2 + 4XY) - 6 = 0$$

$$10X^2 - 15Y^2 = 30$$

$$\frac{X^2}{3} - \frac{Y^2}{3} = 1 \quad \dots (4) \quad \text{which represents hyperbola}$$

$$a^2 = 3, \quad b^2 = 2$$

For center Put $X = 0$, $Y = 0$

$$\Rightarrow \frac{2x + y}{\sqrt{5}} = 0 \Rightarrow 2x + y = 0 \quad \& \quad Y = 0$$

$$\frac{2x - y}{\sqrt{5}} = 0 \Rightarrow -x + 2y = 0$$

Solving $2x + y = 0$ & $-x + 2y = 0$

we get $x = 0$, $y = 0$

Thus center $(0, 0)$

For transverse axis put $Y = 0$ we get

$$x - 2y = 0$$

Conjugate axis put $X = 0$, we get $2x + y = 0$

Vertices

$$(X, Y) = (\pm \sqrt{3}, 0)$$

$$\frac{2x + y}{\sqrt{5}} = \pm \sqrt{15} \quad (i) \quad \frac{2y - x}{\sqrt{5}} = 0 \Rightarrow x = 2y \quad \dots (ii)$$



By solution of (i) & (ii) we get

$$\left(2\sqrt{\frac{3}{5}}, \sqrt{\frac{3}{5}} \right), \left(-2\sqrt{\frac{3}{5}}, -\sqrt{\frac{3}{5}} \right)$$

Foci (X, Y) = $(\pm \sqrt{5}, 0)$

$$X = \pm \sqrt{5} \quad Y = 0$$

$$2x + y = \pm 5, \quad -x + 2y = 0$$

After solution of above equations

$$(2, 1) \text{ \& } (-2, -1)$$

$$(ix) \quad x^2 - 4xy - 2y^2 + 10x + 4y = 0 \quad (1)$$

Solution:

$$a = 1, \quad b = -2 \quad 2h = -4$$

$$\tan 2\theta = \frac{2h}{a-b} = \frac{-4}{3}$$

$$\frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{-4}{3} \quad \text{After cross multiplication we get}$$

$$2 \tan^2 \theta - 3 \tan \theta - 2 = 0$$

$$2 \tan^2 \theta - 4 \tan \theta + \tan \theta - 2 = 0$$

$$2 \tan \theta (\tan \theta - 2) + 1(\tan \theta - 2) = 0$$

$$\tan \theta - 2 = 0 \quad 2 \tan \theta + 1 = 0$$

$$\tan \theta = 0 \quad \cot \theta = \frac{1}{2}$$

$$\sec \theta = \sqrt{5} \quad \cos \theta = \frac{1}{\sqrt{5}} \quad \sin \theta = \frac{2}{\sqrt{5}}$$

Now equations of transformation are

$$\left. \begin{aligned} x &= X \cos \theta - Y \sin \theta = \frac{X - 2Y}{\sqrt{5}} \\ x &= X \sin \theta + Y \cos \theta = \frac{2X - Y}{\sqrt{5}} \end{aligned} \right\} \quad (2) \quad \text{Put in (1)}$$

$$\left(\frac{X - 2Y}{\sqrt{5}} \right)^2 - 4 \left(\frac{X - 2Y}{\sqrt{5}} \right) \left(\frac{2X - Y}{\sqrt{5}} \right) - 2 \left(\frac{2X - Y}{\sqrt{5}} \right)^2 + 10 \left(\frac{X - 2Y}{\sqrt{5}} \right) + 4 \left(\frac{2X - Y}{\sqrt{5}} \right) = 0$$

$$\Rightarrow -15X^2 + 10Y^2 + 18\sqrt{5}X - 16\sqrt{5}Y = 0$$

$$\Rightarrow 10 \left(Y^2 - \frac{16\sqrt{5}}{10} Y \right) - 15 \left(X^2 - \frac{18\sqrt{5}}{15} X \right) = 0$$



$$10 \left(Y^2 - \frac{8\sqrt{5}}{5} Y \right) - 15 \left(X^2 - \frac{6\sqrt{5}}{5} X \right) = 0$$

$$10 \left[Y^2 - \frac{8\sqrt{5}}{5} Y + \left(\frac{4}{\sqrt{5}} \right)^2 \right] - 15 \left[X^2 - \frac{6\sqrt{5}}{5} X + \left(\frac{3}{\sqrt{5}} \right)^2 \right] = 10 \left(\frac{16}{5} \right) - 15 \left(\frac{9}{5} \right)$$

$$10 \left(Y - \frac{4}{\sqrt{5}} \right)^2 - 15 \left(X - \frac{3}{\sqrt{5}} \right)^2 = 5$$

$$\frac{\left(Y - \frac{4}{\sqrt{5}} \right)^2}{\frac{1}{2}} - \frac{\left(X - \frac{3}{\sqrt{5}} \right)^2}{\frac{1}{3}} = 1 \quad \dots\dots (3) \quad \text{Hyperbola}$$

$$a^2 = \frac{1}{2} \quad \& \quad b^2 = \frac{1}{3}$$

$$e^2 = \frac{a^2 + b^2}{a^2} = \frac{5}{3} \Rightarrow e = \sqrt{\frac{5}{3}}$$

Center put

$$Y - \frac{4}{\sqrt{5}} = 0 \Rightarrow Y = \frac{4}{\sqrt{5}}, \quad X - \frac{3}{\sqrt{5}} = 0 \Rightarrow X = \frac{3}{\sqrt{5}}$$

$$X - 2Y = \sqrt{5} x, \quad X - 2Y = \sqrt{5} x$$

$$2X + Y = \sqrt{5} y, \quad 2X + Y = \sqrt{5} y$$

Multiplying 2nd equation by 2

$$X - 2Y = \sqrt{5} x$$

$$4X + 2Y = 2\sqrt{5} y$$

$$5X = \sqrt{5} (x + 2y)$$

$$X = \frac{1}{\sqrt{5}} (x + 2y)$$

$$\frac{3}{\sqrt{5}} = \frac{1}{\sqrt{5}} (x + 2y)$$

$$x + 2y = 3$$

$$-2x + y = 4$$

After solution we get

$$\text{Center } (-1, 2)$$

$$\text{Foci } = (0, \pm ae)$$

Multiplying equation 1st by 2

$$2X - 4Y = 2\sqrt{5} x$$

$$-2X + Y = -\sqrt{5} y$$

$$-5y = \sqrt{5} (2x - y)$$

$$Y = -\frac{1}{\sqrt{5}} (2x - y)$$

$$-\frac{1}{\sqrt{5}} (2x - y) = \frac{4}{\sqrt{5}}$$

$$-2x + y = 4$$



$$\begin{aligned} \left(X - \frac{3}{\sqrt{5}}, Y - \frac{4}{\sqrt{5}} \right) &= \left(0, \pm \frac{\sqrt{5}}{\sqrt{6}} \right) \\ X - \frac{3}{\sqrt{5}} &= 0 & Y - \frac{4}{\sqrt{5}} &= \pm \frac{\sqrt{5}}{\sqrt{6}} \\ X &= \frac{3}{\sqrt{5}} & Y &= \frac{4}{\sqrt{5}} \pm \frac{\sqrt{5}}{\sqrt{6}} \\ \frac{1}{\sqrt{5}}(x+2y) &= \frac{3}{\sqrt{5}} & \frac{-1}{\sqrt{5}}(2x-2y) &= \frac{4}{\sqrt{5}} + \frac{\sqrt{5}}{\sqrt{6}} \\ x+2y &= 3 & -2x+y &= 4 \pm \frac{5}{\sqrt{6}} \end{aligned}$$

After simplification foci are

$$\left(-1 - \frac{2}{\sqrt{6}}, 2 + \frac{1}{\sqrt{6}} \right) \quad \& \quad \left(-1 + \frac{2}{\sqrt{6}}, 2 - \frac{1}{\sqrt{6}} \right)$$

Q.2: Show that

$10xy + 8x - 15y - 12 = 0$ represents a pair of straight lines and find an equation of each line.

Solution:

$$10xy + 8x - 15y - 12 = 0 \quad \dots\dots (1)$$

$$\text{Compare it with } ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$a = 0, \quad h = 5, \quad b = 0, \quad g = 4, \quad f = \frac{-15}{2}, \quad C = -12$$

$$\text{Now consider determinant} = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \begin{vmatrix} 0 & 5 & 4 \\ 5 & 0 & \frac{-15}{2} \\ 4 & \frac{-15}{2} & -12 \end{vmatrix}$$

$$= 0 \begin{vmatrix} 0 & \frac{-15}{2} \\ -\frac{15}{2} & -12 \end{vmatrix} - 5 \begin{vmatrix} 5 & \frac{-15}{2} \\ 4 & -12 \end{vmatrix} + 4 \begin{vmatrix} 5 & 0 \\ 4 & \frac{-15}{2} \end{vmatrix}$$

$$= 0 - 5(-60 + 30) + 4\left(\frac{-75}{2}\right) \Rightarrow 150 - 150 = 0$$

This shows that equation (1) represents a pair of straight line, now we'll solve (1) as



$$10xy + 8x - 15y - 12 = 0$$

$$2x(5y + 4) - 3(5y + 4) = 0$$

$$(5y + 4)(2x - 3) = 0$$

$$5y + 4 = 0, \quad 2x - 3 = 0$$

$$(ii) \quad 6x^2 + xy - y^2 - 21x - 8y + 9 = 0 \quad (1)$$

Solution:

$$\text{Comparing by } ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$a = 6, \quad h = \frac{1}{2}, \quad b = -1, \quad g = \frac{-21}{2}, \quad f = -4, \quad c = 9$$

$$\text{Now consider determinant} = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$$\begin{aligned} &= \begin{vmatrix} 6 & \frac{1}{2} & \frac{-21}{2} \\ \frac{1}{2} & -1 & -4 \\ \frac{-21}{2} & -4 & 9 \end{vmatrix} \\ &= 6 \begin{vmatrix} -1 & -4 \\ -4 & 9 \end{vmatrix} - \frac{1}{2} \begin{vmatrix} \frac{1}{2} & -4 \\ \frac{-21}{2} & 9 \end{vmatrix} - \frac{21}{2} \begin{vmatrix} \frac{1}{2} & -1 \\ \frac{-21}{2} & -4 \end{vmatrix} \\ &= 6(-9 - 16) - \frac{1}{2} \left(\frac{9}{2} - \frac{84}{2} \right) - \frac{21}{2} \left(\frac{-4}{2} - \frac{21}{2} \right) \\ &= 6(-25) - \frac{1}{2} \left(\frac{-75}{2} \right) - \frac{21}{2} \left(\frac{-25}{2} \right) \\ &= -150 + \frac{75}{4} + \frac{525}{4} \\ &= -150 + 150 = 0 \end{aligned}$$

This shows that given equation (1) represents a pair of straight lines. Now rearranging (1) we have

$$6x^2 + (y - 21)x - (y^2 + 8y - 9) = 0$$



By Quadratic formula

$$x = \frac{-(y-21) \pm \sqrt{(y-21)^2 + 4(6)(y^2 + 8y - 9)}}{2(6)}$$

$$x = \frac{-(y-21) \pm \sqrt{y^2 + 441 - 42y + 24y^2 + 192y - 216}}{12}$$

$$x = \frac{-y + 21 \pm \sqrt{25y^2 + 150y + 225}}{12}$$

$$x = \frac{-y + 21 \pm \sqrt{(5y + 15)^2}}{12} = \frac{-y + 21 \pm (5y + 15)}{12}$$

$$x = \frac{-y + 21 + 5y + 15}{12}, \quad \frac{-y + 21 - 5y - 15}{12}$$

$$x = \frac{4y + 36}{12} \quad x = \frac{-6y + 6}{12}$$

$$x = \frac{4(y + 9)}{12} \quad x = \frac{6(-y + 1)}{12}$$

$$x = \frac{y + 9}{3} \quad x = \frac{-y + 1}{2}$$

$$3x - y - 9 = 0, \quad 2x + y - 1 = 0 \quad \text{Ans.}$$

Q.3: Find an equation of tangent to each of the given conics at indicated point.

(i) $3x^2 - 7y^2 + 2x - y - 48 = 0$ at $(4, 1)$

Solution:

$$3x^2 - 7y^2 + 2x - y - 48 = 0 \quad \text{at } (4, 1)$$

Diff. w.r.t 'x'

$$6x - 14y \frac{dy}{dx} + 2 - \frac{dy}{dx} = 0$$

$$(-14y - 1) \frac{dy}{dx} = -6x - 2$$

$$\frac{dy}{dx} = \frac{-6x - 2}{-14y - 1} = \frac{6x + 2}{14y + 1}$$

$$m = \left. \frac{dy}{dx} \right|_{(4, 1)} = \frac{6(4) + 2}{14(1) + 1} = \frac{26}{15}$$

Equation of tangent at the point $(4, 1)$ is

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{26}{15} (x - 4)$$



$$15y - 15 = 26x - 104$$

$$\Rightarrow \boxed{26x - 15y - 89 = 0}$$

(ii) $x^2 + 5xy - 4y^2 + 4 = 0$ at $y = -1$

Solution:

$$x^2 + 5xy - 4y^2 + 4 = 0$$

$$\text{Put } y = -1$$

$$x^2 - 5x - 4 + 4 = 0$$

$$x(x - 5) = 0$$

$$x = 0, \quad x = 5$$

We have two points $(0, -1)$ & $(5, -1)$

Now diff (1) w.r.t 'x'

$$2x + 5 \left[y + \frac{dy}{dx} x \right] - 8y \frac{dy}{dx} = 0$$

$$2x + 5y + 5x \frac{dy}{dx} - 8y \frac{dy}{dx} = 0$$

$$(5x - 8y) \frac{dy}{dx} = -2x - 5y$$

$$\frac{dy}{dx} = \frac{-(2x + 5y)}{-(8y - 5x)}$$

$$m = \frac{dy}{dx} \Big|_{(0, -1)} = \frac{2(0) + 5(-1)}{8(-1) - 5(0)} = \frac{-5}{-8} = \frac{5}{8}$$

Equation of tangent at point $(0, -1)$ is

$$y - y_1 = m(x - x_1)$$

$$y + 1 = \frac{5}{8} (x - 0)$$

$$8y + 8 = 5x \Rightarrow \boxed{5x - 8y - 8 = 0} \quad \text{Ans}$$

Next

$$m = \frac{dy}{dx} \Big|_{(5, -1)} = \frac{2(5) + 5(-1)}{8(-1) - 5(5)} = \frac{10 - 5}{-8 - 25} = \frac{5}{-33}$$

Equation of tangent at $(5, -1)$ is

$$y - y_1 = m(x - x_1)$$

$$y + 1 = \frac{-5}{33} (x - 5)$$

$$33y + 33 = -5x + 25$$

$$\boxed{5x + 33y = -8} \quad \text{Ans}$$



(iii) $x^2 + 4xy - 3y^2 - 5x - 9y + 6 = 0$ at $x = 3$

Solution:

$$x^2 + 4xy - 3y^2 - 5x - 9y + 6 = 0 \quad \dots (1)$$

Put $x = 3$

$$9 + 12y - 3y^2 - 15 - 9y + 6 = 0$$

$$-3y^2 + 3y = 0$$

$$-3y(y - 1) = 0$$

$$\Rightarrow y = 0, \quad y = 1$$

Required points are (3, 0) & (3, 1)

Diff. (1) w.r.t 'x'

$$2x + 4[y + x \frac{dy}{dx}] - 3(2y \frac{dy}{dx}) - 5 - 9 \frac{dy}{dx} = 0$$

$$2x + 4y + 4x \frac{dy}{dx} - 6y \frac{dy}{dx} - 5 - 9 \frac{dy}{dx} = 0$$

$$(4x - 6y - 9) \frac{dy}{dx} = -2x - 4y + 5$$

$$\frac{dy}{dx} = \frac{-2x - 4y + 5}{4x - 6y - 9}$$

$$m = \frac{dy}{dx} \Big|_{(3,0)} = \frac{-2(3) - 4(0) + 5}{4(3) - 6(0) - 9}$$

$$m = \frac{-6 + 5}{12 - 9} = \frac{-1}{3}$$

Equation of tangent at (3, 0)

$$y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{-1}{3} (x - 3)$$

$$3y = -x + 3 \Rightarrow \boxed{x + 3y - 3 = 0}$$

Ans.

$$\text{Next, } m = \frac{dy}{dx} \Big|_{(3,1)} = \frac{-2(3) - 4(1) + 5}{4(3) - 6(1) - 9} = \frac{-6 - 4 + 5}{12 - 6 - 9} = \frac{5}{3}$$

Equation of tangent at (3, 1)

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{5}{3} (x - 3)$$

$$3y - 3 = 5x - 15$$

$$\boxed{5x - 3y - 12 = 0}$$

